

ROB-GY 6103

Advanced Mechatronics

Lecture 8

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Software Installation Steps

- **Run jor-setup1.sh shell script (see script details in following slides)**
- **Update package database on your Ubuntu system by downloading from repos information about newest packages**
 - No installation or upgrade of packages
 - Update info about latest Ver. and dependencies for various packages
- **Auto-setup for visudo: obviate password entry on invocation of sudo**
- **Auto-install following packages**
 - nano, idle, libcanberra.gtk-module, libcanberra.gtk3-module, chromium, jtop, visual studio code, conda, ...
- **Fix some user permissions**

Shell Script: jonSetup1-DWNLD.sh

```
#!/bin/bash

# --- Download a folder with various edited files ---
echo "Setting up some path aliases..."
HOME_DIR="/home/vk-jn-or"
cd "$HOME_DIR"
DWNLD_PARENT="$HOME_DIR/Downloads/Jetson-Orin-Nano"
mkdir -p $DWNLD_PARENT
# Correct format for wget to download from goggle drive
echo "Downloading and unzipping a folder with various pre-edited files..."
wget --no-check-certificate
'https://docs.google.com/uc?export=download&id=1tI9cJS2W500hivBIsbwpRjh5MJDFygCk
' -O "$DWNLD_PARENT/AM_JON_2SHARE.zip"
cd "$DWNLD_PARENT"
unzip AM_JON_2SHARE.zip
rm AM_JON_2SHARE.zip
cd "$HOME_DIR"

echo "Converting shell scripts to executables..."
find "$DWNLD_PARENT/AM_JON_2SHARE/shellScripts/" -type f -name "*.sh" -exec
chmod +x {} \;

echo "Download and unzip completed..."
```

Download a resources folder

Convert shell scripts into executables

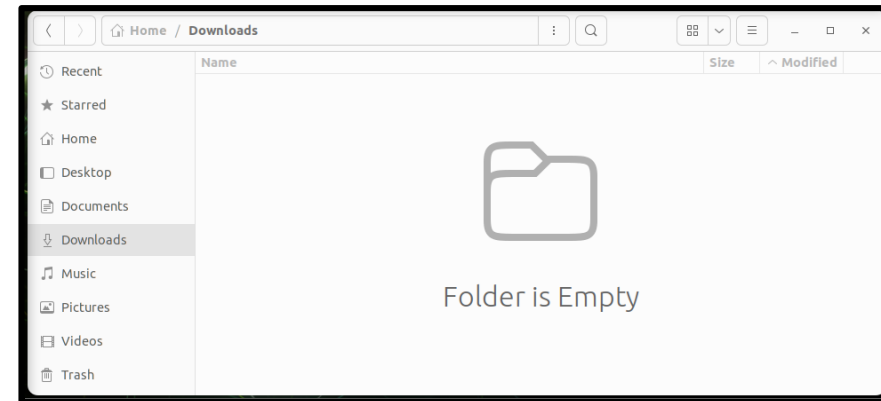
Terminal O/P: jonSetup1-DWNLD.sh

```
Open  jonSetup1-DWNLD.sh  Save
~/Desktop

1 #!/bin/bash
2
3 # --- Download a folder with various edited files ---
4 echo "Setting up some path aliases..."
5 HOME_DIR="/home/vk-jn-or"
6 cd "$HOME_DIR"
7 DWNLD_PARENT="$HOME_DIR/Downloads/Jetson-Orin-Nano"
8 mkdir -p $DWNLD_PARENT
9 # Correct format for wget to download from goggle drive
10 echo "Downloading and unzipping a folder with various pre-edited files..."
11 wget --no-check-certificate 'https://docs.google.com/uc?
    export=download&id=1tI9cJS2W500hivBIsbwpRjh5MJDFygCk' -O "$DWNLD_PARENT/AM_JON_2SHARE.zip"
12 cd "$DWNLD_PARENT"
13 unzip AM_JON_2SHARE.zip
14 rm AM_JON_2SHARE.zip
15 cd "$HOME_DIR"
16
17 echo "Converting shell scripts to executables..."
18 find "$DWNLD_PARENT/AM_JON_2SHARE/shellScripts/" -type f -name "*.sh" -exec chmod +x {} \;
19
20 echo "Download and unzip completed..."
```

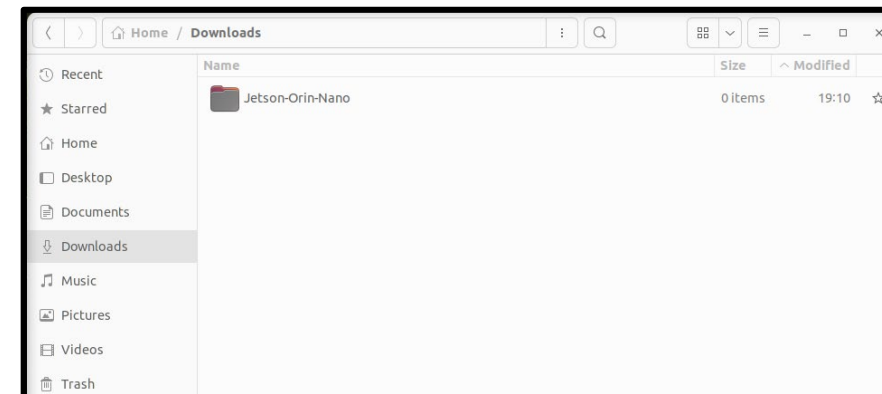
```
vk-jn-or@vkjnor-desktop: ~/Desktop

vk-jn-or@vkjnor-desktop:~$ cd Desktop/
vk-jn-or@vkjnor-desktop:~/Desktop$ chmod +x jonSetup1-DWNLD.sh
vk-jn-or@vkjnor-desktop:~/Desktop$ ./jonSetup1-DWNLD.sh
```



```
vk-jn-or@vkjnor-desktop: ~/Desktop

vk-jn-or@vkjnor-desktop:~$ cd Desktop/
vk-jn-or@vkjnor-desktop:~/Desktop$ chmod +x jonSetup1-DWNLD.sh
vk-jn-or@vkjnor-desktop:~/Desktop$ ./jonSetup1-DWNLD.sh
Setting up some path aliases...
Downloading and unzipping a folder with various pre-edited files...
--2026-02-09 19:10:55-- https://docs.google.com/uc?export=download&id=1tI9cJS2W500hivBIsbwpRjh5MJDFygCk
Resolving docs.google.com (docs.google.com)... 2607:f8b0:4006:80e::200e, 142.250.217.142
Connecting to docs.google.com (docs.google.com)[2607:f8b0:4006:80e::200e]:443... connected.
HTTP request sent, awaiting response... 303 See Other
Location: https://drive.usercontent.google.com/download?id=1tI9cJS2W500hivBIsbwpRjh5MJDFygCk&export=download [following]
--2026-02-09 19:10:56-- https://drive.usercontent.google.com/download?id=1tI9cJS2W500hivBIsbwpRjh5MJDFygCk&export=download
Resolving drive.usercontent.google.com (drive.usercontent.google.com)... 2607:f8b0:4006:815::2001, 142.250.80.65
Connecting to drive.usercontent.google.com (drive.usercontent.google.com)[2607:f8b0:4006:815::2001]:443... connected.
HTTP request sent, awaiting response...
```



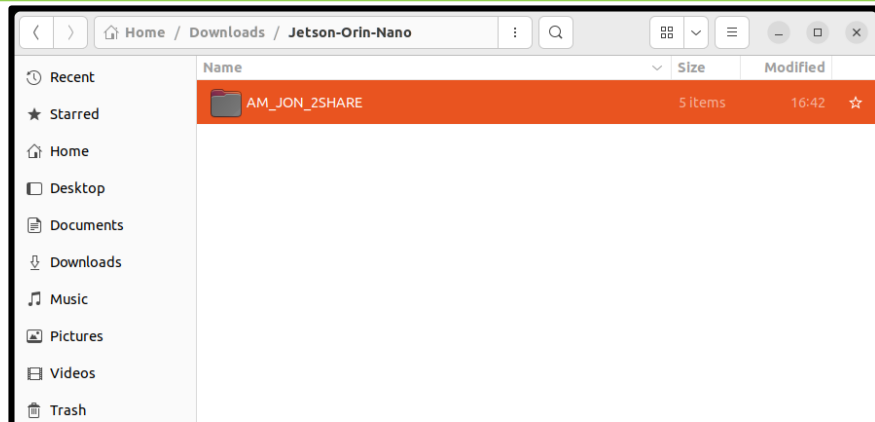
Terminal O/P: jonSetup1-DWNLD.sh

```
vk-jn-or@vkjnor-desktop: ~/Desktop
vk-jn-or@vkjnor-desktop: ~/Desktop$ ./jonSetup1-DWNLD.sh
Setting up some path aliases...
Downloading and unzipping a folder with various pre-edited files...
--2026-02-19 17:28:07-- https://docs.google.com/uc?export=download&id=1tI9cJS2W500hVBIswprJh5MJDFygc
Resolving docs.google.com (docs.google.com)... 2607:f8b0:4006:80c::200e, 142.251.210.46
Connecting to docs.google.com (docs.google.com)[2607:f8b0:4006:80c::200e]:443... connected.
HTTP request sent, awaiting response... 303 See Other
Location: https://drive.usercontent.google.com/download?id=1tI9cJS2W500hVBIswprJh5MJDFygc&export=download [following]
--2026-02-19 17:28:07-- https://drive.usercontent.google.com/download?id=1tI9cJS2W500hVBIswprJh5MJDFygc&export=download
Resolving drive.usercontent.google.com (drive.usercontent.google.com)... 2607:f8b0:4006:800::2001, 142.251.35.161
Connecting to drive.usercontent.google.com (drive.usercontent.google.com)[2607:f8b0:4006:800::2001]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13306347 (13M) [application/octet-stream]
Saving to: '/home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE.zip'

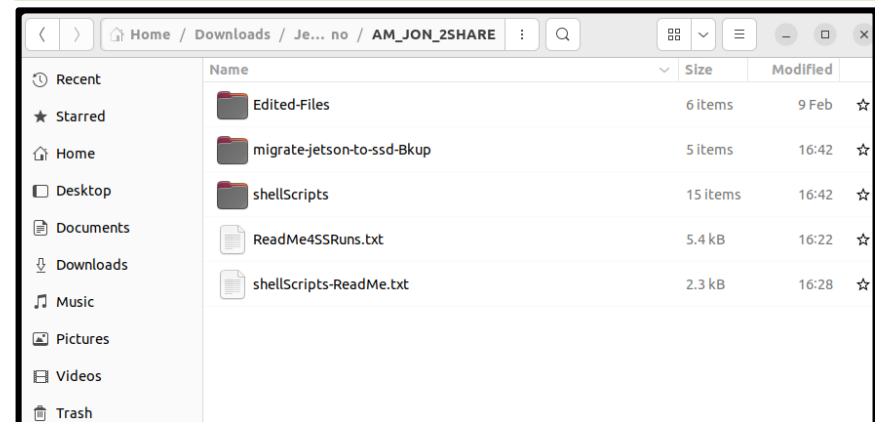
/home/vk-jn-or/Downloads/Jetson- 100%[=====] 12.69M 32.5MB/s in 0.4s

2026-02-19 17:28:10 (32.5 MB/s) - '/home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE.zip' saved [13306347/13306347]

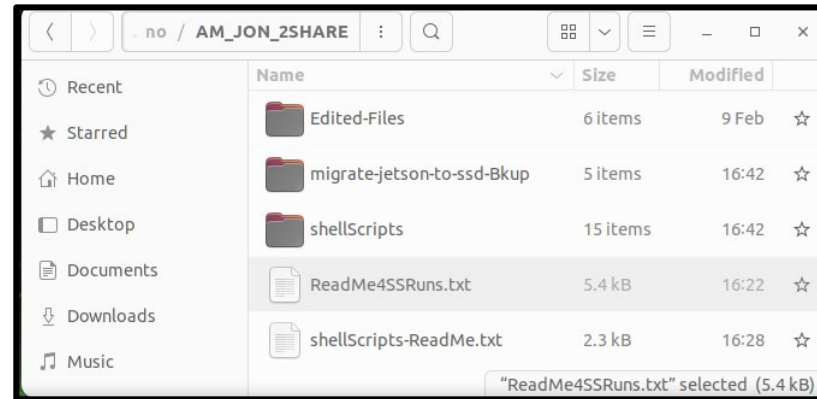
Archive: AM_JON_2SHARE.zip
  creating: AM_JON_2SHARE/
  creating: AM_JON_2SHARE/Edited-Files/
  creating: AM_JON_2SHARE/Edited-Files/Face_Recg/
  inflating: AM_JON_2SHARE/Edited-Files/Face_Recg/doorbell_camera_1.py
  creating: AM_JON_2SHARE/Edited-Files/home_vk-jn-or/
  inflating: AM_JON_2SHARE/Edited-Files/home_vk-jn-or/condaActivate.sh
  inflating: AM_JON_2SHARE/Edited-Files/home_vk-jn-or/condaActivate2.sh
  inflating: AM_JON_2SHARE/Edited-Files/home_vk-jn-or/condaDeactivate.sh
  inflating: AM_JON_2SHARE/Edited-Files/home_vk-jn-or/condaDeactivate2.sh
  inflating: AM_JON_2SHARE/Edited-Files/home_vk-jn-or/launch_cameras.sh
  creating: AM_JON_2SHARE/Edited-Files/home_vk-jn-or/Pictures/
  inflating: AM_JON_2SHARE/Edited-Files/home_vk-jn-or/Pictures/mpvCam.png
  creating: AM_JON_2SHARE/Edited-Files/jetson-inference/
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/dockerOn.sh
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/mem_maint_4_dockerOn.sh
  creating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/
  creating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backupcppFiles/
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backupcppFiles/camera-capture.cpp
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backupcppFiles/detectnet.cpp
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backupcppFiles/imagenet.cpp
  creating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backupcppFiles/updatedFiles/
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backupcppFiles/updatedFiles/camera-capture.cpp
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backupcppFiles/updatedFiles/detectnet.cpp
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backupcppFiles/updatedFiles/imagenet.cpp
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backupcppFiles/updatedFiles/video-viewer.cpp
  creating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backup_pyFiles/
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backup_pyFiles/detectnet_BAB.py
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backup_pyFiles/detectnet_PE.py
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backup_pyFiles/imagenet_BAB.py
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backup_pyFiles/imagenet_PE.py
  creating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backup_pyFiles/updatedFiles/
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backup_pyFiles/updatedFiles/detectnet_BAB.py
  inflating: AM_JON_2SHARE/Edited-Files/jetson-inference/usrFolder/backup_pyFiles/updatedFiles/detectnet_PE.py
```



```
vk-jn-or@vkjnor-desktop: ~/Desktop
creating: AM_JON_2SHARE/Edited-Files/VSC/VSC-VK/VK/vkFolder/vkPyExamples/pyVSC/sourceData1/
creating: AM_JON_2SHARE/Edited-Files/VSC/VSC-VK/VK/vkFolder/vkPyExamples/pyVSC/sourceData1/Advance/
creating: AM_JON_2SHARE/Edited-Files/VSC/VSC-VK/VK/vkFolder/vkPyExamples/pyVSC/sourceData1/Basics/
creating: AM_JON_2SHARE/Edited-Files/VSC/VSC-VK/VK/vkFolder/vkPyExamples/pyVSC/sourceData1/Basics1/
creating: AM_JON_2SHARE/Edited-Files/VSC/VSC-VK/VK/vkFolder/vkPyExamples/pyVSC/sourceData1/Basics2/
creating: AM_JON_2SHARE/Edited-Files/VSC/VSC-VK/VK/vkFolder/vkPyExamples/pyVSC/sourceData1/Basics3/
creating: AM_JON_2SHARE/Edited-Files/VSC/VSC-VK/VK/vkFolder/vkPyExamples/pyVSC/sourceData1/GPUTest/
creating: AM_JON_2SHARE/Edited-Files/VSC/VSC-VK/VK/vkFolder/vkPyExamples/pyVSC/sourceData1/GPUTest/DestImages
-SingleImgTest/
  inflating: AM_JON_2SHARE/Edited-Files/VSC/VSC-VK/VK/vkFolder/vkPyExamples/pyVSC/sourceData1/GPUTest/DestImages
-SingleImgTest/Image1.jpg
  inflating: AM_JON_2SHARE/Edited-Files/VSC/VSC-VK/VK/vkFolder/vkPyExamples/pyVSC/sourceData1/GPUTest/DestImages
-SingleImgTest/Image2.jpg
  creating: AM_JON_2SHARE/Edited-Files/VSC/VSC-VK/VK/vkFolder/vkPyExamples/Testing/
  creating: AM_JON_2SHARE/Edited-Files/VSC/VSC-VK/VK/vkFolder/vkPyExamples/Testing/Face-Recognition/
  inflating: AM_JON_2SHARE/Edited-Files/VSC/VSC-VK/VK/vkFolder/vkPyExamples/VK-OR-Samples.code1-workspace.code-w
orkspace
  creating: AM_JON_2SHARE/shellScripts/
  creating: AM_JON_2SHARE/shellScripts/Script1/
  inflating: AM_JON_2SHARE/shellScripts/Script1/jonSetup1-DWNLD.sh
  creating: AM_JON_2SHARE/shellScripts/Script10/
  inflating: AM_JON_2SHARE/shellScripts/Script10/jonSetup10-conJI-Apps.sh
  creating: AM_JON_2SHARE/shellScripts/Script11/
  inflating: AM_JON_2SHARE/shellScripts/Script11/jonSetup11-dckr.sh
  creating: AM_JON_2SHARE/shellScripts/Script12/
  inflating: AM_JON_2SHARE/shellScripts/Script12/jonSetup12-dckr.sh
  creating: AM_JON_2SHARE/shellScripts/Script13/
  inflating: AM_JON_2SHARE/shellScripts/Script13/jonSetup13-dckr.sh
  creating: AM_JON_2SHARE/shellScripts/Script14/
  inflating: AM_JON_2SHARE/shellScripts/Script14/jonSetup14.sh
  creating: AM_JON_2SHARE/shellScripts/Script2/
  inflating: AM_JON_2SHARE/shellScripts/Script2/jonSetup2-status.sh
  creating: AM_JON_2SHARE/shellScripts/Script2/jonSetup2.sh
  inflating: AM_JON_2SHARE/shellScripts/Script3/
  inflating: AM_JON_2SHARE/shellScripts/Script3/jonSetup3-sound.sh
  creating: AM_JON_2SHARE/shellScripts/Script4/
  inflating: AM_JON_2SHARE/shellScripts/Script4/jonSetup4-nvcc-ptsh.sh
  creating: AM_JON_2SHARE/shellScripts/Script5/
  inflating: AM_JON_2SHARE/shellScripts/Script5/jonSetup5-CS1.sh
  creating: AM_JON_2SHARE/shellScripts/Script6/
  inflating: AM_JON_2SHARE/shellScripts/Script6/jonSetup6-CV-Pkgs.sh
  inflating: AM_JON_2SHARE/shellScripts/Script6/jonSetup6-status.sh
  creating: AM_JON_2SHARE/shellScripts/Script7/
  inflating: AM_JON_2SHARE/shellScripts/Script7/jonSetup7-doorCam.sh
  inflating: AM_JON_2SHARE/shellScripts/Script7/jonSetup7-status.sh
  creating: AM_JON_2SHARE/shellScripts/Script8/
  inflating: AM_JON_2SHARE/shellScripts/Script8/jonSetup8-status.sh
  inflating: AM_JON_2SHARE/shellScripts/Script8/josSetup8-camApps.sh
  creating: AM_JON_2SHARE/shellScripts/Script9/
  inflating: AM_JON_2SHARE/shellScripts/Script9/jonSetup9-condaII.sh
  inflating: AM_JON_2SHARE/shellScripts/Script9/jonSetup9-status.sh
  inflating: AM_JON_2SHARE/shellScripts/Script9/ReadMe.txt
Converting shell Scripts to executables...
Download and unzip completed...
vk-jn-or@vkjnor-desktop: ~/Desktop$
```



ReadMe4SSRuns.txt



```
Open  ReadMe4SSRuns.txt  Save
~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE

1
2 ##### jonSetup1-DWNLD.sh #####
3 cd Desktop/
4 chmod +x jonSetup1-DWNLD.sh
5 ./jonSetup1-DWNLD.sh
6
7
8 ##### jonSetup2.sh #####
9 cd /home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script2
10 ./jonSetup2.sh
11
12 # jonSetup2-status.sh
13 ./jonSetup2-status.sh
14
15 ##### jonSetup3-sound.sh #####
16 cd /home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script3
17 # Check current sound o/p device:
18 pacmd list-sinks | grep -E '\* index:name:'
19 # Run the script to update to HDMI
20 ./jonSetup3-sound.sh
21 # Check current sound o/p device:
22 pacmd list-sinks | grep -E '\* index:name:'
23
24 ##### jonSetup4-nvcc-ptsh.sh #####
25 cd /home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script4
26 ./jonSetup4-nvcc-ptsh.sh
27
28 ##### jonSetup5-CSI.sh #####
29 cd /home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script5/
30 ./jonSetup5-CSI.sh
31
32 # conda and jtop trials
33 condaon1
34 condaoff1
35 jtop
36
37 # jetson clocks and jtop trials
```

**Copy and paste cd
path and
./filename.sh
instructions in
terminal using this
.txt doc**

Shell Script: jonSetup2.sh

```
#!/bin/bash
```

```
# --- JETSON ORIN NANO SETUP SCRIPT ---
```

```
echo "Starting fresh install jonSetup2.sh..."
```

```
# --- 1. AUTOMATE VISUDO (No Password Access) ---
```

```
echo "[1/7] Obviate the need to enter pswd for sudo..."
```

```
sudo apt-get update
```

```
echo "$USER ALL=(ALL) NOPASSWD: ALL" | sudo tee /etc/sudoers.d/$USER
```

```
sudo chmod 0440 /etc/sudoers.d/$USER
```

Use sudo without pswd

```
# --- 2. INSTALL NANO & IDLE ---
```

```
echo "[2/7] Installing nano and idle..."
```

```
sudo dpkg --configure -a
```

```
sudo apt --fix-broken install -y
```

```
sudo apt-get install -y nano idle libcanberra.gtk-module libcanberra.gtk3-module
```

Install nano, idle, ...

```
# --- 3. CHROMIUM & SNAP FIX ---
```

```
echo "[3/7] Installing Chromium and fixing Snapd..."
```

```
sudo apt-get update
```

```
sudo snap install chromium
```

Chromium browser

```
mkdir -p ~/Downloads/Jetson-Orin-Nano
```

```
cd ~/Downloads/Jetson-Orin-Nano
```

```
snap download snapd --revision=24724
```

```
sudo snap ack snapd_24724.assert
```

```
sudo snap install snapd_24724.snap
```

```
sudo snap refresh --hold snapd
```

```
cd ~
```

```
# --- 4. VISUAL STUDIO CODE (ARM64) ---
echo "[4/7] Installing VS Code..."
cd ~/Downloads/Jetson-Orin-Nano
wget -O vscode_arm64.deb "https://code.visualstudio.com/sha/download?build=stable&os=linux-deb-arm64"
sudo dpkg -i vscode_arm64.deb
sudo apt-get install -f -y
```

Install VSC

```
# --- 5. CONDA (MINIFORGE) - FRESH INSTALL ---
echo "[5/7] Downloading and installing Miniforge..."
cd ~/Downloads/Jetson-Orin-Nano

# Download fresh for this new install
wget https://github.com/conda-forge/miniforge/releases/latest/download/Miniforge3-Linux-aarch64.sh
bash Miniforge3-Linux-aarch64.sh -b -p $HOME/miniforge3
```

Adds a block of code to `~/.bashrc`
that points to `conda.sh`

```
cd ~

# Initialize Conda using the full path for a new install
$HOME/miniforge3/bin/conda init
$HOME/miniforge3/bin/conda config --set auto_activate_base false
$HOME/miniforge3/bin/conda create -y -n vk1_env python=3.10.12
```

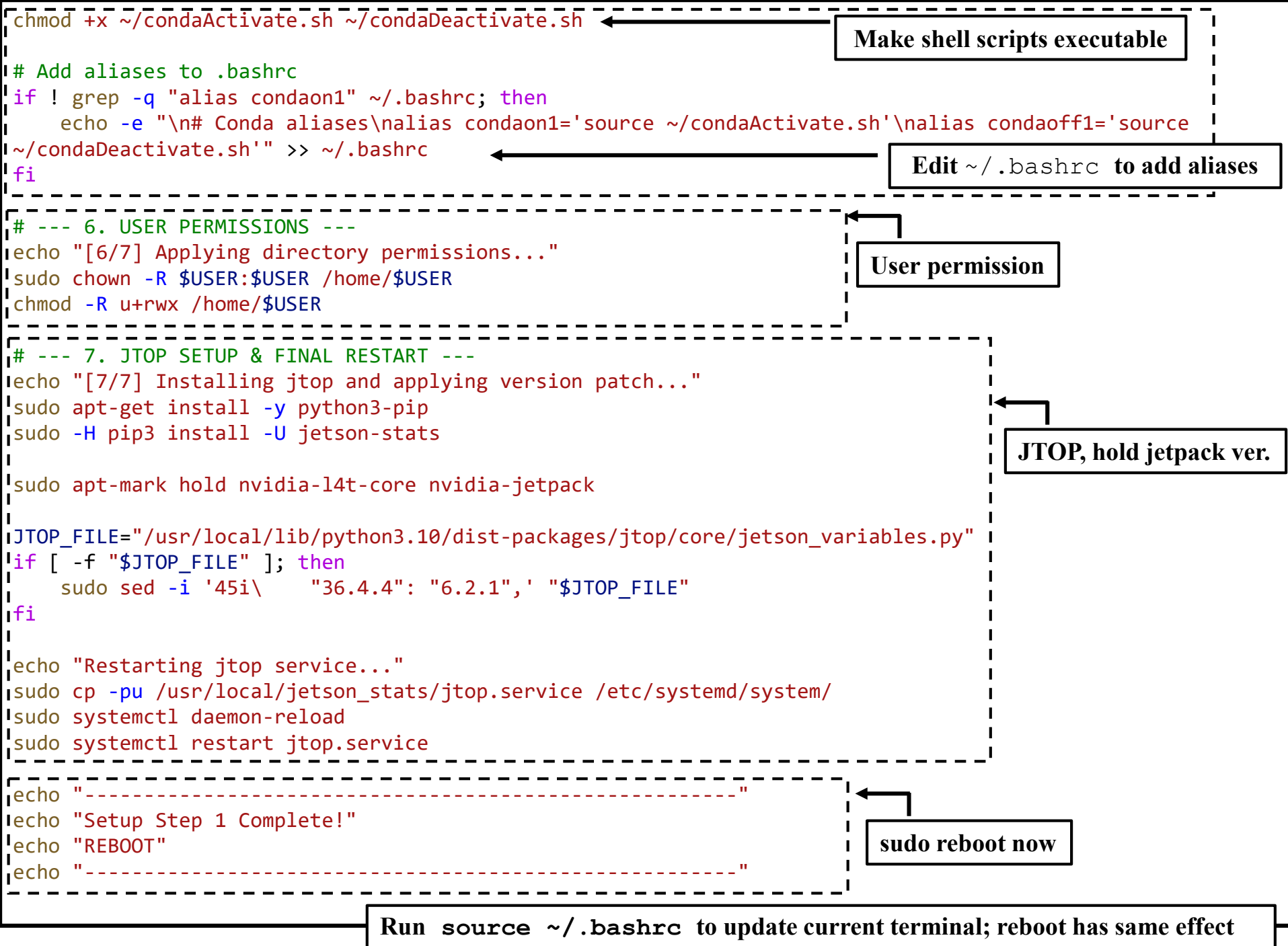
Install conda

```
# Create activation helper scripts
cat <<EOF > ~/condaActivate.sh
#!/bin/bash
. "$HOME/miniforge3/etc/profile.d/conda.sh"
conda activate vk1_env
EOF
```

Conda activate/
deactivate shell
scripts

```
cat <<EOF > ~/condaDeactivate.sh
#!/bin/bash
conda deactivate
source ~/.bashrc
EOF
```

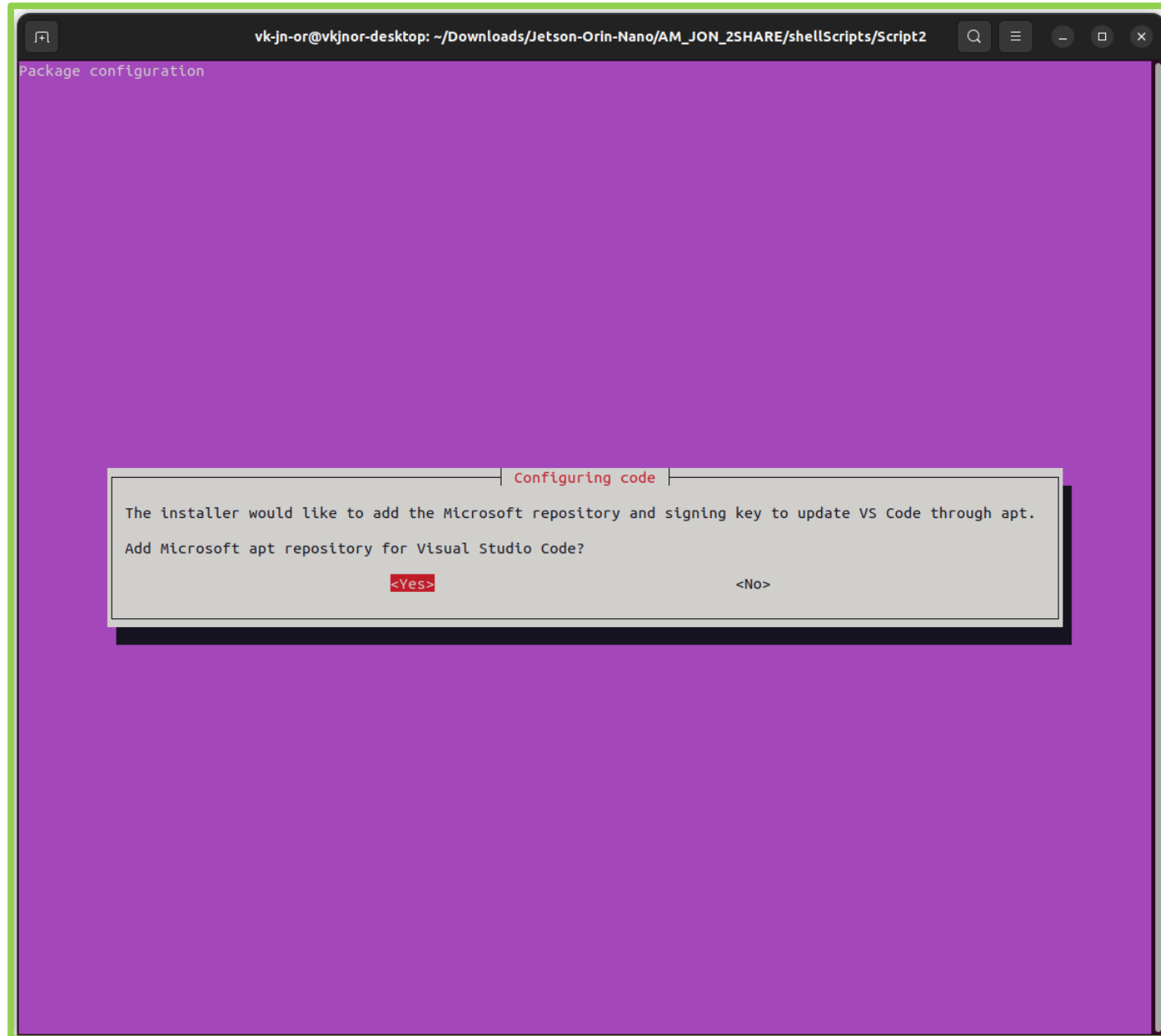
`~/.bashrc`: hidden file in home dir. of user, executed upon launching new Linux Terminal
`source ~/.bashrc`: executes this file in current terminal subshell session



Terminal O/P: jonSetup2.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script2
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script2$ ./jonSetup2.sh
Starting fresh install jonSetup2.sh...
[1/7] Obviate the need to enter pswd for sudo...
[sudo] password for vk-jn-or:
Get:1 file:/var/cudnn-local-tegra-repo-ubuntu2204-9.3.0 InRelease [1,572 B]
Get:1 file:/var/cudnn-local-tegra-repo-ubuntu2204-9.3.0 InRelease [1,572 B]
Get:2 file:/var/l4t-cuda-tegra-repo-ubuntu2204-12-6-local InRelease [1,572 B]
Get:2 file:/var/l4t-cuda-tegra-repo-ubuntu2204-12-6-local InRelease [1,572 B]
Get:3 file:/var/nv-tensorrt-local-tegra-repo-ubuntu2204-10.3.0-cuda-12.5 InRelease [1,572 B]
Get:3 file:/var/nv-tensorrt-local-tegra-repo-ubuntu2204-10.3.0-cuda-12.5 InRelease [1,572 B]
Hit:4 http://ports.ubuntu.com/ubuntu-ports jammy InRelease
Hit:5 http://ports.ubuntu.com/ubuntu-ports jammy-updates InRelease
Hit:6 http://ports.ubuntu.com/ubuntu-ports jammy-backports InRelease
Hit:7 https://download.docker.com/linux/ubuntu jammy InRelease
Hit:8 http://ports.ubuntu.com/ubuntu-ports jammy-security InRelease
Hit:9 https://repo.download.nvidia.com/jetson/common r36.4 InRelease
Hit:10 https://repo.download.nvidia.com/jetson/t234 r36.4 InRelease
Hit:11 https://repo.download.nvidia.com/jetson/ffmpeg r36.4 InRelease
Reading package lists... Done
vk-jn-or ALL=(ALL) NOPASSWD: ALL
[2/7] Installing nano and idle...
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
0 upgraded, 0 newly installed, 0 to remove and 310 not upgraded.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Note, selecting 'libcanberra-gtk-module' for regex 'libcanberra.gtk-module'
Note, selecting 'libcanberra-gtk3-module' for regex 'libcanberra.gtk3-module'
libcanberra-gtk3-module is already the newest version (0.30-10ubuntu1.22.04.1).
libcanberra-gtk3-module set to manually installed.
The following additional packages will be installed:
  fonts-mathjax idle-python3.10 libcanberra-gtk0 libjs-mathjax
Suggested packages:
  fonts-mathjax-extras fonts-stix libjs-mathjax-doc hunspell
The following NEW packages will be installed:
  fonts-mathjax idle-python3.10 libcanberra-gtk-module libcanberra-gtk0 libjs-mathjax nano
0 upgraded, 7 newly installed, 0 to remove and 310 not upgraded.
Need to get 8,508 kB of archives.
After this operation, 49.4 MB of additional disk space will be used.
Get:1 http://ports.ubuntu.com/ubuntu-ports jammy-updates/main arm64 nano arm64 6.2-1ubuntu0.1 [277 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports jammy/main arm64 fonts-mathjax all 2.7.9+dfsg-1 [2,208 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports jammy/main arm64 libjs-mathjax all 2.7.9+dfsg-1 [5,665 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports jammy-updates/universe arm64 idle-python3.10 all 3.10.12-1~22.04.14 [337 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports jammy-updates/universe arm64 idle all 3.10.6-1~22.04.1 [2,730 B]
Get:6 http://ports.ubuntu.com/ubuntu-ports jammy-updates/universe arm64 libcanberra-gtk0 arm64 0.30-10ubuntu1.22.04.1 [7,730 B]
Get:7 http://ports.ubuntu.com/ubuntu-ports jammy-updates/universe arm64 libcanberra-gtk-module arm64 0.30-10ubuntu1.22.04.1 [10.1 kB]
Fetched 8,508 kB in 2s (3,765 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package nano.
(Reading database ... 200022 files and directories currently installed.)
Preparing to unpack .../0-nano_6.2-1ubuntu0.1_arm64.deb ...
```

Shell Script Execution: VSC Setup Message

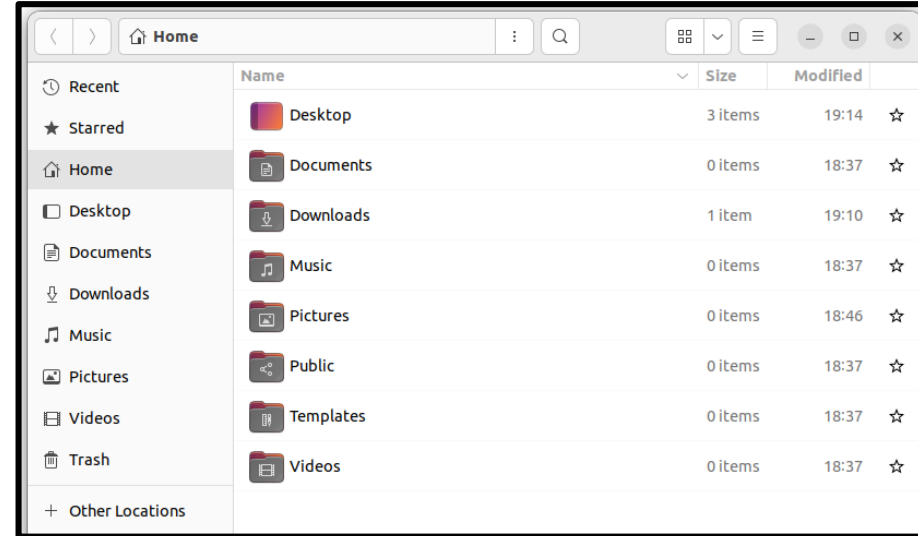
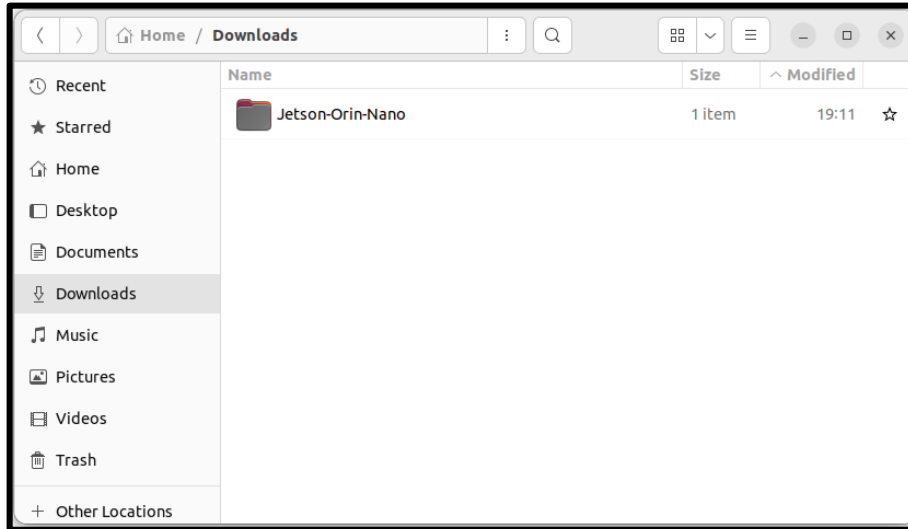


Terminal O/P: jonSetup2.sh

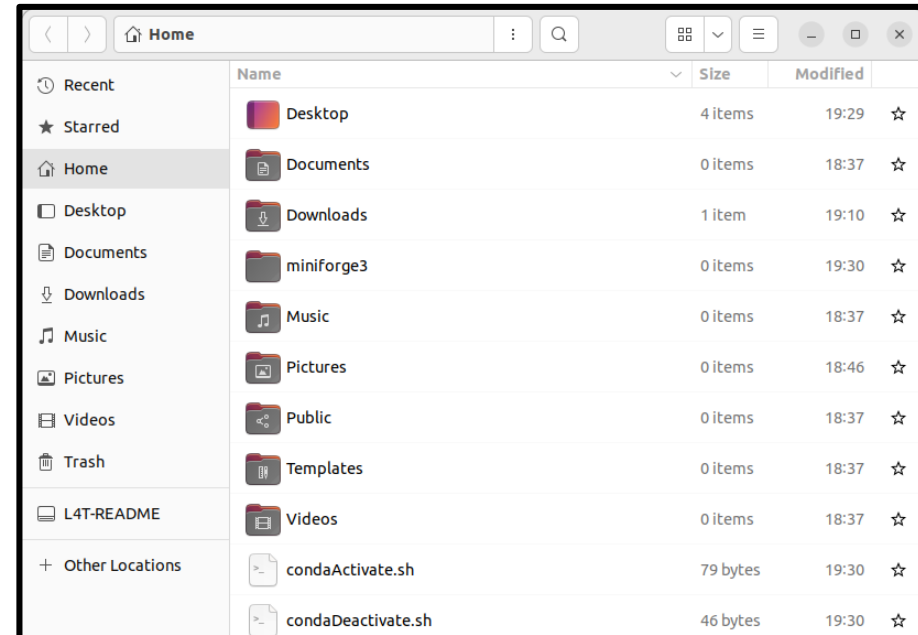
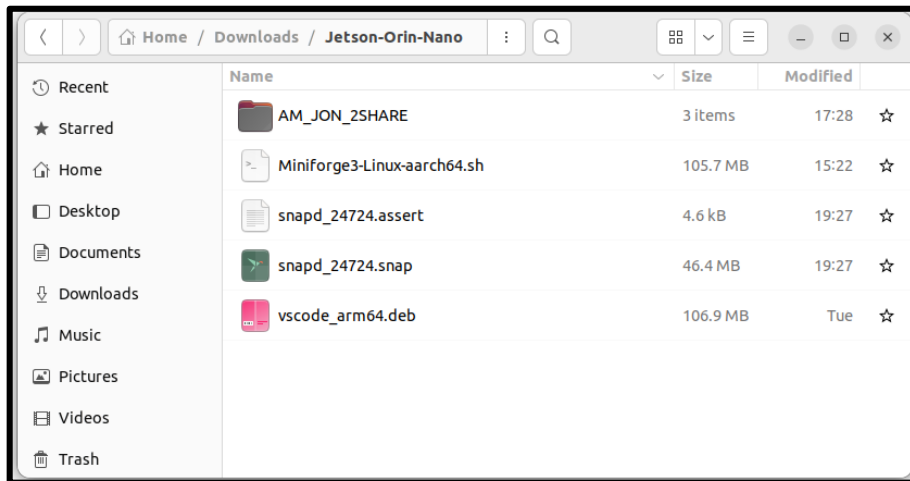
```
python-setuptools-doc
The following NEW packages will be installed:
python3-pip python3-setuptools python3-wheel
The following packages will be upgraded:
python3-pkg-resources
1 upgraded, 3 newly installed, 0 to remove and 309 not upgraded.
Need to get 1,678 kB/1,811 kB of archives.
After this operation, 9,005 kB of additional disk space will be used.
Get:1 http://ports.ubuntu.com/ubuntu-ports jammy-updates/main arm64 python3-setuptools all 59.6.0-1.2ubuntu0.22.04.3 [340 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports jammy-updates/universe arm64 python3-wheel all 0.37.1-2ubuntu0.22.04.1 [32.0 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports jammy-updates/universe arm64 python3-pip all 22.0.2+dfsg-1ubuntu0.7 [1,306 kB]
Fetched 1,678 kB in 1s (1,359 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
(Reading database ... 209420 files and directories currently installed.)
Preparing to unpack .../python3-pkg-resources_59.6.0-1.2ubuntu0.22.04.3_all.deb ...
Unpacking python3-pkg-resources (59.6.0-1.2ubuntu0.22.04.3) over (59.6.0-1.2ubuntu0.22.04.2) ...
Selecting previously unselected package python3-setuptools.
Preparing to unpack .../python3-setuptools_59.6.0-1.2ubuntu0.22.04.3_all.deb ...
Unpacking python3-setuptools (59.6.0-1.2ubuntu0.22.04.3) ...
Selecting previously unselected package python3-wheel.
Preparing to unpack .../python3-wheel_0.37.1-2ubuntu0.22.04.1_all.deb ...
Unpacking python3-wheel (0.37.1-2ubuntu0.22.04.1) ...
Selecting previously unselected package python3-pip.
Preparing to unpack .../python3-pip_22.0.2+dfsg-1ubuntu0.7_all.deb ...
Unpacking python3-pip (22.0.2+dfsg-1ubuntu0.7) ...
Setting up python3-pkg-resources (59.6.0-1.2ubuntu0.22.04.3) ...
Setting up python3-setuptools (59.6.0-1.2ubuntu0.22.04.3) ...
Setting up python3-wheel (0.37.1-2ubuntu0.22.04.1) ...
Setting up python3-pip (22.0.2+dfsg-1ubuntu0.7) ...
Processing triggers for man-db (2.10.2-1) ...
Collecting jetson-stats
  Downloading jetson-stats-4.3.2.tar.gz (118 kB)
    118.7/118.7 KB 1.4 MB/s eta 0:00:00
  Preparing metadata (setup.py) ... done
Requirement already satisfied: distro in /usr/lib/python3/dist-packages (from jetson-stats) (1.7.0)
Collecting smbus2
  Downloading smbus2-0.6.0-py2.py3-none-any.whl (11 kB)
Building wheels for collected packages: jetson-stats
  Building wheel for jetson-stats (setup.py) ... done
  Created wheel for jetson-stats: filename=jetson_stats-4.3.2-py3-none-any.whl size=157755 sha256=91829edc2955fe1d5d197116dda8ed9a2ffdfb27dfd6ab7b7cf3cc19ee6553cb
  Stored in directory: /root/.cache/pip/wheels/71/fb/63/2649a533f80a9b3b2ad770eb9b68373e3359674796778f0c31
Successfully built jetson-stats
Installing collected packages: smbus2, jetson-stats
Successfully installed jetson-stats-4.3.2 smbus2-0.6.0
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment instead: https://pip.pypa.io/warnings/venv
nvidia-l4t-core set on hold.
nvidia-jetpack set on hold.
Restarting jtop service...
-----
Setup Step 1 Complete!
REBOOT
-----
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script2$
```

Shell Script Execution: Download/Home

Initial contents before executing the shell script



New contents after executing the shell script



Terminal O/P: jonSetup2.sh

vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script2

```
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script2$ source ~/.bashrc
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script2$ cat ~/.bashrc
# ~/.bashrc: executed by bash(1) for non-login shells.
# see /usr/share/doc/bash/examples/startup-files (in the package bash-doc)
# for examples

# If not running interactively, don't do anything
case $- in
    *i*) ;;
    *) return;;
esac
```

~/.bashrc listing

⋮

~/.bashrc listing

vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script2

```
# >>> conda initialize >>>
# !! Contents within this block are managed by 'conda init' !!
__conda_setup="$(('/home/vk-jn-or/miniforge3/bin/conda' 'shell.bash' 'hook' 2> /dev/null)"
if [ $? -eq 0 ]; then
    eval "$__conda_setup"
else
    if [ -f "/home/vk-jn-or/miniforge3/etc/profile.d/conda.sh" ]; then
        . "/home/vk-jn-or/miniforge3/etc/profile.d/conda.sh"
    else
        export PATH="/home/vk-jn-or/miniforge3/bin:$PATH"
    fi
fi
unset __conda_setup
# <<< conda initialize <<<
```

~/.bashrc listing

Items managed by conda init

```
# Conda aliases
alias condaon1='source ~/.condaActivate.sh'
alias condaoff1='source ~/.condaDeactivate.sh'
```

Items added to ~/.bashrc

vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script2\$

Shell Script: jonSetup2-status.sh

- Run `jonSetup2-status.sh` to assess JON status after executing `jonSetup2.sh`

```
#!/bin/bash

# --- JETSON ORIN NANO SETUP STATUS SCRIPT: jonSetup2.sh ---
echo "Ubuntu Version ..."
lsb_release -a

echo "nvidia-jetpack ..."
apt-cache show nvidia-jetpack | grep Version

echo "nvidia L4T core package ..."
dpkg-query --show nvidia-l4t-core

echo "Jetpack mapping ..."
apt-cache policy nvidia-jetpack

echo "Current power mode..."
sudo nvpmodel -q

echo "Python Version ..."
python3 --version

echo "OpenCV ..."
dpkg -l | grep libopencv

echo "numpy ..."
dpkg -l | grep numpy

python -c "import cv2; print('CV2 Version          = ', cv2.__version__); build_info =
cv2.getBuildInformation(); print('Number CUDA-enabled devices =
',cv2.cuda.getCudaEnabledDeviceCount());    print('CUDA enabled          = ', 'CUDA' in build_info)"
```

Terminal O/P: jonSetup2-status.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script2
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script2$ ./jonSetup2-status.sh
Ubuntu Version ...
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.5 LTS
Release:        22.04
Codename:       jammy
nvidia-jetpack ...
Version: 6.2.1+b38
Version: 6.2+b77
Version: 6.1+b123
nvidia L4T core package ...
nvidia-l4t-core 36.4.4-20250616085344
Jetpack mapping ...
nvidia-jetpack:
  Installed: (none)
  Candidate: 6.2.1+b38
  Version table:
   6.2.1+b38 600
        600 https://repo.download.nvidia.com/jetson/common r36.4/main arm64 Packages
   6.2+b77 600
        600 https://repo.download.nvidia.com/jetson/common r36.4/main arm64 Packages
   6.1+b123 600
        600 https://repo.download.nvidia.com/jetson/common r36.4/main arm64 Packages
Current power mode...
NV Power Mode: MAXN_SUPER
2
Python Version ...
Python 3.10.12
OpenCV ...
ii libopencv                4.8.0-1-g6371ee1                arm64                Open Computer Vision Library
ii libopencv-core4.5d:arm64 4.5.4+dfsg-9ubuntu4             arm64                computer vision core library
ii libopencv-dev            4.8.0-1-g6371ee1                arm64                Open Computer Vision Library
ii libopencv-python         4.8.0-1-g6371ee1                arm64                Open Computer Vision Library
ii libopencv-samples        4.8.0-1-g6371ee1                arm64                Open Computer Vision Library
numpy ...
ii python3-numpy            1:1.21.5-1ubuntu22.04.1         arm64                Fast array facility to the Python 3 language
CV2 Version                  = 4.8.0
Number CUDA-enabled devices = 0
CUDA_enabled                 = False
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script2$
```

Shell Script: jonSetup3-sound.sh

- Fix JON to use HDMI sound O/P

```
#!/bin/bash
```

```
# --- Fix sound setting to use HDMI ---
```

```
# --- 1: Change sound setting ---
```

```
echo "[1/3] Changing sound output to HDMI"
```

```
pacmd set-default-sink alsa_output.platform-3510000.hda.hdmi-stereo
```

Immediate change to
HDMI sound o/p

```
# --- 2: Persistent setting: Add to /etc/pulse/default.pa ---
```

```
# This replicates manually opening nano and adding the line to the bottom
```

```
echo "[2/3] Adding persistent sink to /etc/pulse/default.pa..."
```

For persistent HDMI,
edit /etc/pulse/default.pa

```
# --- Auto-comment out the device-restore module to prevent override of manual setting ---
```

```
sudo sed -i '/^load-module module-default-device-restore/ s/^/# /' /etc/pulse/default.pa
```

```
# --- Add the correct sink command ---
```

```
# Remove any prior attempts at this so that we don't keep appending this, then add it fresh.
```

```
sudo sed -i '/.*alsa_output\.platform-3510000\.hda\.hdmi-stereo/d' /etc/pulse/default.pa
```

```
sudo bash -c "echo 'set-default-sink alsa_output.platform-3510000.hda.hdmi-stereo' >>
```

```
/etc/pulse/default.pa"
```

Shell Script: jonSetup3-sound.sh

```
# --- 3: Add Startup Command: Create the Startup Application entry ---
# This replicates manually adding 'pulseaudio -k' to Startup Applications
echo "[3/3] Creating Startup Application entry: Restart PulseAudio..."
mkdir -p ~/.config/autostart

cat <<EOF > ~/.config/autostart/pulseaudio.desktop
[Desktop Entry]
Type=Application
Exec=bash -c "sleep 2 && pulseaudio -k"
Hidden=false
NoDisplay=false
X-GNOME-Autostart-enabled=true
Name[en_US]=Restart PulseAudio
Name=Restart PulseAudio
Comment[en_US]=
Comment=
EOF

# Make the startup file executable
chmod +x ~/.config/autostart/pulseaudio.desktop
```

**Add to Startup
Application**

```
# --- Play a success sound ---
echo "Testing audio output..."
# Plays a short 1-second sine wave beep
speaker-test -t sine -f 440 -l 1 -p 0.5 > /dev/null 2>&1
```

Test HDMI O/P

Sudo reboot now

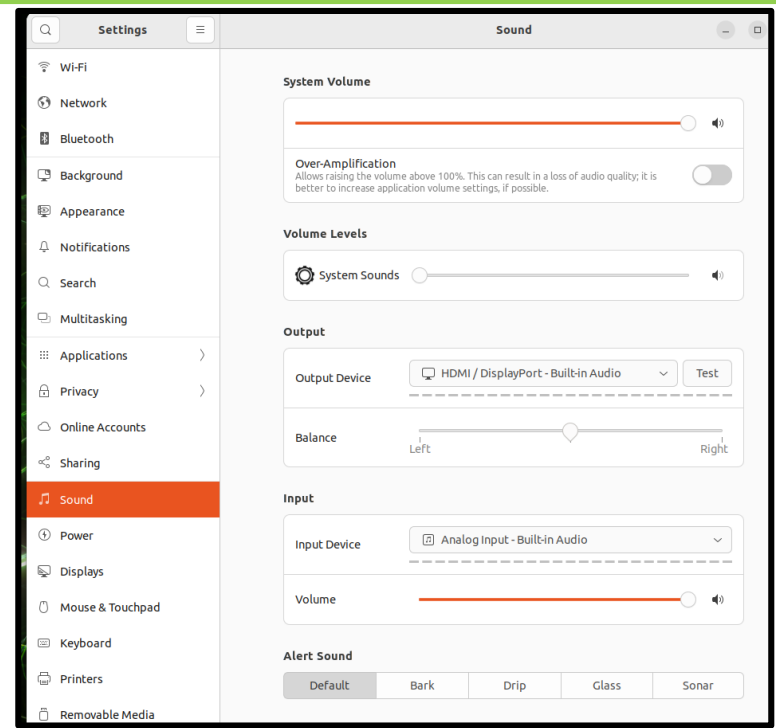
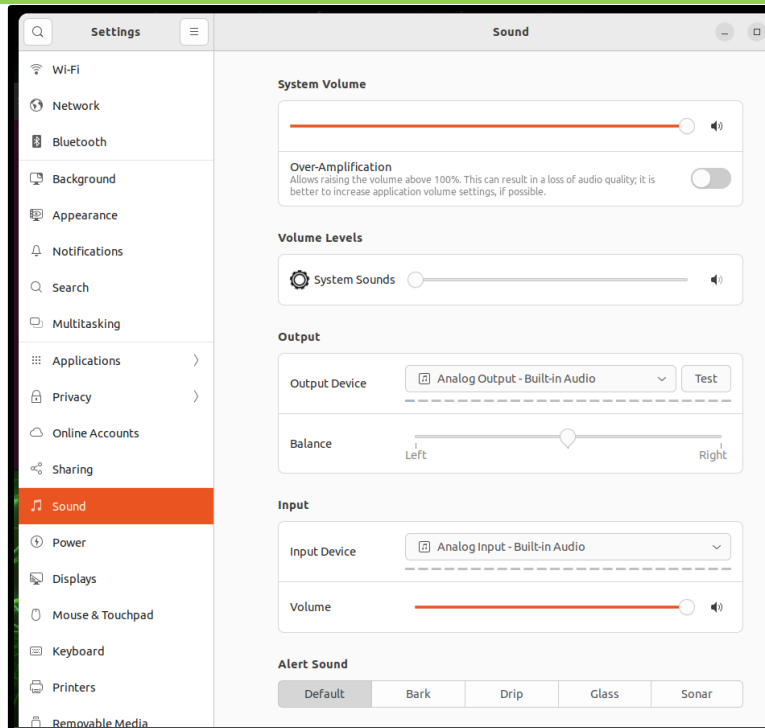
```
echo "Done. Settings applied to config and Startup Applications."
echo "Run -- pacmd list-sinks | grep -E '\* index:name:' "
echo "If the * index: is directly above name: <alsa_output.platform-3510000.hda.hdmi-stereo>, then HDMI is active"
echo "Done. Please reboot to verify the startup application trigger."
```

Terminal O/P: jonSetup3-sound.sh

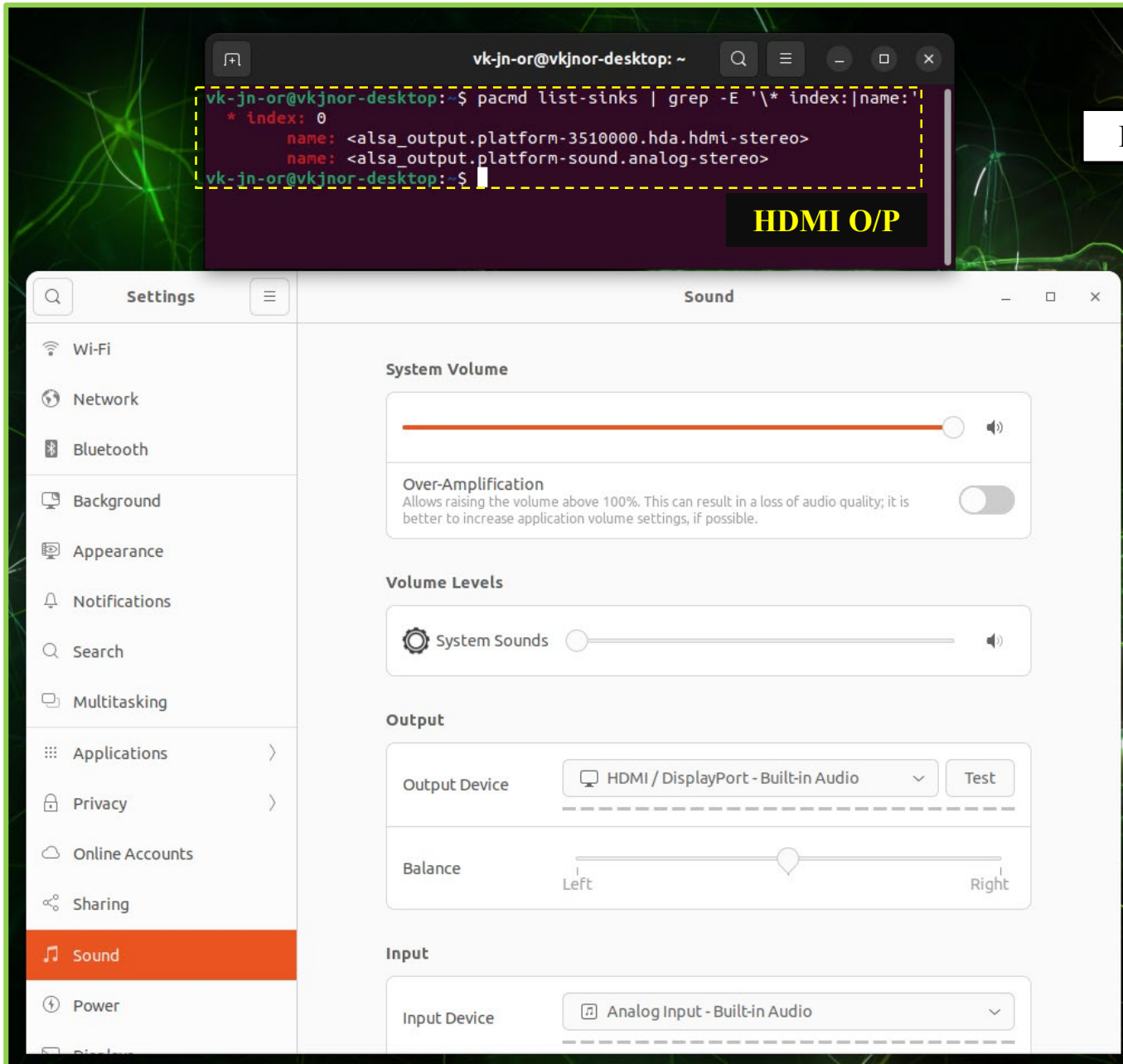
```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script3
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script3$ pacmd list-sinks | grep -E '\* index:|name:'
    name: <alsa_output.platform-3510000.hda.hdmi-stereo>
    * index: 1
    name: <alsa_output.platform-sound.analog-stereo>
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script3$ ./jonSetup3-sound.sh
[1/3] Changing sound output to HDMI
[2/3] Adding persistent sink to /etc/pulse/default.pa...
[3/3] Creating Startup Application entry: Restart PulseAudio...
Testing audio output...
Done. Settings applied to config and Startup Applications.
Run -- pacmd list-sinks | grep -E '\* index:|name:'
If the * index: is directly above name: <alsa output.platform-3510000.hda.hdmi-stereo>, then HDMI is active
Done. Please reboot to verify the startup application trigger.
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script3$ pacmd list-sinks | grep -E '\* index:|name:'
    * index: 0
    name: <alsa_output.platform-3510000.hda.hdmi-stereo>
    name: <alsa_output.platform-sound.analog-stereo>
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script3$
```

----- Analog O/P

----- HDMI O/P



Terminal O/P: jonSetup3-sound.sh



Post reboot

Shell Script: jonSetup4-nvcc-pth.sh

- Fix nvcc compiler path

```
#!/bin/bash
```

```
# --- Path for nvcc compiler ---
```

```
# Define the lines to be added
```

```
PATH_LINE='export PATH=/usr/local/cuda/bin:$PATH'
```

```
LD_LINE='export LD_LIBRARY_PATH=/usr/local/cuda/lib64:$LD_LIBRARY_PATH'
```

Aliases for paths

```
echo "Updating ~/.bashrc with CUDA paths..."
```

```
# Check if the PATH line already exists to avoid duplicates
```

```
if ! grep -qF "$PATH_LINE" ~/.bashrc; then
```

```
    echo "$PATH_LINE" >> ~/.bashrc
```

```
    echo "Added PATH to .bashrc"
```

```
fi
```

Edit ~/.bashrc

```
# Check if the LD_LIBRARY_PATH line already exists
```

```
if ! grep -qF "$LD_LINE" ~/.bashrc; then
```

```
    echo "$LD_LINE" >> ~/.bashrc
```

```
    echo "Added LD_LIBRARY_PATH to .bashrc"
```

```
fi
```

```
# Exporting to the current subshell so the script can verify immediately
```

```
export PATH=/usr/local/cuda/bin:$PATH
```

```
export LD_LIBRARY_PATH=/usr/local/cuda/lib64:$LD_LIBRARY_PATH
```

Immediate export to verify

```
echo "-----"
```

```
echo "Verification:"
```

```
nvcc --version
```

```
echo "-----"
```

```
echo "DONE. Please run 'source ~/.bashrc' in your open terminals to apply changes."
```

Terminal O/P: jonSetup4-nvcc-pth.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script4
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script3$ cd /home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script4
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script4$ ./jonSetup4-nvcc-pth.sh
Updating ~/.bashrc with CUDA paths...
Added PATH to .bashrc
Added LD_LIBRARY_PATH to .bashrc
-----
Verification:
nvcc: NVIDIA (R) Cuda compiler driver
Copyright (c) 2005-2024 NVIDIA Corporation
Built on Wed Aug 14 10:14:07 PDT 2024
Cuda compilation tools, release 12.6, V12.6.68
Build cuda_12.6.r12.6/compiler.34714021_0
-----
DONE. Please run 'source ~/.bashrc' in your open terminals to apply changes.
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script4$ source ~/.bashrc
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script4$ cat ~/.bashrc
# ~/.bashrc: executed by bash(1) for non-login shells.
# see /usr/share/doc/bash/examples/startup-files (in the package bash-doc)
# for examples

# If not running interactively, don't do anything
case $- in
    *(*) ;;
    *) return;;
esac
```

nvcc located

...

Newly added items

```
# >>> conda initialize >>>
# !! Contents within this block are managed by 'conda init' !!
__conda_setup="$('/home/vk-jn-or/miniforge3/bin/conda' 'shell.bash' 'hook' 2> /dev/null)"
if [ $? -eq 0 ]; then
    eval "$__conda_setup"
else
    if [ -f "/home/vk-jn-or/miniforge3/etc/profile.d/conda.sh" ]; then
        . "/home/vk-jn-or/miniforge3/etc/profile.d/conda.sh"
    else
        export PATH="/home/vk-jn-or/miniforge3/bin:$PATH"
    fi
fi
unset __conda_setup
# <<< conda initialize <<<

# Conda aliases
alias condaon1='source ~/condaActivate.sh'
alias condaoff1='source ~/condaDeactivate.sh'
export PATH=/usr/local/cuda/bin:$PATH
export LD_LIBRARY_PATH=/usr/local/cuda/lib64:$LD_LIBRARY_PATH
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script4$
```

Shell Script: jonSetup5-CSI.sh

- Enable CSI camera – run shell script jonSetup5-CSI.sh

```
#!/bin/bash

# --- JETSON ORIN NANO CSI CAMERA SETUP SCRIPT ---

# Clear screen for readability
clear

echo "=====
echo "          JETSON-IO CAMERA CONFIGURATION GUIDE"
echo "=====
echo "Follow these exact steps in the application:"
echo ""
echo "1. Select: 'Configure Jetson 24pin CSI Connector'"
echo "2. Select: 'Configure compatible hardware'"
echo "3. Select: 'Camera IMX219-A' (or 'Camera IMX219' for slot 0)"
echo "4. Select: 'Save pin changes'"
echo "5. Select: 'Save and reboot to reconfigure pins'"
echo ""
echo "CONTROLS:"
echo " - Use ARROW KEYS to move"
echo " - Use ENTER to select"
echo " - Use BACKSPACE to go back"
echo "=====
```

Shell Script: jonSetup5-CSI.sh

```
echo ""
echo ">>> AFTER REBOOT IS FINISHED:"
echo "    You must verify the hardware is active by running:"
echo "    ls /dev/video0"
echo "    nvgstcapture-1.0"
echo "    gst-launch-1.0 nvarguscamerasrc sensor-id=0 ! nvvidconv ! 'video/x-raw,format=BGRx' ! videoconvert ! ximagesink"
echo "===== "
echo ""
read -p "Press ENTER to launch jetson-io.py..."

# Launch the actual application
sudo python3 /opt/nvidia/jetson-io/jetson-io.py
```

Terminal O/P: jonSetup5-CSI.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script5
vk-jn-or@vkjnor-desktop:~$ cd /home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script5/
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script5$ ./jonSetup5-CSI.sh
```



```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script5

=====
      JETSON-IO CAMERA CONFIGURATION GUIDE
=====
Follow these exact steps in the application:

1. Select: 'Configure Jetson 24pin CSI Connector'
2. Select: 'Configure compatible hardware'
3. Select: 'Camera IMX219-A' (or 'Camera IMX219' for slot 0)
4. Select: 'Save pin changes'
5. Select: 'Save and reboot to reconfigure pins'

CONTROLS:
- Use ARROW KEYS to move
- Use ENTER to select
- Use BACKSPACE to go back

=====

>>> AFTER REBOOT IS FINISHED:
You must verify the hardware is active by running:
ls /dev/video0
nvgstcapture-1.0
gst-launch-1.0 nvarguscamerasrc sensor-id=0 ! nvvidconv ! 'video/x-raw,format=BGRx' ! videoconvert ! ximagesink
=====

Press ENTER to launch jetson-io.py...
```

Terminal O/P: jonSetup5-CSL.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_
1
===== Jetson Expansion Header Tool =====

Select one of the following:

Configure Jetson 40pin Header
Configure Jetson 24pin CSI Connector
Configure Jetson M.2 Key E Slot
Exit
```

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_
2
===== Jetson Expansion Header Tool =====

3.3V ( 1) .. ( 2) i2c3
i2c3 ( 3) .. ( 4) GND
GND ( 7) .. ( 8) NA
NA ( 9) .. (10) GND
GND (13) .. (14) NA
NA (15) .. (16) GND
GND (19) .. (20) NA
NA (21) .. (22) GND
GND (23) .. (24) GND

Jetson 24pin CSI Connector:

Configure for compatible hardware
Back
```

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_
3
===== Jetson Expansion Header Tool =====

Select one of the following options:

Camera IMX219 Dual
Camera IMX219-A
Camera IMX219-A and IMX477-C
Camera IMX219-C
Camera IMX477 Dual
Camera IMX477 Dual 4 lane
Camera IMX477-A
Camera IMX477-A and IMX219-C
Camera IMX477-C

Back
```

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_
4
===== Jetson Expansion Header Tool =====

3.3V ( 1) .. ( 2) i2c3
i2c3 ( 3) .. ( 4) GND
GND ( 7) .. ( 8) NA
NA ( 9) .. (10) GND
GND (13) .. (14) NA
NA (15) .. (16) GND
GND (19) .. (20) NA
NA (21) .. (22) GND
GND (23) .. (24) GND

Jetson 24pin CSI Connector:

Save pin changes
Discard pin changes
```

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_
5
===== Jetson Expansion Header Tool =====

Select one of the following:

Configure Jetson 40pin Header
Re-configure Jetson 24pin CSI Connector
Configure Jetson M.2 Key E Slot
Save and reboot to reconfigure pins
Save and exit without rebooting
Discard all pin changes
Exit
```

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_
6
===== Jetson Expansion Header Tool =====

Modified /boot/extlinux/extlinux.conf to add following DTBO entries:

/boot/tegra234-p3767-camera-p3768-imx219-A.dtbo

Press any key to reboot the system now or Ctrl-C to abort
```

Test Run of Some Packages: conda, jtop

```
vk-jn-or@vkjnor-desktop: ~  
vk-jn-or@vkjnor-desktop:~$ condaon1  
(vk1 env) vk-jn-or@vkjnor-desktop:~$ condaoff1  
vk-jn-or@vkjnor-desktop:~$ jtop
```

jtop MAXN SUPER|CPU 10.9%|GPU 0.0%

Model: NVIDIA Jetson Orin Nano Engineering Reference Developer Kit Super - Jetpack 6.2.1 [L4T 36.4.4]

1 [|||||] 5.9%] 729MHz 4 [|||||] 5.0%] 729MHz
2 [|||||] 6.0%] 729MHz 5 [|||||] 6.0%] 729MHz
3 [|||||] 3.0%] 729MHz 6 [|||||] 6.0%] 729MHz

Mem [|||||] 1.7G/7.4G] FAN [|||||] 33.3%] 1829RPM
Swp [|||||] 0k/3.7G]
Emc [204MHz] 3.2GHz] 2.1GHz 0%

GPU [|||||] 0.0%] 306MHz
Dsk [|||||] 26.6G/115G]

Jetson Clocks: inactive
NV Power[2]: MAXN_SUPER
Uptime: 0 days 0:2:47

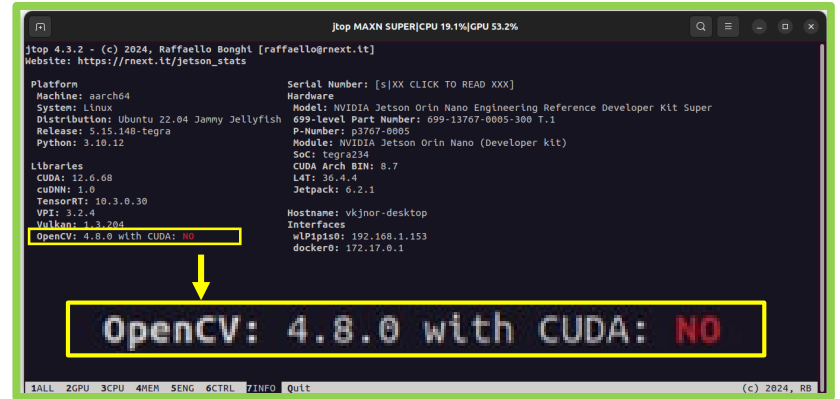
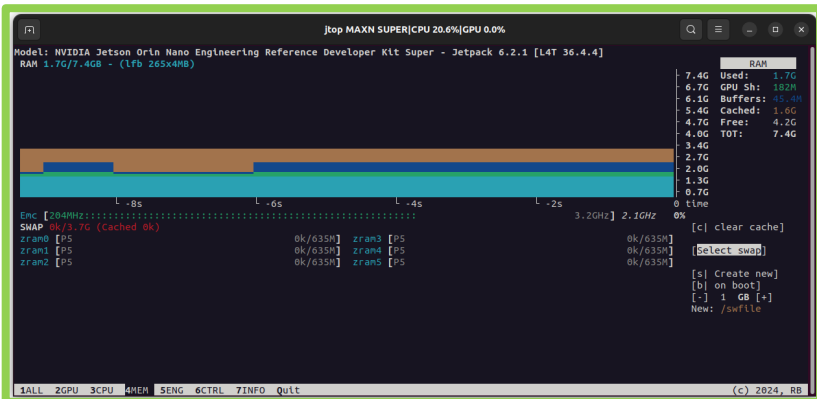
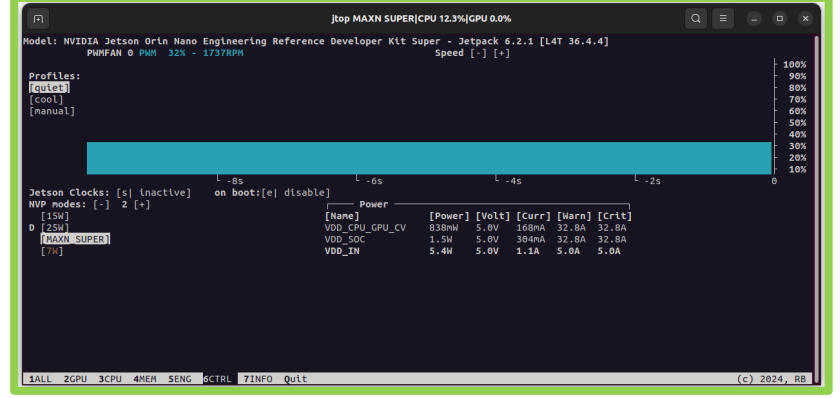
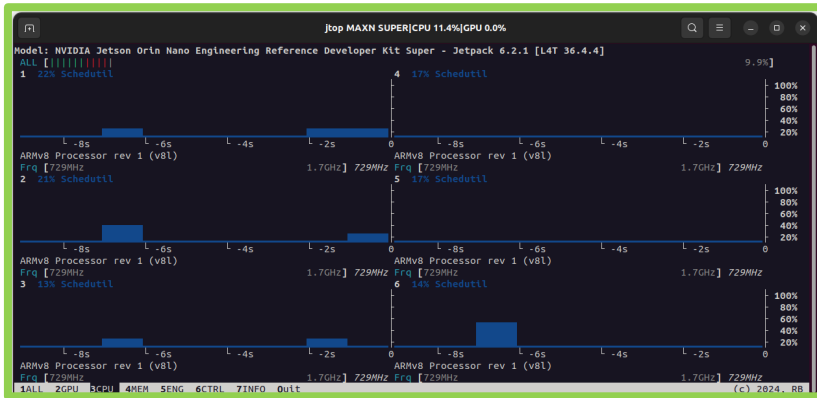
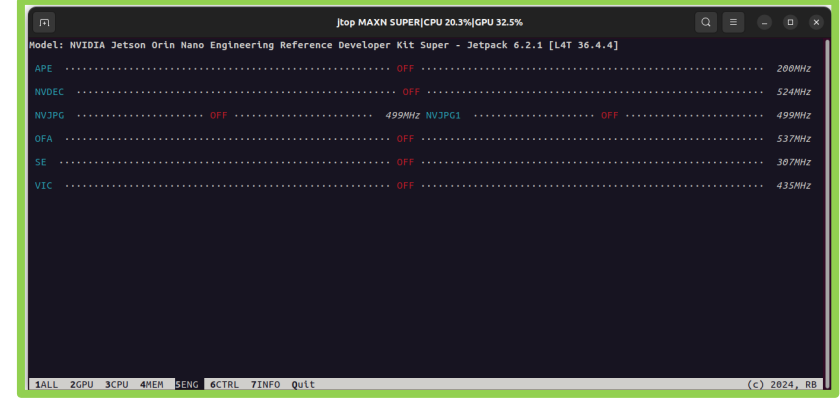
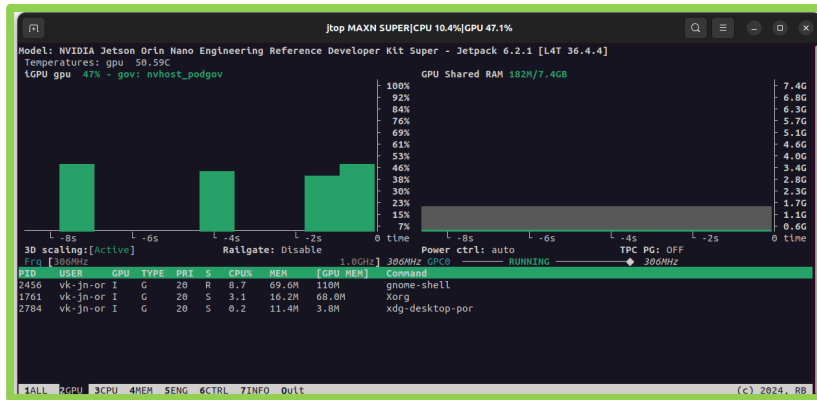
PID	USER	GPU	TYPE	PRI	S	CPU%	MEM	[GPU MEM]	Command
2456	vk-jn-or	I	G	20	S	8.7	68.8M	107M	gnome-shell
1761	vk-jn-or	I	G	20	S	2.8	16.1M	67.6M	Xorg
2784	vk-jn-or	I	G	20	S	0.2	11.4M	3.8M	xdg-desktop-por

[HW engines]		[Sensor]		[Temp]		[Power]		[Inst]		[Avg]	
APE: [OFF]		cpu		51.38C		CPU	GPU	CV	599mW	618mW	
NVDEC: [OFF]		cv0		Offline		SOC			1.5W	1.5W	
NVJPG: [OFF]	NVJPG1: [OFF]	cv1		Offline		VDD_IN			5.1W	5.1W	
SE: [OFF]	VIC: [OFF]	cv2		Offline							
		gpu		50.66C							
		soc0		51.50C							
		soc1		51.62C							
		soc2		50.22C							
		tj		51.62C							

1ALL 2GPU 3CPU 4MEM 5ENG 6CTRL 7INFO Quit

(c) 2024, RB

JTOP Screens

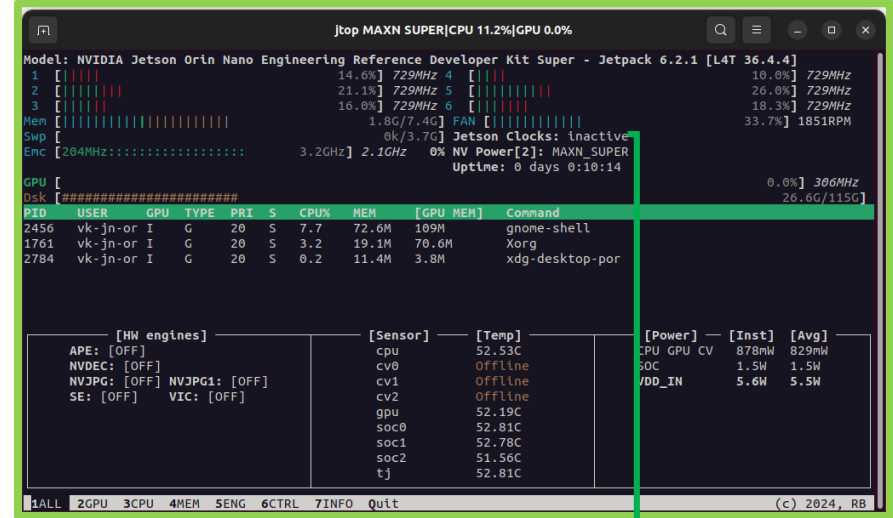
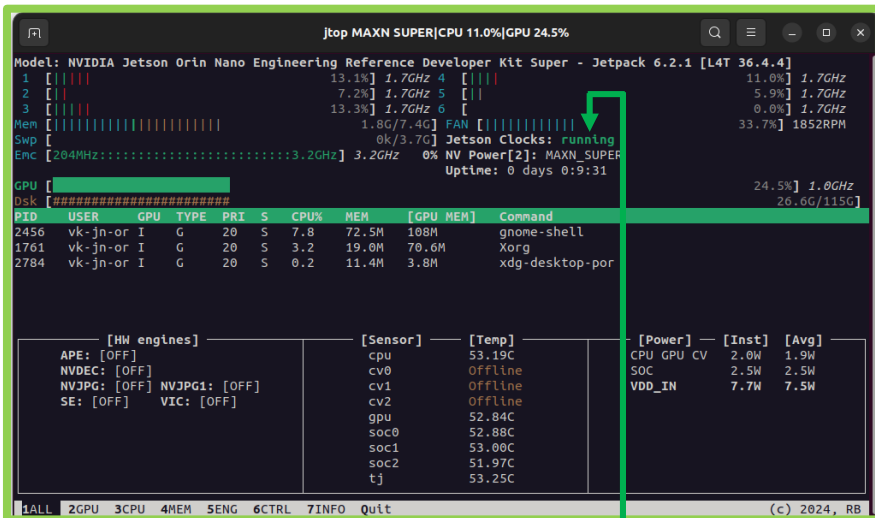


Note: JTOP reports OpenCV version “only” for the system-wide install

Setup for jetson_clocks

```
vk-jn-or@vkjnor-desktop:~$ sudo jetson_clocks --store
Storing system configuration in /root/.jetsonclocks_conf.txt
vk-jn-or@vkjnor-desktop:~$ sudo jetson_clocks
vk-jn-or@vkjnor-desktop:~$ jtop
```

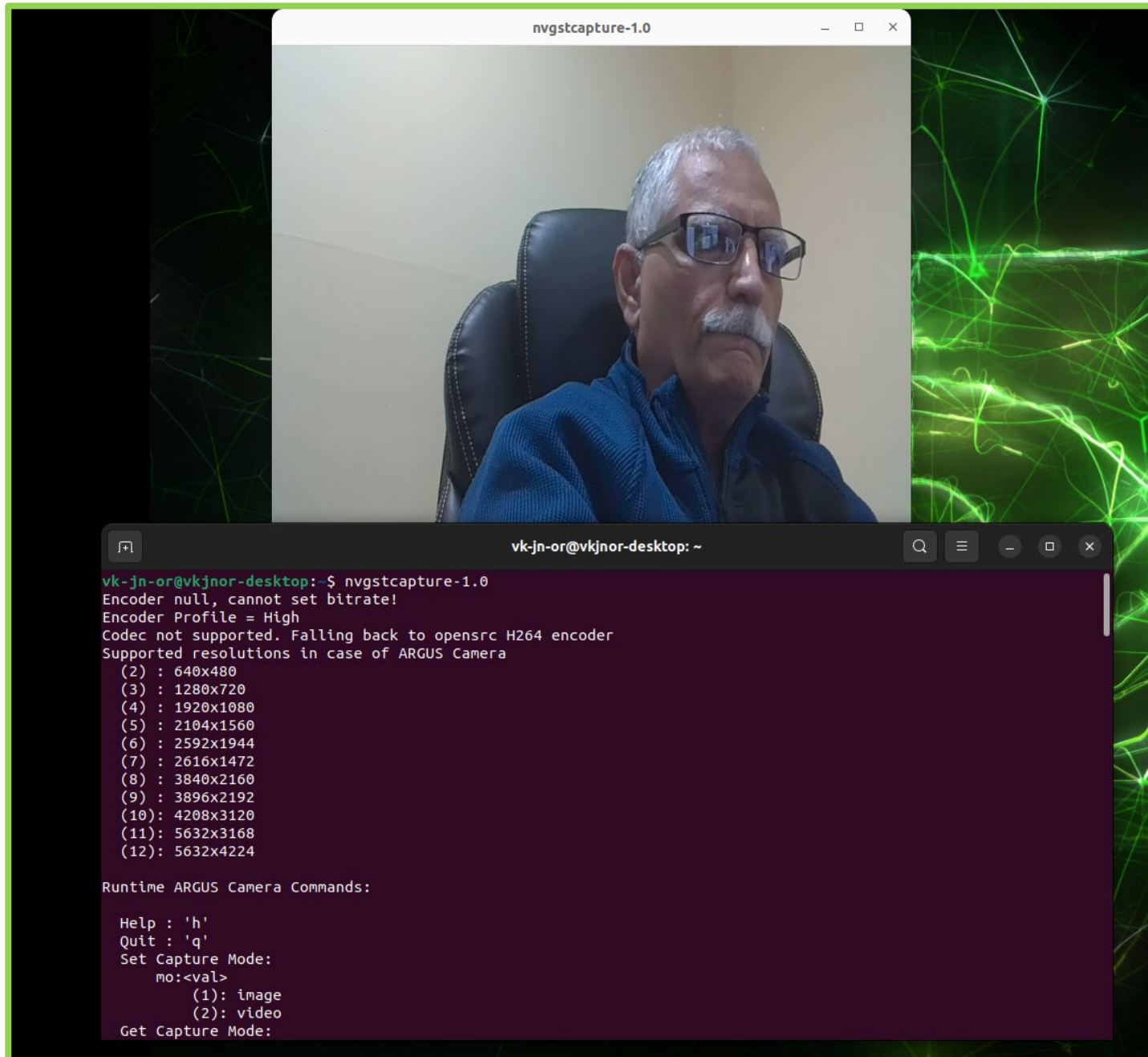
```
vk-jn-or@vkjnor-desktop:~$ sudo jetson_clocks --store
Storing system configuration in /root/.jetsonclocks_conf.txt
vk-jn-or@vkjnor-desktop:~$ sudo jetson_clocks
vk-jn-or@vkjnor-desktop:~$ jtop
vk-jn-or@vkjnor-desktop:~$ sudo jetson_clocks --restore
Restoring system configuration from /root/.jetsonclocks_conf.txt
vk-jn-or@vkjnor-desktop:~$ jtop
vk-jn-or@vkjnor-desktop:~$
```



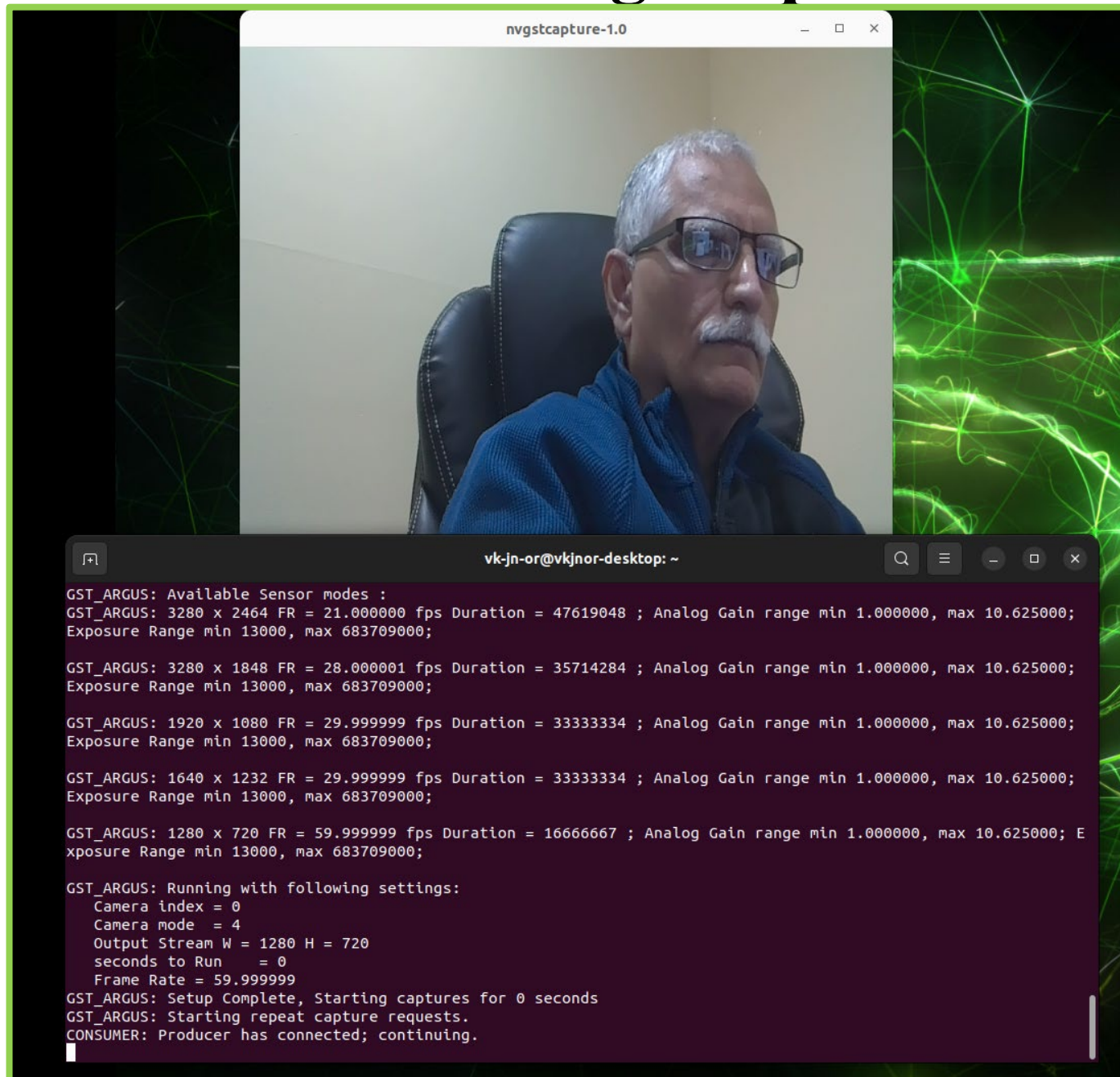
Jetson Clocks: running

Jetson Clocks: inactive

Test Run of nvgstcapture-1.0



Test Run of nvgstcapture-1.0



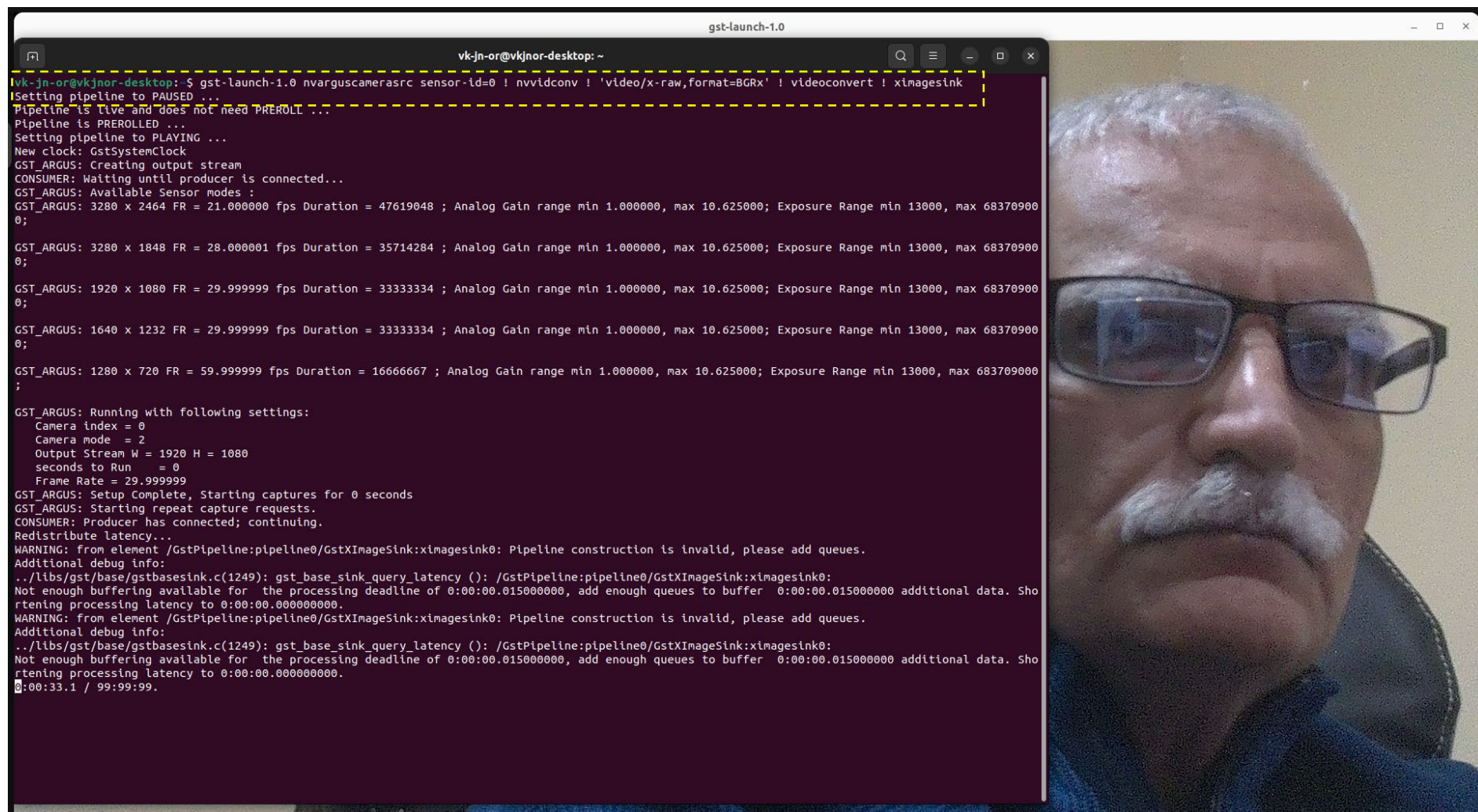
Test Run of nvgstcapture-1.0

```
vk-jn-or@vkjnor-desktop: ~  
Runtime encoder configuration options:  
  
Set Encoding Bit-rate(in bytes):  
  br:<val> e.g., br:4000000  
Get Encoding Bit-rate(in bytes):  
  gbr  
Set Encoding Profile(only for H.264):  
  ep:<val> e.g., ep:1  
    (0): Baseline  
    (1): Main  
    (2): High  
Get Encoding Profile(only for H.264):  
  gep  
Force IDR Frame on video Encoder(only for H.264):  
  Enter 'f'  
  
bitrate = 4000  
  
Using winsys: x11  
** Message: 20:19:37.343: <main:4732> iterating capture loop ....  
GST_ARGUS: Creating output stream  
CONSUMER: Waiting until producer is connected...  
GST_ARGUS: Available Sensor modes :  
GST_ARGUS: 3280 x 2464 FR = 21.000000 fps Duration = 47619048 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;  
GST_ARGUS: 3280 x 1848 FR = 28.000001 fps Duration = 35714284 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;  
GST_ARGUS: 1920 x 1080 FR = 29.999999 fps Duration = 33333334 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;  
GST_ARGUS: 1640 x 1232 FR = 29.999999 fps Duration = 33333334 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;  
GST_ARGUS: 1280 x 720 FR = 59.999999 fps Duration = 16666667 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;  
  
GST_ARGUS: Running with following settings:  
  Camera index = 0  
  Camera mode  = 4  
  Output Stream W = 1280 H = 720  
  seconds to Run   = 0  
  Frame Rate = 59.999999  
GST_ARGUS: Setup Complete, Starting captures for 0 seconds  
GST_ARGUS: Starting repeat capture requests.  
CONSUMER: Producer has connected; continuing.  
-----  
ERROR on bus: by /GstPipeline:capture_native_pipeline: Output window was closed  
debug info:  
/dvs/git/dirty/git-master_linux/tests-multimedia/nvgstapps/nvgstcapture-1.0/nvgstcapture.c(4160): nvgst_handle_xevents (): /GstPipeline:capture_native_pipeline  
GST_ARGUS: Cleaning up  
CONSUMER: Done Success  
GST_ARGUS: Done Success  
** Message: 20:21:23.251: <main:4742> Capture completed  
** Message: 20:21:23.254: <main:4793> Camera application will now exit  
-----  
vk-jn-or@vkjnor-desktop:~$
```

Terminal o/p after closing the camera feed

Test Run of gst-launch-1.0 nvarguscamerasrc

gst-launch-1.0 nvarguscamerasrc sensor-id=0 ! nvvidconv ! 'video/x-raw,format=BGRx' ! videoconvert ! ximagesink



vk-jn-or@vkjnor-desktop: ~

```
vk-jn-or@vkjnor-desktop:~$ gst-launch-1.0 nvarguscamerasrc sensor-id=0 ! nvvidconv ! 'video/x-raw,format=BGRx' ! videoconvert ! ximagesink
Setting pipeline to PAUSED ...
Pipeline is live and does not need PREROLL ...
Pipeline is PREROLLED ...
Setting pipeline to PLAYING ...
New clock: GstSystemClock
GST_ARGUS: Creating output stream
CONSUMER: Waiting until producer is connected...
GST_ARGUS: Available Sensor modes :
GST_ARGUS: 3280 x 2464 FR = 21.000000 fps Duration = 47619048 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 68370900
0;
GST_ARGUS: 3280 x 1848 FR = 28.000001 fps Duration = 35714284 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 68370900
0;
GST_ARGUS: 1920 x 1080 FR = 29.999999 fps Duration = 33333334 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 68370900
0;
GST_ARGUS: 1640 x 1232 FR = 29.999999 fps Duration = 33333334 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 68370900
0;
GST_ARGUS: 1280 x 720 FR = 59.999999 fps Duration = 16666667 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 68370900
;
GST_ARGUS: Running with following settings:
Camera index = 0
Camera mode = 2
Output Stream W = 1920 H = 1080
seconds to Run = 0
Frame Rate = 29.999999
GST_ARGUS: Setup Complete, Starting captures for 0 seconds
GST_ARGUS: Starting repeat capture requests.
CONSUMER: Producer has connected; continuing.
Redistribute latency...
WARNING: from element /GstPipeline:pipeline0/GstXImageSink:ximagesink0: Pipeline construction is invalid, please add queues.
Additional debug info:
../libs/gst/base/gstbasesink.c(1249): gst_base_sink_query_latency (): /GstPipeline:pipeline0/GstXImageSink:ximagesink0:
Not enough buffering available for the processing deadline of 0:00:00.015000000, add enough queues to buffer 0:00:00.015000000 additional data. Sho
rtening processing latency to 0:00:00.000000000.
WARNING: from element /GstPipeline:pipeline0/GstXImageSink:ximagesink0: Pipeline construction is invalid, please add queues.
Additional debug info:
../libs/gst/base/gstbasesink.c(1249): gst_base_sink_query_latency (): /GstPipeline:pipeline0/GstXImageSink:ximagesink0:
Not enough buffering available for the processing deadline of 0:00:00.015000000, add enough queues to buffer 0:00:00.015000000 additional data. Sho
rtening processing latency to 0:00:00.000000000.
0:00:33.1 / 99:99:99.
```

The video feed shows a man with glasses and a mustache, wearing a blue jacket, looking directly at the camera.

Shell Script: jonSetup6-CV-Pkgs.sh

- Install swapfile, CUDA-enabled opencv, dlib, face-recognition, matplotlib, scikit-learn, pydot, graphviz, GIMP, ...

```
#!/bin/bash
```

```
# --- JETSON ORIN NANO: CUDA OPENCV, DLIB & APPS ---
```

```
# --- 1. Configuration ---
```

```
echo "[1/7] Setting some dirs and envs ..."
```

```
DOWNLOAD_DIR="/home/vk-jn-or/Downloads/Jetson-Orin-Nano"
```

```
APP_DIR="/home/vk-jn-or/Applications"
```

```
ENV_NAME="vk1_env"
```

```
echo "Sourcing conda profile to enable the 'conda' command..."
```

```
source ~/miniforge3/etc/profile.d/conda.sh
```

```
echo "Making some dirs..."
```

```
mkdir -p "$DOWNLOAD_DIR"
```

```
mkdir -p "$APP_DIR"
```

```
# --- 2. SWAPFILE (8GB) ---
```

```
echo "[2/7] Creating 8GB Swapfile..."
```

```
if [ ! -f /swapfile ]; then
```

```
    sudo fallocate -l 8G /swapfile
```

```
    sudo chmod 600 /swapfile
```

```
    sudo mkswap /swapfile
```

```
    sudo swapon /swapfile
```

```
    echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
```

```
    echo "Swapfile created and verified:"
```

```
    free -h
```

```
    swapon --show
```

```
else
```

```
    echo "Swapfile already exists. Skipping."
```

```
fi
```

Some configuration setup

8GB swapfile to address
memory constraints
Swapfile is needed for dlib
and face-recognition
installation and use in
Step 3

```
# --- 3. CUDA-ENABLED OPENCV ---
echo "[3/7] Installing CUDA-enabled OpenCV..."
conda activate $ENV_NAME # Activate vk1_env
```

Used a wheel to install
opencv with CUDA and
gstreamer support

```
unset PIP_INDEX_URL
export PIP_INDEX_URL=https://pypi.jetson-ai-lab.io/jp6/cu128
```

```
pip install opencv-python #opencv: gstreamer and cuda support using a wheel from PIP_INDEX_URL
```

```
echo "Fixing missing dependencies..."
sudo apt-get update
sudo apt install libtesseract-dev -y
sudo apt install libatlas-base-dev -y
```

```
# --- 4. DLIB AND FACE RECOGNITION ---
```

```
echo "[4/7] Installing dlib and face_recognition..."
conda deactivate
```

```
sudo apt-get install python3-pip cmake libopenblas-dev liblapack-dev libjpeg-dev -y
```

```
conda activate $ENV_NAME # Activate vk1_env
pip3 install numpy
```

```
# Downgrade setuptools
pip install "setuptools<81"
```

Added to address face-recognition installation failure

```
# Check if dlib or face_recognition are already partially installed
if pip show dlib > /dev/null 2>&1 || pip show face_recognition > /dev/null 2>&1; then
    echo "Found existing or partial installation. Cleaning up for a fresh build..."
    pip uninstall dlib face_recognition -y
    rm -rf ~/.cache/pip/wheels/
fi
```

```
pip3 install dlib face_recognition
```

```
# --- 5. ADDITIONAL PACKAGES & SYSTEM FIXES ---  
echo "[5/7] Installing additional Python packages..."  
pip3 install matplotlib scikit-learn pydot graphviz
```

**Various python packages
for ML and some fixes for
pre-loading some libs**

```
conda deactivate  
sudo apt-get install graphviz -y
```

```
# Apply LD_PRELOAD fix to .bashrc for libffi  
FIX_LINE="export LD_PRELOAD=/usr/lib/aarch64-linux-gnu/libffi.so.8"  
if ! grep -q "libffi.so.8" ~/.bashrc; then  
    echo -e "\n# Fix for libffi\n$FIX_LINE" >> ~/.bashrc  
fi
```

**GNU image manipulation
program (GIMP): a tool
to work with
photos/images**

```
# --- 6. GIMP APPIMAGE & ICON DOWNLOAD ---  
echo "[6/7] Setting up GIMP AppImage and Icon..."  
cd "$DOWNLOAD_DIR"
```

```
GIMP_URL="https://download.gimp.org/gimp/v3.0/linux/GIMP-3.0.4-aarch64.AppImage"  
if [ ! -f "GIMP-3.0.4-aarch64.AppImage" ]; then  
    wget "$GIMP_URL"  
fi  
chmod +x GIMP-3.0.4-aarch64.AppImage  
mv GIMP-3.0.4-aarch64.AppImage "$APP_DIR/"
```

```
ICON_URL="https://upload.wikimedia.org/wikipedia/commons/4/45/The_GIMP_icon_-_gnome.svg"  
wget -O "$APP_DIR/GIMP.png" "$ICON_URL"
```

```
# --- 7. CREATE GIMP DESKTOP SHORTCUT ---
echo "[7/7] Creating GIMP desktop shortcut..."
cat <<EOF > ~/.local/share/applications/gimp.desktop
[Desktop Entry]
Name=GIMP
Comment=Create and edit images
Exec=$APP_DIR/GIMP-3.0.4-aarch64.AppImage %U
Icon=$APP_DIR/GIMP.png
Terminal=false
Type=Application
Categories=Graphics;2DGraphics;
EOF

chmod +x ~/.local/share/applications/gimp.desktop
update-desktop-database ~/.local/share/applications/
```

GIMP: Create a desktop shortcut

```
# FINAL VERIFICATION
echo ""
echo "=====
echo "VERIFYING INSTALLATION..."
echo "=====
```

Confirm some installs

```
conda activate $ENV_NAME

python -c "import dlib; print('Dlib version:', dlib.__version__); print('CUDA support enabled:',
dlib.DLIB_USE_CUDA); print('CUDA devices found:', dlib.cuda.get_num_devices()); import face_recognition;
print('Face Recognition version:', face_recognition.__version__)"

conda deactivate

echo "=====
echo "SETUP COMPLETE. PLEASE RUN: source ~/.bashrc"
echo "=====
```

Terminal O/P: jonSetup6-CV-Pkgs.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script6
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script6$ ./jonSetup6-CV-Pkgs.sh
[1/7] Setting some dirs and envs ...
Sourcing conda profile to enable the 'conda' command...
Making some dirs...
[2/7] Creating 8GB Swapfile...
Setting up swapspace version 1, size = 8 GiB (8589930496 bytes)
no label, UUID=de064a9b-a84f-4889-b77e-3a73d941cb04
/swapfile none swap sw 0 0
Swapfile created and verified:

```

	total	used	free	shared	buff/cache	available
Mem:	7.4Gi	1.8Gi	4.1Gi	31Mi	1.6Gi	5.4Gi
Swap:	11Gi	0B	11Gi			

```

NAME      TYPE      SIZE USED PRIO
/dev/zram0 partition 635M  0B   5
/dev/zram1 partition 635M  0B   5
/dev/zram2 partition 635M  0B   5
/dev/zram3 partition 635M  0B   5
/dev/zram4 partition 635M  0B   5
/dev/zram5 partition 635M  0B   5
/swapfile_file _8G_ 0B _2_
[3/7] Installing CUDA-enabled OpenCV...
Looking in indexes: https://pypi.jetson-ai-lab.io/jp6/cu128
Collecting opencv-python
  Downloading https://pypi.jetson-ai-lab.io/jp6/cu128/%2Bf/395/7b5d0bec560ae/opencv_python-4.11.0-py3-none-any.whl (1.1 kB)
Collecting opencv-contrib-python==4.11.0 (from opencv-python)
  Downloading https://pypi.jetson-ai-lab.io/jp6/cu128/%2Bf/921/d2951292c79e7/opencv_contrib_python-4.11.0.86-cp310-cp310-linux_aarch64.whl (43.5 MB)
43.5/43.5 MB 3.0 MB/s 0:00:14
Collecting numpy>=1.21.2 (from opencv-contrib-python==4.11.0->opencv-python)
  Downloading https://pypi.jetson-ai-lab.io/root/pypi/%2Bf/efd/28d4e9cd7d7a8/numpy-2.2.6-cp310-cp310-manylinux_2_17_aarch64.manylinux2014_aarch64.whl (14.3 MB)
14.3/14.3 MB 3.4 MB/s 0:00:04
Installing collected packages: numpy, opencv-contrib-python, opencv-python
Successfully installed numpy-2.2.6 opencv-contrib-python-4.11.0.86 opencv-python-4.11.0
Fixing missing dependencies...
Get:1 file:/var/cudnn-local-tegra-repo-ubuntu2204-9.3.0 InRelease [1,572 B]
Get:2 file:/var/l4t-cuda-tegra-repo-ubuntu2204-12-6-local InRelease [1,572 B]
Get:1 file:/var/cudnn-local-tegra-repo-ubuntu2204-9.3.0 InRelease [1,572 B]
Get:3 file:/var/nv-tensorrt-local-tegra-repo-ubuntu2204-10.3.0-cuda-12.5 InRelease [1,572 B]
Get:2 file:/var/l4t-cuda-tegra-repo-ubuntu2204-12-6-local InRelease [1,572 B]
Get:3 file:/var/nv-tensorrt-local-tegra-repo-ubuntu2204-10.3.0-cuda-12.5 InRelease [1,572 B]
Hit:4 https://repo.download.nvidia.com/jetson/common r36.4 InRelease
Hit:5 https://download.docker.com/linux/ubuntu jammy InRelease
Hit:6 https://repo.download.nvidia.com/jetson/t234 r36.4 InRelease
Hit:7 https://packages.microsoft.com/repos/code stable InRelease
Hit:8 https://repo.download.nvidia.com/jetson/ffmpeg r36.4 InRelease
Hit:9 http://ports.ubuntu.com/ubuntu-ports jammy InRelease
Hit:10 http://ports.ubuntu.com/ubuntu-ports jammy-updates InRelease
Hit:11 http://ports.ubuntu.com/ubuntu-ports jammy-security InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libarchive-dev libarchive13 liblpt5 liblptonica-dev libtesseract4
```

swapfile

opencv from wheel

Terminal O/P: jonSetup6-CV-Pkgs.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script6
Fetched 3,635 kB in 3s (1,215 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libjpeg-turbo8-dev:arm64.
(Reading database ... 210658 files and directories currently installed.)
Preparing to unpack .../libjpeg-turbo8-dev_2.1.2-0ubuntu1_arm64.deb ...
Unpacking libjpeg-turbo8-dev:arm64 (2.1.2-0ubuntu1) ...
Selecting previously unselected package libjpeg8-dev:arm64.
Preparing to unpack .../libjpeg8-dev_8c-2ubuntu10_arm64.deb ...
Unpacking libjpeg8-dev:arm64 (8c-2ubuntu10) ...
Selecting previously unselected package libjpeg-dev:arm64.
Preparing to unpack .../libjpeg-dev_8c-2ubuntu10_arm64.deb ...
Unpacking libjpeg-dev:arm64 (8c-2ubuntu10) ...
Selecting previously unselected package liblapack-dev:arm64.
Preparing to unpack .../liblapack-dev_3.10.0-2ubuntu1_arm64.deb ...
Unpacking liblapack-dev:arm64 (3.10.0-2ubuntu1) ...
Setting up libjpeg-turbo8-dev:arm64 (2.1.2-0ubuntu1) ...
Setting up liblapack-dev:arm64 (3.10.0-2ubuntu1) ...
Setting up libjpeg8-dev:arm64 (8c-2ubuntu10) ...
Setting up libjpeg-dev:arm64 (8c-2ubuntu10) ...
Looking in indexes: https://pypi.jetson-ai-lab.io/jp6/cu128
Requirement already satisfied: numpy in /home/vk-jn-or/miniforge3/envs/vk1_env/lib/python3.10/site-packages (2.2.6)
Looking in indexes: https://pypi.jetson-ai-lab.io/jp6/cu128
Collecting setuptools<81
  Downloading https://pypi.jetson-ai-lab.io/root/pypi/%2Bf/95b/30ddfb717250e/setuptools-80.10.2-py3-none-any.whl (1.1 MB)
    1.1/1.1 MB 1.5 MB/s 0:00:00
Installing collected packages: setuptools
  Attempting uninstall: setuptools
    Found existing installation: setuptools 82.0.0
    Uninstalling setuptools-82.0.0:
      Successfully uninstalled setuptools-82.0.0
Successfully installed setuptools-80.10.2
Looking in indexes: https://pypi.jetson-ai-lab.io/jp6/cu128
Collecting dlib
  Downloading https://pypi.jetson-ai-lab.io/root/pypi/%2Bf/9ab/6a6fe113cc36a/dlib-20.0.0.tar.gz (3.3 MB)
    3.3/3.3 MB 3.1 MB/s 0:00:01
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing metadata (pyproject.toml) ... done
Collecting face_recognition
  Downloading https://pypi.jetson-ai-lab.io/root/pypi/%2Bf/c54/3e91c8cfbf24d/face_recognition-1.3.0-py2.py3-none-any.whl (15 kB)
Collecting face-recognition-models>=0.3.0 (from face_recognition)
  Downloading https://pypi.jetson-ai-lab.io/root/pypi/%2Bf/b79/bd200a88c87c9/face_recognition_models-0.3.0.tar.gz (100.1 MB)
    100.1/100.1 MB 2.7 MB/s 0:00:35
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing metadata (pyproject.toml) ... done
Collecting Click>=6.0 (from face_recognition)
  Downloading https://pypi.jetson-ai-lab.io/root/pypi/%2Bf/981/153a64e25f12d/click-8.3.1-py3-none-any.whl (108 kB)
Requirement already satisfied: numpy in /home/vk-jn-or/miniforge3/envs/vk1_env/lib/python3.10/site-packages (from face_recognition) (2.2.6)
Collecting Pillow (from face_recognition)
  Downloading https://pypi.jetson-ai-lab.io/root/pypi/%2Bf/3a5/cbcdcdad0af3d/pillow-12.1.1-cp310-cp310-manylinux_2_27_aarch64.manylinux_2_28_aarch64.whl (6.3 MB)
    6.3/6.3 MB 3.0 MB/s 0:00:02
Building wheels for collected packages: dlib, face-recognition-models
  Building wheel for dlib (pyproject.toml) ... |
```

Terminal O/P: jonSetup6-CV-Pkgs.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script6

Setting up libcdt5:arm64 (2.42.2-6ubuntu0.1) ...
Setting up libcgraph6:arm64 (2.42.2-6ubuntu0.1) ...
Setting up libgts-bin (0.7.6+darcs121130-5) ...
Setting up libgvc6 (2.42.2-6ubuntu0.1) ...
Setting up libgvpr2:arm64 (2.42.2-6ubuntu0.1) ...
Setting up graphviz (2.42.2-6ubuntu0.1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.9) ...
[6/7] Setting up GIMP AppImage and Icon...
--2026-02-19 18:47:36-- https://download.gimp.org/gimp/v3.0/linux/GIMP-3.0.4-aarch64.AppImage
Resolving download.gimp.org (download.gimp.org)... 140.211.167.248
Connecting to download.gimp.org (download.gimp.org)|140.211.167.248|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://cofractal-ewr.mm.fcix.net/gimp/gimp/v3.0/linux/GIMP-3.0.4-aarch64.AppImage [following]
--2026-02-19 18:47:38-- https://cofractal-ewr.mm.fcix.net/gimp/gimp/v3.0/linux/GIMP-3.0.4-aarch64.AppImage
Resolving cofractal-ewr.mm.fcix.net (cofractal-ewr.mm.fcix.net)... 2606:7940:52:10::10, 199.255.19.134
Connecting to cofractal-ewr.mm.fcix.net (cofractal-ewr.mm.fcix.net)|2606:7940:52:10::10|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 155933192 (149M) [application/octet-stream]
Saving to: 'GIMP-3.0.4-aarch64.AppImage'

GIMP-3.0.4-aarch64.AppImage      100%[=====] 148.71M  56.3MB/s   in 2.6s

2026-02-19 18:47:41 (56.3 MB/s) - 'GIMP-3.0.4-aarch64.AppImage' saved [155933192/155933192]

--2026-02-19 18:47:41-- https://upload.wikimedia.org/wikipedia/commons/4/45/The_GIMP_icon_-_gnome.svg
Resolving upload.wikimedia.org (upload.wikimedia.org)... 2620:0:861:edia::2:b, 208.80.154.240
Connecting to upload.wikimedia.org (upload.wikimedia.org)|2620:0:861:edia::2:b|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [image/svg+xml]
Saving to: '/home/vk-jn-or/Applications/GIMP.png'

/home/vk-jn-or/Applications/GIMP.png      [ <=> ] 21.36K  --.-KB/s   in 0.004s

2026-02-19 18:47:41 (5.66 MB/s) - '/home/vk-jn-or/Applications/GIMP.png' saved [21877]

[7/7] Creating GIMP desktop shortcut...

-----
VERIFYING INSTALLATION...
=====
Dlib version: 20.0.0
CUDA support enabled: True
CUDA devices found: 1
/home/vk-jn-or/miniforge3/envs/vk1_env/lib/python3.10/site-packages/face_recognition_models/_init__.py:7: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to setuptools<81.
  from pkg_resources import resource_filename
Face Recognition version: 1.2.3
=====
SETUP COMPLETE. PLEASE RUN: source ~/.bashrc
=====
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script6$ source ~/.bashrc
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script6$
```

GIMP download

GIMP desktop shortcut

Face recognition install → success

Terminal O/P: jonSetup6-CV-Pkgs.sh

vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script6



```
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script6$ cat ~/.bashrc
# ~/.bashrc: executed by bash(1) for non-login shells.
# see /usr/share/doc/bash/examples/startup-files (in the package bash-doc)
# for examples

# If not running interactively, don't do anything
case $- in
    *i*) ;;
    *) return;;
esac

# don't put duplicate lines or lines starting with space in the history.
# See bash(1) for more options
HISTCONTROL=ignoreboth
```

...

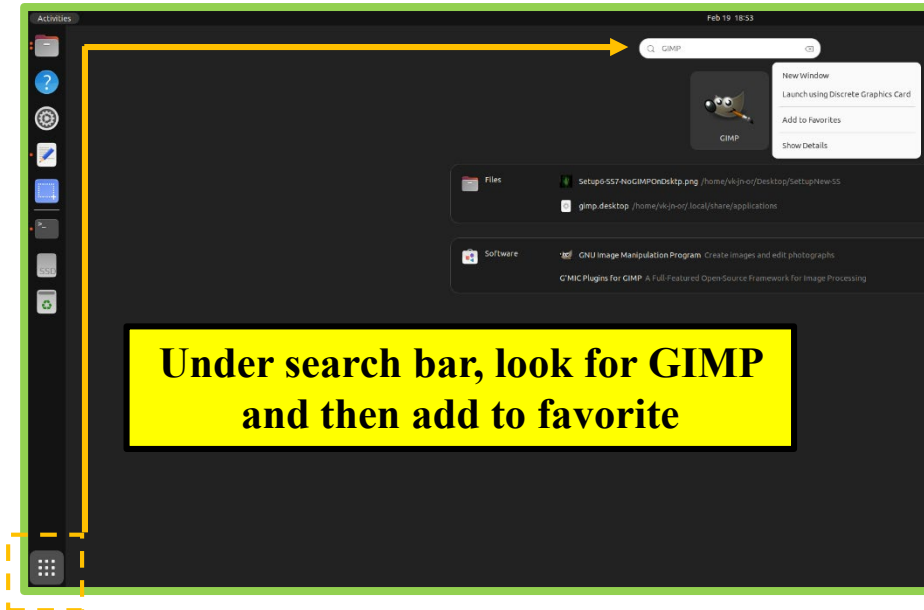
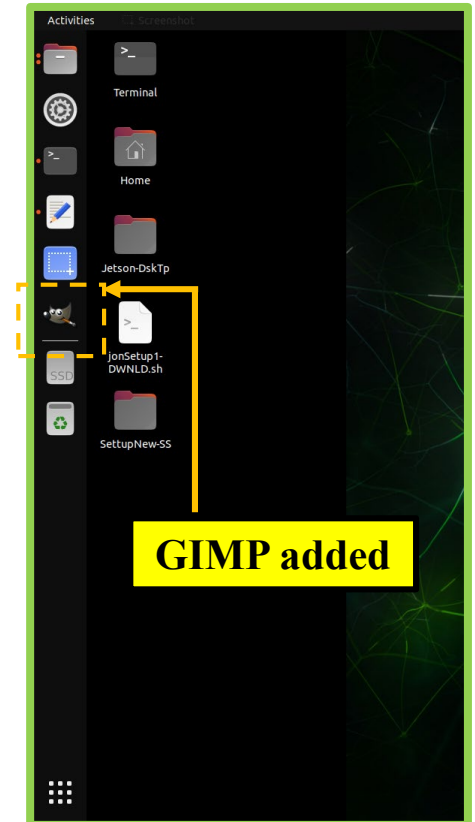
```
# >>> conda initialize >>>
# !! Contents within this block are managed by 'conda init' !!
__conda_setup="$(('/home/vk-jn-or/miniforge3/bin/conda' 'shell.bash' 'hook' 2> /dev/null)"
if [ $? -eq 0 ]; then
    eval "$__conda_setup"
else
    if [ -f "/home/vk-jn-or/miniforge3/etc/profile.d/conda.sh" ]; then
        . "/home/vk-jn-or/miniforge3/etc/profile.d/conda.sh"
    else
        export PATH="/home/vk-jn-or/miniforge3/bin:$PATH"
    fi
fi
unset __conda_setup
# <<< conda initialize <<<

# Conda aliases
alias condaon1='source ~/condaActivate.sh'
alias condaoff1='source ~/condaDeactivate.sh'
export PATH=/usr/local/cuda/bin:$PATH
export LD_LIBRARY_PATH=/usr/local/cuda/lib64:$LD_LIBRARY_PATH

# Fix for libffi
export LD_PRELOAD=/usr/lib/aarch64-linux-gnu/libffi.so.8
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script6$
```

New item added to ~/.bashrc

Add GIMP to Favorites



Shell Script: jonSetup6-status

- Check JON status after executing jonSetup6-CV-Pkgs

```
#!/bin/bash
```

```
# --- JETSON ORIN NANO SETUP STATUS SCRIPT - jonSetup6-status.sh ---
```

```
ENV_NAME="vk1_env"
```

```
echo "Sourcing conda profile to enable the 'conda' command..."
```

```
source ~/miniforge3/etc/profile.d/conda.sh
```

```
conda activate $ENV_NAME      # Activate vk1_env
```

```
echo "Python Version ..."
```

```
python3 --version
```

```
echo "OpenCV ..."
```

```
dpkg -l | grep libopencv
```

```
echo "numpy ..."
```

```
dpkg -l | grep numpy
```

```
python -c "import cv2; print('CV2 Version          = ', cv2.__version__); build_info =  
cv2.getBuildInformation(); print('Number CUDA-enabled devices =  
,cv2.cuda.getCudaEnabledDeviceCount());   print('CUDA enabled          = ', 'CUDA' in build_info)"
```

```
python -c "import dlib; print('Dlib version:', dlib.__version__); print('CUDA support enabled:',  
dlib.DLIB_USE_CUDA); print('CUDA devices found:', dlib.cuda.get_num_devices()); import face_recognition;  
print('Face Recognition version:', face_recognition.__version__)"
```

```
conda deactivate
```

Terminal O/P: jonSetup6-status.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script6
vk-jn-or@vkjnor-desktop:~$ cd /home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script6/
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script6$ ./jonSetup6-status.sh
Sourcing conda profile to enable the 'conda' command...
Python Version ...
Python 3.10.12
OpenCV ...
ii libopencv 4.8.0-1-g6371ee1 arm64 Open Computer Vision Library
ii libopencv-core4.5d:arm64 4.5.4+dfsg-9ubuntu4 arm64 computer vision core library
ii libopencv-dev 4.8.0-1-g6371ee1 arm64 Open Computer Vision Library
ii libopencv-python 4.8.0-1-g6371ee1 arm64 Open Computer Vision Library
ii libopencv-samples 4.8.0-1-g6371ee1 arm64 Open Computer Vision Library
numpy ...
ii python3-numpy 1:1.21.5-1ubuntu22.04.1 arm64 Fast array facility to the Python 3 language
CV2 Version = 4.11.0
Number CUDA-enabled devices = 1
CUDA enabled = True
Dlib version: 20.0.0
CUDA support enabled: True
CUDA devices found: 1
/home/vk-jn-or/miniforge3/envs/vk1_env/lib/python3.10/site-packages/face_recognition_models/_init_.py:7: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to Setuptools<81.
from pkg_resources import resource_filename
Face Recognition version: 1.2.3
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script6$
```

Note:
CUDA-enabled devices: 1
CUDA enabled: True
dlib version: 20.0.0
Ignore warning about face_recognition package slated for removal

System-wide OpenCV from Jetpack—different from OpenCV in conda environment

```
CV2 Version = 4.8.0
Number CUDA-enabled devices = 0
CUDA enabled = False
```

Script: jonSetup7-doorCam.sh

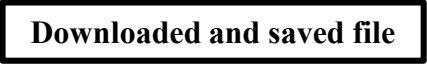
- Run `jonSetup7-doorCam.sh` to download `doorcam.zip` and copy an edited file

```
#!/bin/bash

# Configuration
DOWNLOAD_PARENT_DIR="/home/vk-jn-or/Downloads/Jetson-Orin-Nano"
FACE_RECG_DIR="$DOWNLOAD_PARENT_DIR/Face_Recg"
DOWNLOAD_FACE_RECG_Edit_DIR="/home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/Edited-Files/Face_Recg"
Doorcam_URL="https://gist.github.com/ageitgey/84943a12dd0d9f54e90f824b94e4c2a9/archive/906b921481a0f18ddf1fcbb1e8a45d58fbde3b5e.zip"

echo "Setting up Face Recognition directory..."
mkdir -p "$FACE_RECG_DIR"
cd "$FACE_RECG_DIR"

echo "Downloading doorcam.zip..."
# Using the raw URL to ensure a clean python file download
wget -O doorcam.zip "$Doorcam_URL"
echo "Unzipping downloaded file..."
unzip doorcam.zip
echo "Removing zip file..."
rm doorcam.zip
echo "Changing to uncompressed folder..."
cd 8*
echo "Copying downloaded Python file into $FACE_RECG_DIR folder..."
cp *.py ../doorbell_camera.py
echo "Removing downloaded file and folder..."
rm *.py
cd ..
rmdir 8*
```



Script: jonSetup7-doorCam.sh

Use this file and compare
with doorbell_camera.py



```
# Save as a new version
echo "Copying an edited Python file..."
cp "$DOWNLOAD_FACE_RECG_Edit_DIR/doorbell_camera_1.py" "$FACE_RECG_DIR/doorbell_camera_1.py"

echo "Use $FACE_RECG_DIR/doorbell_camera_1.py in next step..."
# Commented out from $FACE_RECG_DIR/doorbell_camera.py the original line 134 and inserted the cv2.cvtColor
line
# Original:      rgb_small_frame = small_frame[:, :, :-1]
# Replacement:  rgb_small_frame = cv2.cvtColor(small_frame, cv2.COLOR_BGR2RGB)

echo "=====
echo "PROCESS COMPLETE"
echo "File saved as: $FACE_RECG_DIR/doorbell_camera_1.py"
echo "=====
```

```
1  #!/bin/bash
2
3  # Configuration
4  DOWNLOAD_PARENT_DIR="/home/vk-jn-or/Downloads/Jetson-Orin-Nano"
5  FACE_RECG_DIR="$DOWNLOAD_PARENT_DIR/Face_Recg"
6  DOWNLOAD_FACE_RECG_Edit_DIR="/home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/Edited-Files/Face_Recg"
7  Doorcam_URL="https://gist.github.com/ageitgey/84943a12dd0d9f54e90f824b94e4c2a9/archive/906b921481a0f18ddf1fcbb1e8a45d58fbde3b5e.zip"
8
9  echo "Setting up Face Recognition directory..."
10 mkdir -p "$FACE_RECG_DIR"
11 cd "$FACE_RECG_DIR"
12
13 echo "Downloading doorcam.zip..."
14 # Using the raw URL to ensure a clean python file download
15 wget -O doorcam.zip "$Doorcam_URL"
16 echo "Unzipping downloaded file..."
17 unzip doorcam.zip
18 echo "Removing zip file..."
19 rm doorcam.zip
20 echo "Changing to uncompressed folder..."
21 cd 8*
22 echo "Copying downloaded Python file into $FACE_RECG_DIR folder..."
23 cp *.py ../doorbell_camera.py
24 echo "Removing downloaded file and folder..."
25 rm *.py
26 cd ..
27 rmdir 8*
28
29 # Save as a new version
30 echo "Copying an edited Python file..."
31 cp "$DOWNLOAD_FACE_RECG_Edit_DIR/doorbell_camera_1.py" "$FACE_RECG_DIR/doorbell_camera_1.py"
32
33 echo "Use $FACE_RECG_DIR/doorbell_camera_1.py in next step..."
34 # Commented out from $FACE_RECG_DIR/doorbell_camera.py the original line 134 and inserted the cv2.cvtColor line
35 # Original:      rgb_small_frame = small_frame[:, :, :-1]
36 # Replacement:  rgb_small_frame = cv2.cvtColor(small_frame, cv2.COLOR_BGR2RGB)
37
38 echo "=====
39 echo "PROCESS COMPLETE"
40 echo "File saved as: $FACE_RECG_DIR/doorbell_camera_1.py"
41 echo "=====
42
```

Terminal O/P: jonSetup7-doorCam.sh

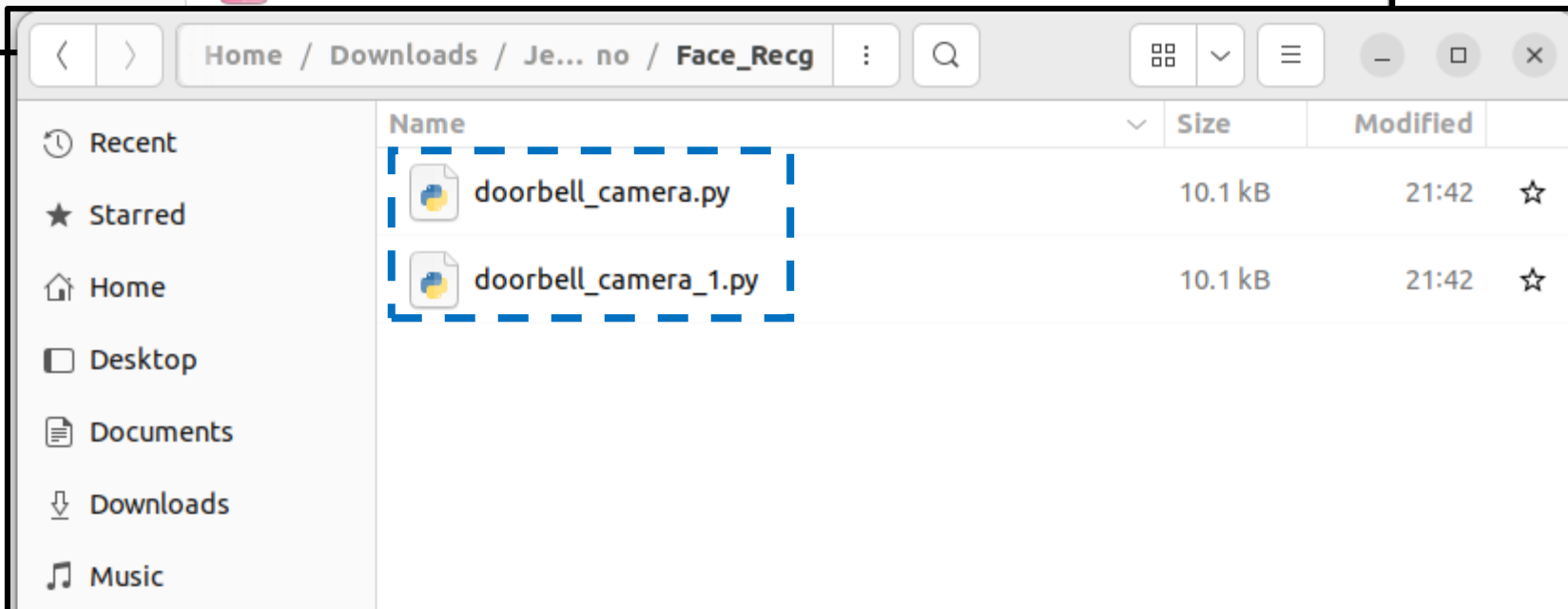
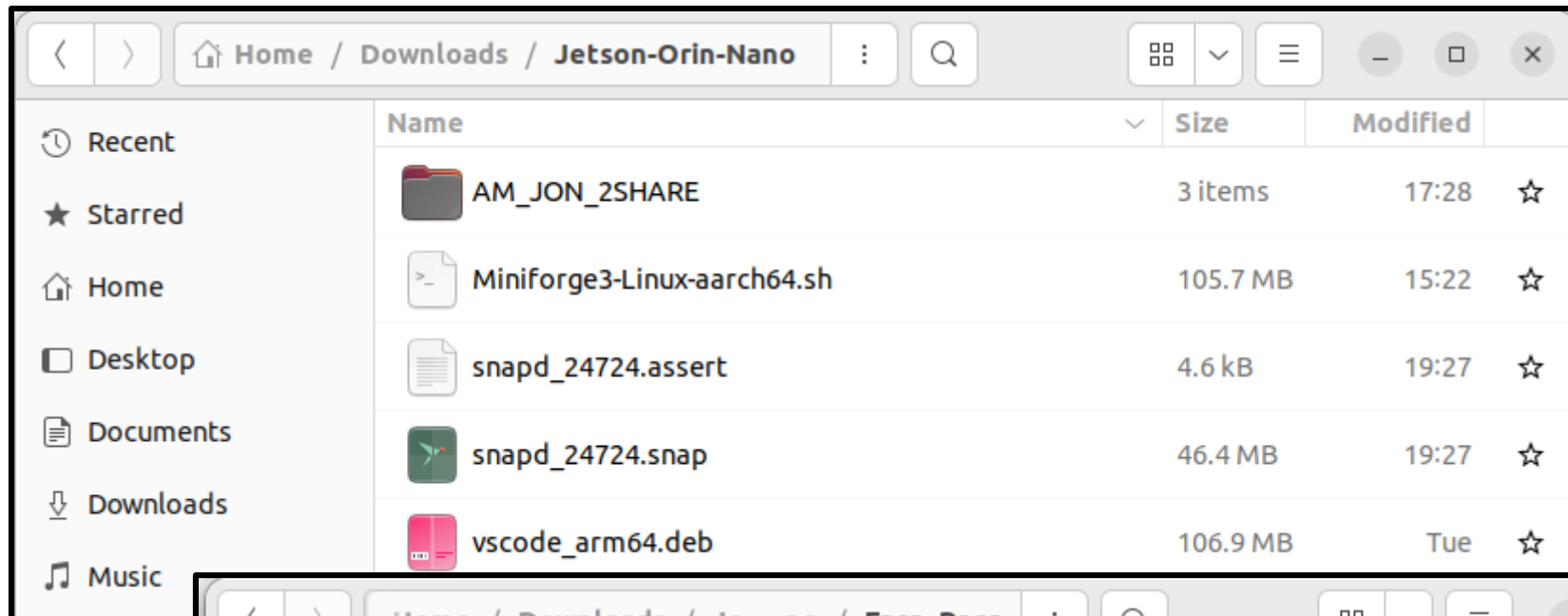
```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script7
vk-jn-or@vkjnor-desktop:~$ cd /home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script7/
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script7$ ./jonSetup7-doorCam.sh
Setting up Face Recognition directory...
Downloading doorcam.zip...
--2026-02-09 21:42:11-- https://gist.github.com/ageitgey/84943a12dd0d9f54e90f824b94e4c2a9/archive/906b921481a0f18ddf1fcb1e8a45d58fbde3b5e.zip
Resolving gist.github.com (gist.github.com)... 140.82.112.3
Connecting to gist.github.com (gist.github.com)|140.82.112.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://codeload.github.com/gist/84943a12dd0d9f54e90f824b94e4c2a9/zip/906b921481a0f18ddf1fcb1e8a45d58fbde3b5e [following]
--2026-02-09 21:42:12-- https://codeload.github.com/gist/84943a12dd0d9f54e90f824b94e4c2a9/zip/906b921481a0f18ddf1fcb1e8a45d58fbde3b5e
Resolving codeload.github.com (codeload.github.com)... 140.82.113.10
Connecting to codeload.github.com (codeload.github.com)|140.82.113.10|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [application/zip]
Saving to: 'doorcam.zip'

doorcam.zip                               [ <=> ] 3.85K --.-KB/s  in 0s

2026-02-09 21:42:12 (17.1 MB/s) - 'doorcam.zip' saved [3942]

Unzipping downloaded file...
Archive: doorcam.zip
906b921481a0f18ddf1fcb1e8a45d58fbde3b5e
  creating: 84943a12dd0d9f54e90f824b94e4c2a9-906b921481a0f18ddf1fcb1e8a45d58fbde3b5e/
    inflating: 84943a12dd0d9f54e90f824b94e4c2a9-906b921481a0f18ddf1fcb1e8a45d58fbde3b5e/doorbell_camera.py
Removing zip file...
Changing to uncompressed folder...
Copying downloaded Python file into /home/vk-jn-or/Downloads/Jetson-Orin-Nano/Face_Recg folder...
Removing downloaded file and folder...
Copying an edited Python file...
Use /home/vk-jn-or/Downloads/Jetson-Orin-Nano/Face_Recg/doorbell_camera_1.py in next step...
=====
PROCESS COMPLETE
File saved as: /home/vk-jn-or/Downloads/Jetson-Orin-Nano/Face_Recg/doorbell_camera_1.py
=====
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script7$
```

jor-setup4-doorCame.sh—Download Folder Status



Script: jonSetup7-status.sh

```
#!/bin/bash

# Configuration
DOWNLOAD_PARENT_DIR="/home/vk-jn-or/Downloads/Jetson-Orin-Nano"
FACE_RECG_DIR="$DOWNLOAD_PARENT_DIR/Face_Recg"
ENV_NAME="vk1_env"

echo "Testing Phase 4..."
echo "Sourcing conda profile to enable the 'conda' command..."
source ~/miniforge3/etc/profile.d/conda.sh
conda activate $ENV_NAME      # Activate vk1_env

echo "Run the dorrbell_camera1.py program..."
python "$FACE_RECG_DIR/doorbell_camera_1.py"

echo
"=====
echo "Capture and save an image of the camera output with face in a box"
echo
"=====

conda deactivate
```

Terminal O/P: jonSetup7-status.sh

```
vk-jn-or@vkJnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script7
Testing Phase 4...
Sourcing conda profile to enable the 'conda' command...
Run the dorrbell_camera1.py program...
/home/vk-jn-or/miniforge3/envs/vk1_env/lib/python3.10/site-packages/face_recognition_models/__init__.py:7: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to Setuptools<81.
from pkg_resources import resource_filename
No previous face data found - starting with a blank known face list.
GST_ARGUS: Creating output stream
CONSUMER: Waiting until producer is connected...
GST_ARGUS: Available Sensor modes :
GST_ARGUS: 3280 x 2464 FR = 21.000000 fps Duration = 47619048 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;
GST_ARGUS: 3280 x 1848 FR = 28.000001 fps Duration = 35714284 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;
GST_ARGUS: 1920 x 1080 FR = 29.999999 fps Duration = 33333334 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;
GST_ARGUS: 1640 x 1232 FR = 29.999999 fps Duration = 33333334 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;
GST_ARGUS: 1280 x 720 FR = 59.999999 fps Duration = 16666667 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;
GST_ARGUS: Running with following settings:
    Camera index = 0
    Camera mode   = 4
    Output Stream W = 1280 H = 720
    seconds to Run   = 0
    Frame Rate      = 59.999999
GST_ARGUS: Setup Complete, Starting captures for 0 seconds
GST_ARGUS: Starting repeat capture requests.
CONSUMER: Producer has connected; continuing.
[ WARN:0@0.989] global_cap_gstreamer.cpp:1777 open OpenCV | GStreamer warning: Cannot query video position: status=0, value=-1, duration=-1
Known faces backed up to disk.
Known faces backed up to disk.
Known faces backed up to disk.
Known faces backed up to disk.
Known faces backed up to disk.
Known faces backed up to disk.
Known faces backed up to disk.
Known faces backed up to disk.
```

Msg. when a face is seen and data stored

O/P from Executing “doorbell_camera_1.py”



Video



Visitors at Door



First visit

At door 20s

jonSetup7-status.sh: Camera O/P

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script7
Testing Phase 4...
Sourcing conda profile to enable the 'conda' command...
Run the dorrbell_camera1.py program...
/home/vk-jn-or/miniforge3/envs/vk1_env/lib/python3.10/site-packages/face_recognition_models/__init__.py:7: UserWarning: pkg_resources is deprecated as an API. See https://
setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to Setuptools<81.
    from pkg_resources import resource_filename
No previous face data found - starting with a blank known face list.
GST_ARGUS: Creating output stream
CONSUMER: Waiting until producer is connected...
GST_ARGUS: Available Sensor modes :
GST_ARGUS: 3280 x 2464 FR = 21.000000 fps Duration = 47619048 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;

GST_ARGUS: 3280 x 1848 FR = 28.000001 fps Duration = 35714284 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;

GST_ARGUS: 1920 x 1080 FR = 29.999999 fps Duration = 33333334 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;

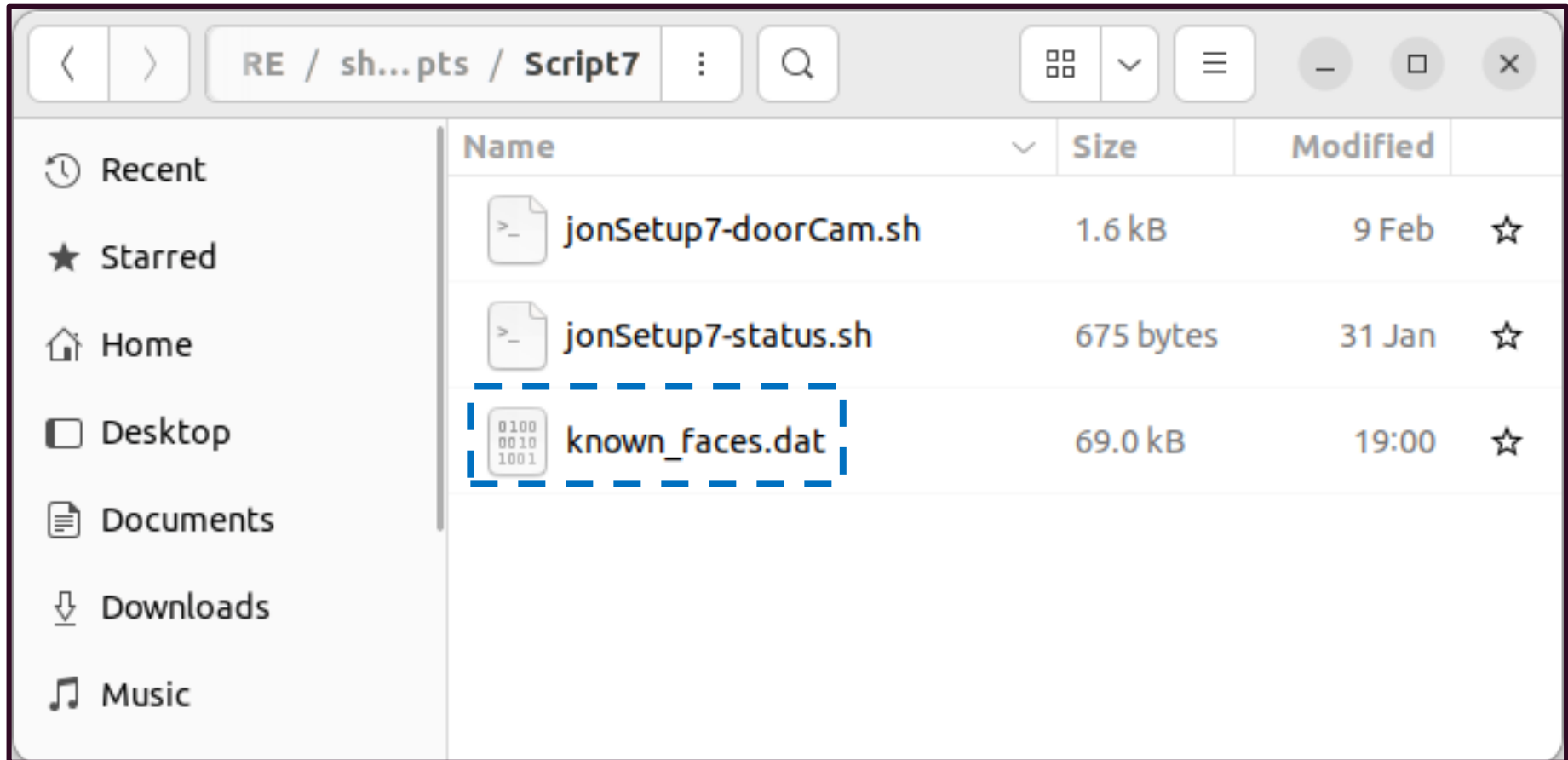
GST_ARGUS: 1640 x 1232 FR = 29.999999 fps Duration = 33333334 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;

GST_ARGUS: 1280 x 720 FR = 59.999999 fps Duration = 16666667 ; Analog Gain range min 1.000000, max 10.625000; Exposure Range min 13000, max 683709000;

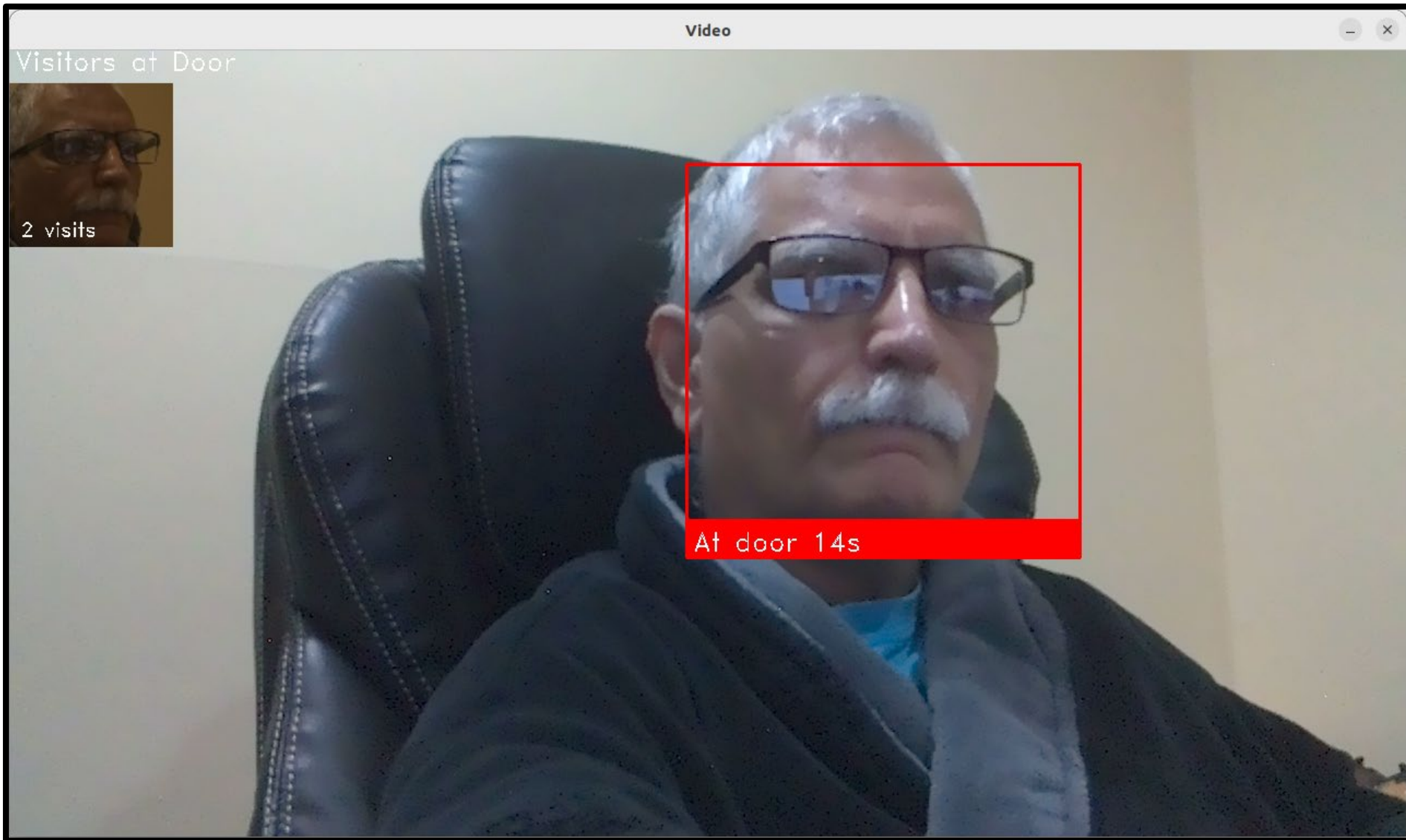
GST_ARGUS: Running with following settings:
    Camera index = 0
    Camera mode = 4
    Output Stream W = 1280 H = 720
    seconds to Run = 0
    Frame Rate = 59.999999
GST_ARGUS: Setup Complete, Starting captures for 0 seconds
GST_ARGUS: Starting repeat capture requests.
CONSUMER: Producer has connected; continuing.
[ WARN:0@0.989] global cap_gstreamer.cpp:1777 open OpenCV | GStreamer warning: Cannot query video position: status=0, value=-1, duration=-1
Known faces backed up to disk.
Known faces backed up to disk.
Known faces backed up to disk.
Known faces backed up to disk.
Known faces backed up to disk.
Known faces backed up to disk.
Known faces backed up to disk.
Known faces backed up to disk.
Known faces backed up to disk.
GST_ARGUS: Cleaning up
CONSUMER: Done Success
GST_ARGUS: Done Success
=====
Capture and save an image of the camera output with face in a box
=====
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script7$
```

jonSetup7-status.sh: Folder

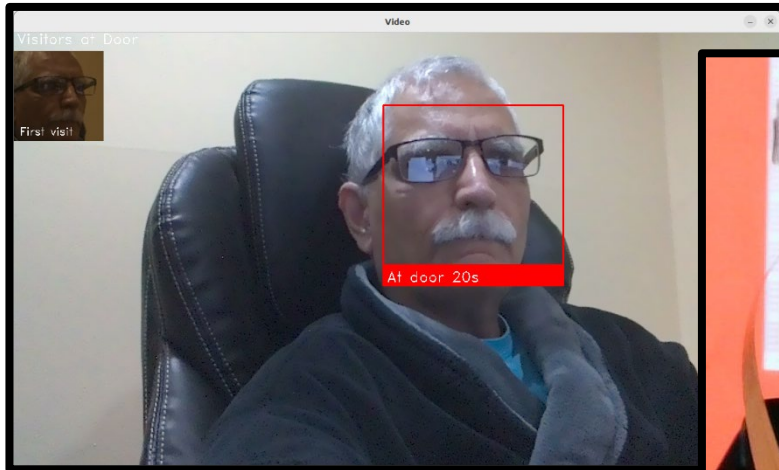
- Status of the folder where the jonSetup7-status.sh file is located



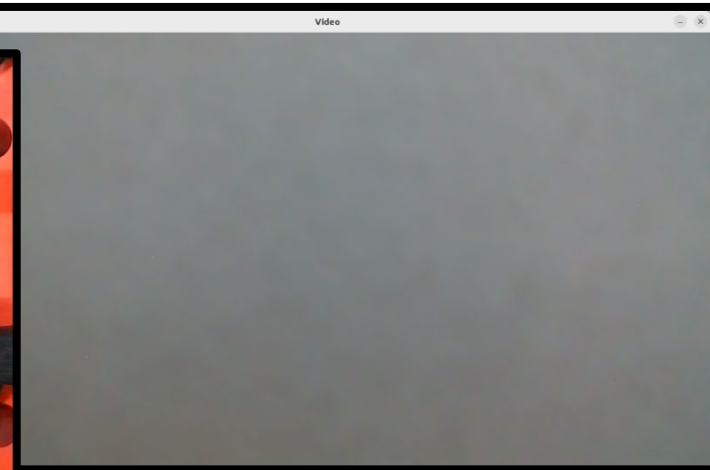
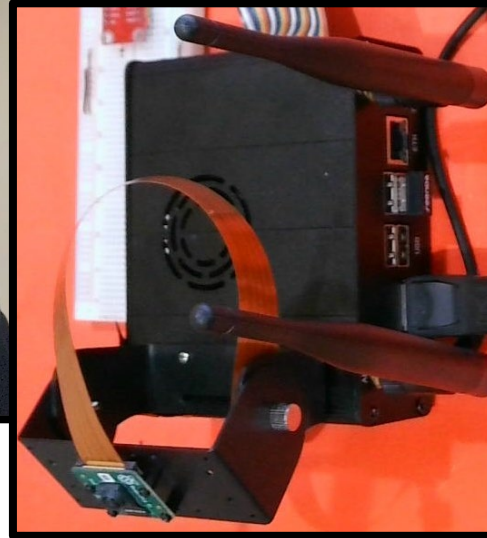
O/P from Executing “doorbell_camera_1.py”



O/P from Executing “doorbell_camera_1.py”



1. Once a face is boxed, first visit is recognized



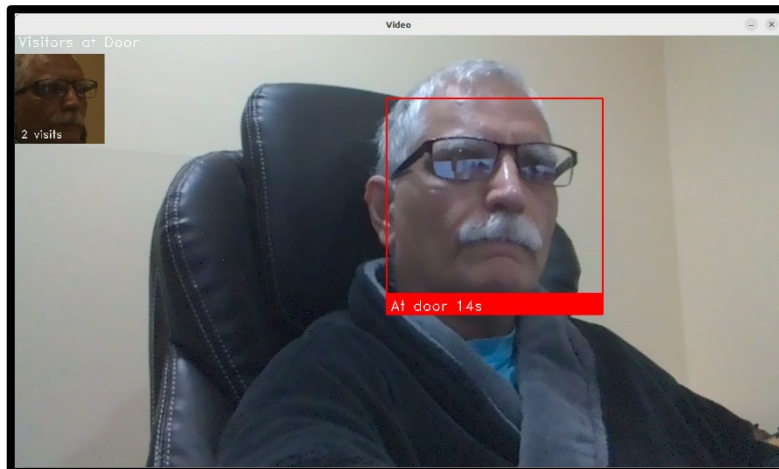
3. Blank image with camera covered



2. Block the camera with hand for a while

4. Remove hand from the front of camera

5. Second visit from same person is recognized



Shell Script: josSetup8-camApps.sh

- Install and configure various image editing and camera apps

```
#!/bin/bash
```

```
# --- INSTALL TOOLS FOR CAMERA ---
```

```
# --- 1. INSTALLATIONS ---
```

```
echo "[1/4] Downloading various applications..."
```

```
sudo apt update
```

```
sudo apt install -y simplescreenrecorder obs-studio v4l-utils mpv imagemagick
```

Screen recording and image editing

```
# --- 2. IMAGE CPYING FOR MPVCam Icon ---
```

```
echo "[2/4] Copying MPV camera image for icon..."
```

```
PICTURE_DIR="/home/vk-jn-or/Pictures"
```

```
PIC_SRC_DIR="/home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/Edited-Files/home_vk-jn-or/Pictures"
```

```
cp "$PIC_SRC_DIR/mpvCam.png" "$PICTURE_DIR/mpvCam.png"
```

```
echo "Icon copied and saved to $PICTURE_DIR/mpvCam.png"
```

```
# --- 3. CREATING CAMERAS LAUNCHER SCRIPT ---
```

```
echo "[3/4] Create and save a camera launch script: ~/launch_cameras.sh"
```

```
cat << 'EOF' > ~/launch_cameras.sh
```

```
#!/bin/bash
```

```
ICON_PATH="/home/vk-jn-or/Pictures/mpvCam.png"
```

Creating an app for launching cameras

```
# Define the choices in a Zenity list
```

```
CHOICE=$(zenity --list --title="Camera Selector" --window-icon=$ICON_PATH \
```

```
--width=350 --height=250 --text="Select configuration:" --column="ID" --column="Type" \
```

```
"1" "USB Camera (video1)" \
```

```
"2" "USB Camera (video3)" \
```

```
"3" "CSI Camera" \
```

```
"4" "Two USB" \
```

```
"5" "Two USB + one CSI" \
```

```
"6" "Close All")
```

```
# Exit if the user hits cancel
[ -z "$CHOICE" ] && exit

# Function to stop any existing camera processes
stop_cameras() {
    pkill mpv
    pkill gst-launch-1.0
}
```

```
# Logic to launch cameras based on choice
case $CHOICE in
```



5)

```
stop_cameras
# Launch both USB cameras and one CSI camera; the first one is sent to background with &
v4l2-ctl --device=/dev/video1 --set-fmt-video=width=640,height=480 && \
mpv av://v4l2:/dev/video1 --profile=low-latency --untimed --geometry=320x240+1+840 &
v4l2-ctl --device=/dev/video3 --set-fmt-video=width=640,height=480 && \
mpv av://v4l2:/dev/video3 --profile=low-latency --untimed --geometry=320x240+395+840 &
gst-launch-1.0 nvarguscamerasrc sensor-id=0 ! 'video/x-raw(memory:NVMM),width=640,height=480' ! \
nvvidconv ! nveglglessink window-x=720 window-y=840 window-width=320 window-height=240
;;
```

Video1, Video2, CSI

6)

```
# Just close everything and exit
stop_cameras
;;
```

esac

EOF

```
chmod +x ~/launch_cameras.sh
```

**Desktop Camera
App Launcher**

```
# --- 4. DESKTOP SHORTCUT ---
```

```
echo "[4/4] Create a desktop App: ~/Desktop/mpv-usb-camera.desktop"
```

```
cat << EOF > ~/Desktop/mpv-usb-camera.desktop
```

```
# For single- or multi-camera:
```

```
[Desktop Entry]
```

```
Name=USB/CSI Camera Suite (MPV 720p)
```

```
Comment=Select and Launch USB/CSI cameras at 640x480 using MPV
```

```
Exec=/home/vk-jn-or/launch_cameras.sh
```

```
Icon=/home/vk-jn-or/Pictures/mpvCam.png
```

```
Terminal=false
```

```
Type=Application
```

```
Categories=AudioVideo;Video;
```

```
EOF
```

```
chmod +x ~/Desktop/mpv-usb-camera.desktop
```

```
echo "Setup complete. Shortcut created on Desktop."
```

```
echo "Click on the file placed on desktop, it will convert to MPV icon!!!"
```

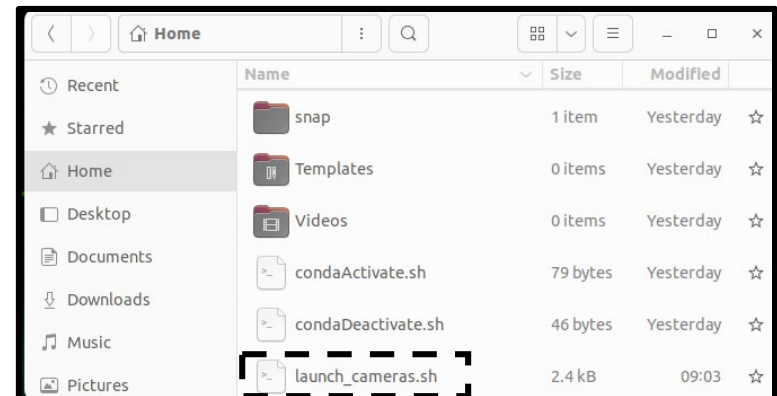
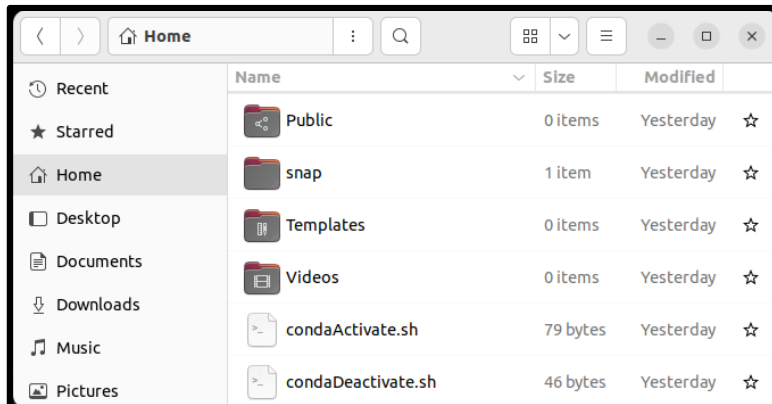
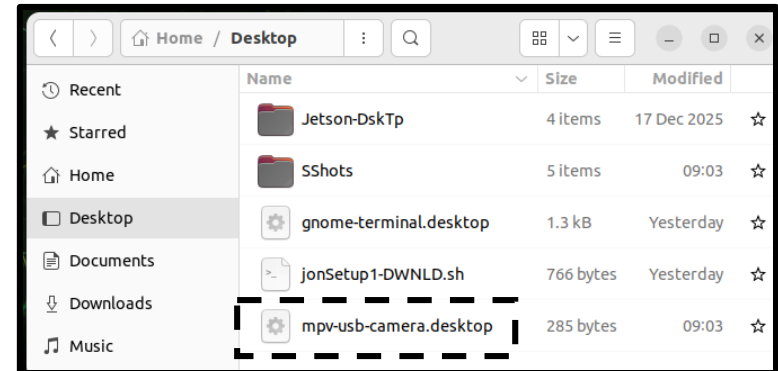
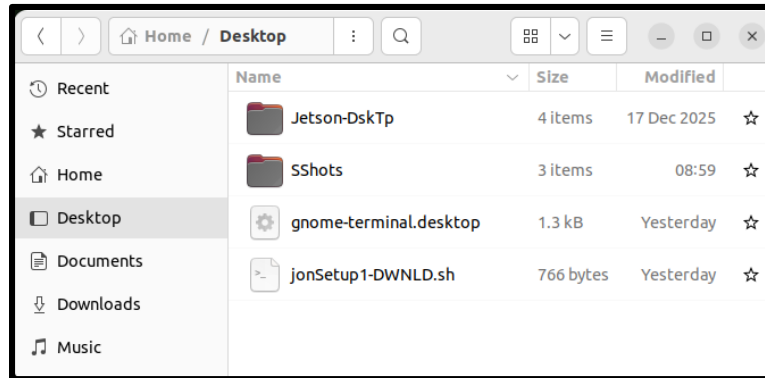
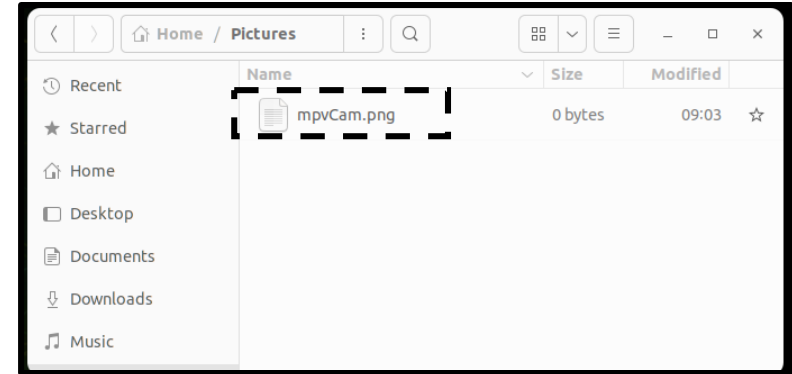
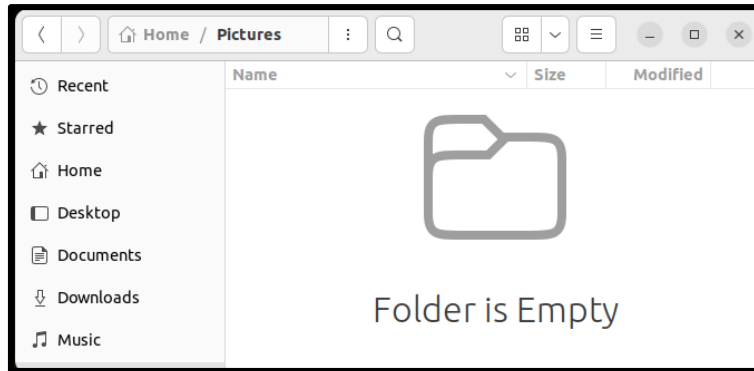
Terminal O/P: jonSetup8-status.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script8
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script8$ ./josSetup8-camApps.sh
[1/4] Downloading various applications...
Get:1 file:/var/cudnn-local-tegra-repo-ubuntu2204-9.3.0 InRelease [1,572 B]
Get:1 file:/var/cudnn-local-tegra-repo-ubuntu2204-9.3.0 InRelease [1,572 B]
Get:2 file:/var/l4t-cuda-tegra-repo-ubuntu2204-12-6-local InRelease [1,572 B]
Get:3 file:/var/nv-tensorrt-local-tegra-repo-ubuntu2204-10.3.0-cuda-12.5 InRelease [1,572 B]
Get:2 file:/var/l4t-cuda-tegra-repo-ubuntu2204-12-6-local InRelease [1,572 B]
Get:3 file:/var/nv-tensorrt-local-tegra-repo-ubuntu2204-10.3.0-cuda-12.5 InRelease [1,572 B]
Hit:4 https://download.docker.com/linux/ubuntu jammy InRelease
Hit:5 https://repo.download.nvidia.com/jetson/common r36.4 InRelease
Hit:6 https://packages.microsoft.com/repos/code stable InRelease
Hit:7 https://repo.download.nvidia.com/jetson/t234 r36.4 InRelease
Hit:8 https://repo.download.nvidia.com/jetson/ffmpeg r36.4 InRelease
Hit:9 http://ports.ubuntu.com/ubuntu-ports jammy InRelease
Hit:10 http://ports.ubuntu.com/ubuntu-ports jammy-updates InRelease
Hit:11 http://ports.ubuntu.com/ubuntu-ports jammy-backports InRelease
Hit:12 http://ports.ubuntu.com/ubuntu-ports jammy-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
308 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
 libavcodec58 libavdevice58 libavfilter7 libavformat58 libavutil56 libpostproc55 libssh-gcrypt-4 libswresample3 libswscale5
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
 ffmpeg gsfnts imagemagick-6-common imagemagick-6.q16 libaribb24-0 libavdevice58 libcddb2 libdouble-conversion3 libdvbpsi10 libebml5
libfftw3-double3 libheif1 libixml10 libjxr-tools liblqr-1-0 liblua5.2-0 libluajit-5.1-2 libluajit-5.1-common libmad0
libmagickcore-6.q16-6 libmagickcore-6.q16-6-extra libmagickwand-6.q16-6 libmatroska7 libmbedcrypto7 libmbdts14 libmbedx509-1
libmd4c0 libmujls1 libnetpbm10 libobs0 libopenmpt-modplug1 libplacebo192 libprotobuf-lite23 libproxy-tools libqt5core5a libqt5dbus5
libqt5gui5 libqt5network5 libqt5svg5 libqt5widgets5 libqt5xml5 libqxt-builder0c2a libsdl-image1.2 libsdl1.2debian
libsdlplay2 libstx11 libspatialaudio0 libstrti.4-openssl libssh2-1 libunpnp13 libv4l2rds0 libva-wayland2 libvlc-bin libvlcs
libvlccore9 libxcb-composite0 libxcb-xinerama0 libxcb-xinput0 netpbm obs-plugins python3-mutagen python3-pycryptodome
python3-pyattr python3-websockets qt5-gtk-platformtheme qtranslatons5-l10n rtmpdump vlc vlc-bin vlc-data vlc-l10n
vlc-plugin-access-extra vlc-plugin-base vlc-plugin-notify vlc-plugin-qt vlc-plugin-samba vlc-plugin-skins2 vlc-plugin-video-output
vlc-plugin-video-splitter vlc-plugin-visualization yt-dlp
Suggested packages:
imagemagick-doc autotrace enscript gimp gnuplot grads hp2xx html2ps libwmf-bin mplayer povray radiance texlive-base-bin transfig
ufraw-batch libfftw3-bin libfftw3-dev inkscape qt5-image-formats-plugins qtwayland5 v4l2loopback-dkms python-mutagen-doc
python3-pyattr-dbg python-pyattr-doc vlc-plugin-fluidsynth vlc-plugin-jack vlc-plugin-svg libvdcss2 libfribidi-bin | bidiv
phantomjs
The following NEW packages will be installed:
ffmpeg gsfnts imagemagick imagemagick-6-common imagemagick-6.q16 libaribb24-0 libavdevice58 libcddb2 libdouble-conversion3
libdvbpsi10 libebml5 libfftw3-double3 libheif1 libixml10 libjxr-tools liblqr-1-0 liblua5.2-0 libluajit-5.1-2 libluajit-5.1-common
libmad0 libmagickcore-6.q16-6 libmagickcore-6.q16-6-extra libmagickwand-6.q16-6 libmatroska7 libmbedcrypto7 libmbdts14
libmbedx509-1 libmd4c0 libmujls1 libnetpbm10 libobs0 libopenmpt-modplug1 libplacebo192 libprotobuf-lite23 libproxy-tools libqt5core5a
```

Terminal O/P: josSetup8-camApps.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script8
update-alternatives: warning: skip creation of /usr/share/man/man1/identify-im6.1.gz because associated file /usr/share/man/man1/identify-im6.q16.1.gz (of link group identify-im6) doesn't exist
update-alternatives: using /usr/bin/stream-im6.q16 to provide /usr/bin/stream (stream) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/stream.1.gz because associated file /usr/share/man/man1/stream-im6.q16.1.gz (of link group stream) doesn't exist
update-alternatives: using /usr/bin/stream-im6.q16 to provide /usr/bin/stream-im6 (stream-im6) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/stream-im6.1.gz because associated file /usr/share/man/man1/stream-im6.q16.1.gz (of link group stream-im6) doesn't exist
update-alternatives: using /usr/bin/display-im6.q16 to provide /usr/bin/display (display) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/display.1.gz because associated file /usr/share/man/man1/display-im6.q16.1.gz (of link group display) doesn't exist
update-alternatives: using /usr/bin/display-im6.q16 to provide /usr/bin/display-im6 (display-im6) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/display-im6.1.gz because associated file /usr/share/man/man1/display-im6.q16.1.gz (of link group display-im6) doesn't exist
update-alternatives: using /usr/bin/montage-im6.q16 to provide /usr/bin/montage (montage) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/montage.1.gz because associated file /usr/share/man/man1/montage-im6.q16.1.gz (of link group montage) doesn't exist
update-alternatives: using /usr/bin/montage-im6.q16 to provide /usr/bin/montage-im6 (montage-im6) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/montage-im6.1.gz because associated file /usr/share/man/man1/montage-im6.q16.1.gz (of link group montage-im6) doesn't exist
update-alternatives: using /usr/bin/mogrify-im6.q16 to provide /usr/bin/mogrify (mogrify) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/mogrify.1.gz because associated file /usr/share/man/man1/mogrify-im6.q16.1.gz (of link group mogrify) doesn't exist
update-alternatives: using /usr/bin/mogrify-im6.q16 to provide /usr/bin/mogrify-im6 (mogrify-im6) in auto mode
update-alternatives: warning: skip creation of /usr/share/man/man1/mogrify-im6.1.gz because associated file /usr/share/man/man1/mogrify-im6.q16.1.gz (of link group mogrify-im6) doesn't exist
Setting up libqt5x11extras5:arm64 (5.15.3-1) ...
Setting up libqt5svg5:arm64 (5.15.3-1) ...
Setting up obs-studio (27.2.3+dfsg1-1) ...
Setting up imagemagick (8:6.9.11.60+dfsg-1.3ubuntu0.22.04.5) ...
Setting up vlc-plugin-qt:arm64 (3.0.16-1build7) ...
Setting up vlc (3.0.16-1build7) ...
Setting up simplescreenrecorder (0.4.4-1) ...
Setting up vlc-plugin-skins2:arm64 (3.0.16-1build7) ...
Processing triggers for libc-bin (2.35-0ubuntu3.9) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for desktop-file-utils (0.26-1ubuntu3) ...
Processing triggers for hicolor-icon-theme (0.17-2) ...
Processing triggers for gnome-menus (3.36.0-1ubuntu3) ...
Processing triggers for libvlc-bin:arm64 (3.0.16-1build7) ...
[2/4] Copying MPV camera image for icon...
Icon copied and saved to /home/vk-jn-or/Pictures/mpvCam.png
[3/4] Create and save a camera launch script: ~/launch_cameras.sh
[4/4] Create a desktop App: ~/Desktop/mpv-usb-camera.desktop
Setup complete. Shortcut created on Desktop.
Click on the file placed on desktop, it will convert to MPV icon!!!.
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script8$
```

josSetup8-camApps.sh—Before/After Status



Shell Script: jonSetup8-status.sh

- Check JON status after installing josSetup8-camApps.sh

```
#!/bin/bash
```

```
echo "Test v4l2-ctl..."
```

```
v4l2-ctl --list-devices
```

```
v4l2-ctl -d /dev/video1 --list-formats-ext #Available framerates MJPG: 30, 15
```

```
echo "Test CSI camera..."
```

```
pkill gst-launch-1.0
```

```
gst-launch-1.0 nvarguscamerasrc sensor-id=0 ! 'video/x-raw(memory:NVMM),width=640,height=480' ! \
nvvidconv ! nveglglessink window-x=1 window-y=840 window-width=320 window-height=240
```

```
pkill gst-launch-1.0
```

```
echo "Test USB cameras using mpv..."
```

```
pkill mpv
```

```
mpv av://v4l2:/dev/video1 --profile=low-latency --untimed --demuxer-lavf-format=video4linux2 --demuxer-
lavf-o-add=pixel_format=mjpeg,video_size=640x480
```

```
pkill mpv
```

```
mpv av://v4l2:/dev/video3 --profile=low-latency --untimed --demuxer-lavf-format=video4linux2 --demuxer-
lavf-o-add=pixel_format=mjpeg,video_size=640x480
```

```
pkill mpv
```

```
echo "Click on mpv-usb-camera.desktop..."
```

CSI

USBCam1, USBCam2

Terminal O/P: jonSetup8-status.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script8
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script8$ ./jonSetup8-status.sh
Test v4l2-ctl...
NVIDIA Tegra Video Input Device (platform:tegra-camrtc-ca):
    /dev/media0

vi-output, imx219 9-0010 (platform:tegra-capture-vi:1):
    /dev/video0

IPEVO V4K: IPEVO V4K (usb-3610000.usb-2.1):
    /dev/video1
    /dev/video2
    /dev/media1

USB Camera: USB Camera (usb-3610000.usb-2.2.1):
    /dev/video3
    /dev/video4
    /dev/media2

ioctl: VIDIOC_ENUM_FMT
    Type: Video Capture

    [0]: 'MJPG' (Motion-JPEG, compressed)
        Size: Discrete 640x480
            Interval: Discrete 0.033s (30.000 fps)
        Size: Discrete 800x600
            Interval: Discrete 0.033s (30.000 fps)
        Size: Discrete 1280x720
            Interval: Discrete 0.033s (30.000 fps)
        Size: Discrete 1600x1200
            Interval: Discrete 0.033s (30.000 fps)
        Size: Discrete 1920x1080
            Interval: Discrete 0.033s (30.000 fps)
        Size: Discrete 2048x1536
            Interval: Discrete 0.067s (15.000 fps)
        Size: Discrete 2592x1944
            Interval: Discrete 0.067s (15.000 fps)
        Size: Discrete 3264x1836
            Interval: Discrete 0.067s (15.000 fps)
        Size: Discrete 3264x2448
            Interval: Discrete 0.067s (15.000 fps)
        Size: Discrete 1024x768
            Interval: Discrete 0.033s (30.000 fps)
    [1]: 'YUVV' (YUYV 4:2:2)
        Size: Discrete 640x480
            Interval: Discrete 0.033s (30.000 fps)
        Size: Discrete 800x600
            Interval: Discrete 0.040s (25.000 fps)
        Size: Discrete 1280x720
            Interval: Discrete 0.100s (10.000 fps)
```

Terminal O/P: jonSetup8-status.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script8

seconds to Run    = 0
Frame Rate = 59.999999
GST_ARGUS: Setup Complete, Starting captures for 0 seconds
GST_ARGUS: Starting repeat capture requests.
CONSUMER: Producer has connected; continuing.
Redistribute latency...
WARNING: from element /GstPipeline:pipeline0/GstEglGlessSink:eglglessink0: Pipeline construction is invalid, please add queues.
Additional debug info:
../libs/gst/base/gstbasesink.c(1249): gst_base_sink_query_latency (): /GstPipeline:pipeline0/GstEglGlessSink:eglglessink0:
Not enough buffering available for the processing deadline of 0:00:00.015000000, add enough queues to buffer 0:00:00.015000000 additional data. Shortening processing latency to 0:00:00.000000000.
WARNING: from element /GstPipeline:pipeline0/GstEglGlessSink:eglglessink0: Pipeline construction is invalid, please add queues.
Additional debug info:
../libs/gst/base/gstbasesink.c(1249): gst_base_sink_query_latency (): /GstPipeline:pipeline0/GstEglGlessSink:eglglessink0:
Not enough buffering available for the processing deadline of 0:00:00.015000000, add enough queues to buffer 0:00:00.015000000 additional data. Shortening processing latency to 0:00:00.000000000.
ERROR: from element /GstPipeline:pipeline0/GstEglGlessSink:eglglessink0: Output window was closed
Additional debug info:
/dvs/git/dirty/git-master_linux/3rdparty/gst/gst-nveglglessink/ext/eglgles/gsteglglessink.c(914): gst_eglglessink_event_thread (): /GstPipeline:pipeline0/GstEglGlessSink:eglglessink0
Execution ended after 0:01:43.175257942
Setting pipeline to NULL ...
GST_ARGUS: Cleaning up
CONSUMER: Done Success
GST_ARGUS: Done Success
Freeing pipeline ...

(gst-launch-1.0:8202): GStreamer-WARNING **: 09:43:04.097: The default memory allocator was freed!

(gst-launch-1.0:8202): GStreamer-CRITICAL **: 09:43:04.098: gst_object_unref: assertion '((GObject *) object)->ref_count > 0' failed
Test USB cameras using mpv...
Passing more than 1 argument to demuxer-lavf-o-add is deprecated!
(+) Video --vid=1 (mjpeg 640x480 30.000fps)
VO: [gpu] 640x480 yuvj422p
V: 00:59:19 / 00:59:19 (100%)
[ffmpeg/demuxer] video4linux2,v4l2: Some buffers are still owned by the caller on close.
[ffmpeg] ioctl(VIDIOC_QBUF): Bad file descriptor

Exiting... (Quit)
Passing more than 1 argument to demuxer-lavf-o-add is deprecated!
(+) Video --vid=1 (mjpeg 640x480 30.000fps)
VO: [gpu] 640x480 yuvj422p
V: 01:01:02 / 01:01:02 (100%)
[ffmpeg/demuxer] video4linux2,v4l2: Some buffers are still owned by the caller on close.
[ffmpeg] ioctl(VIDIOC_QBUF): Bad file descriptor

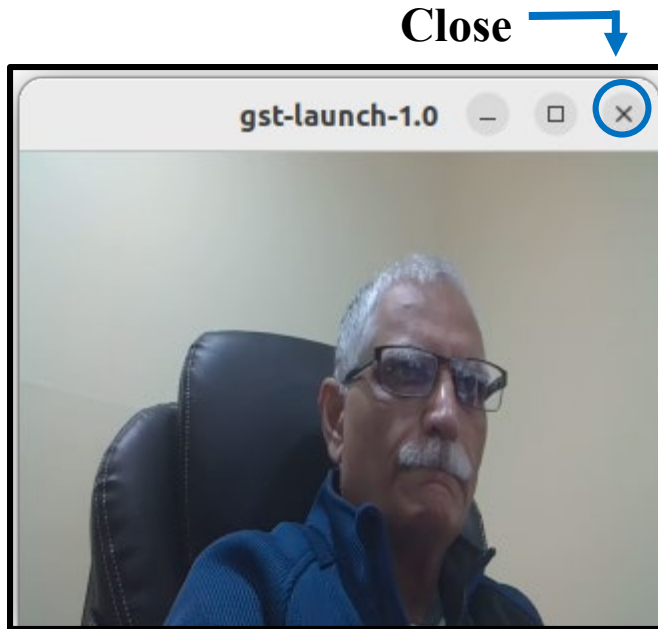
Exiting... (Quit)
Click on mpv-usb-camera.desktop...
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script8$
```

CSI

USBCam1

USBCam2

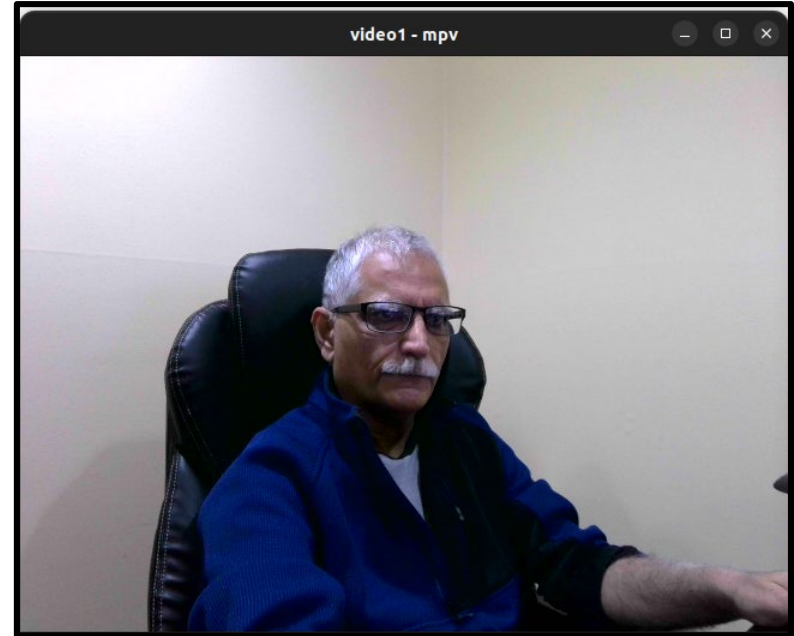
Terminal O/P: josSetup8-camApps.sh



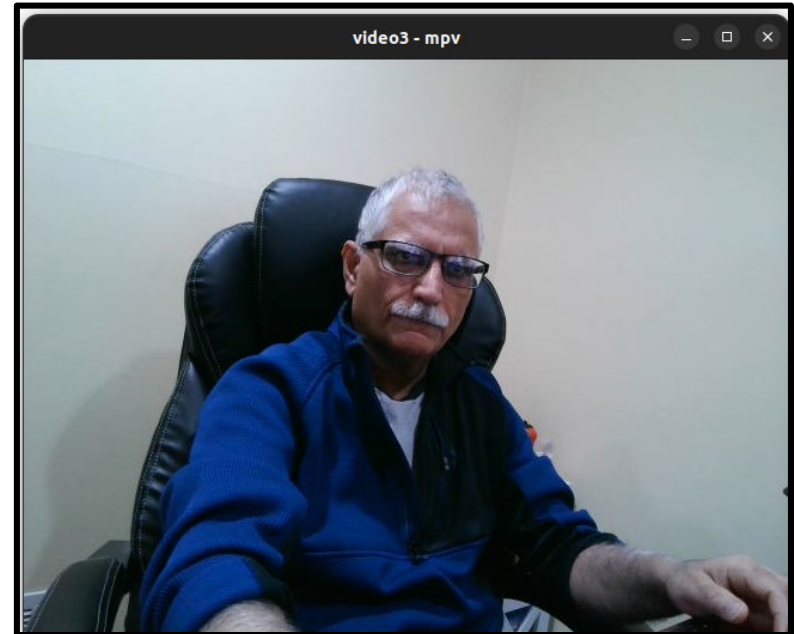
Win1: CSI Camera

Scaled 60%

Win2: Video1 (USB)

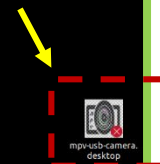
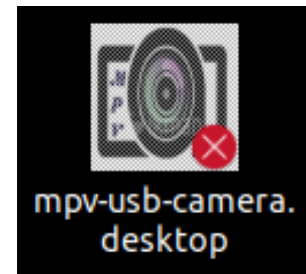


Win3: Video2 (USB)

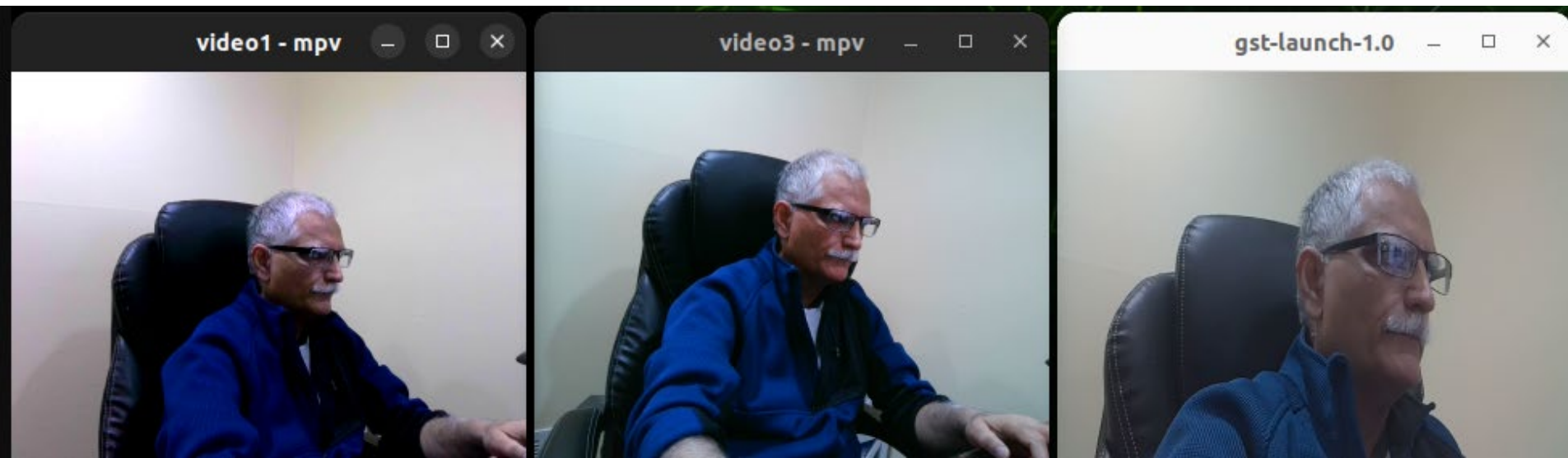
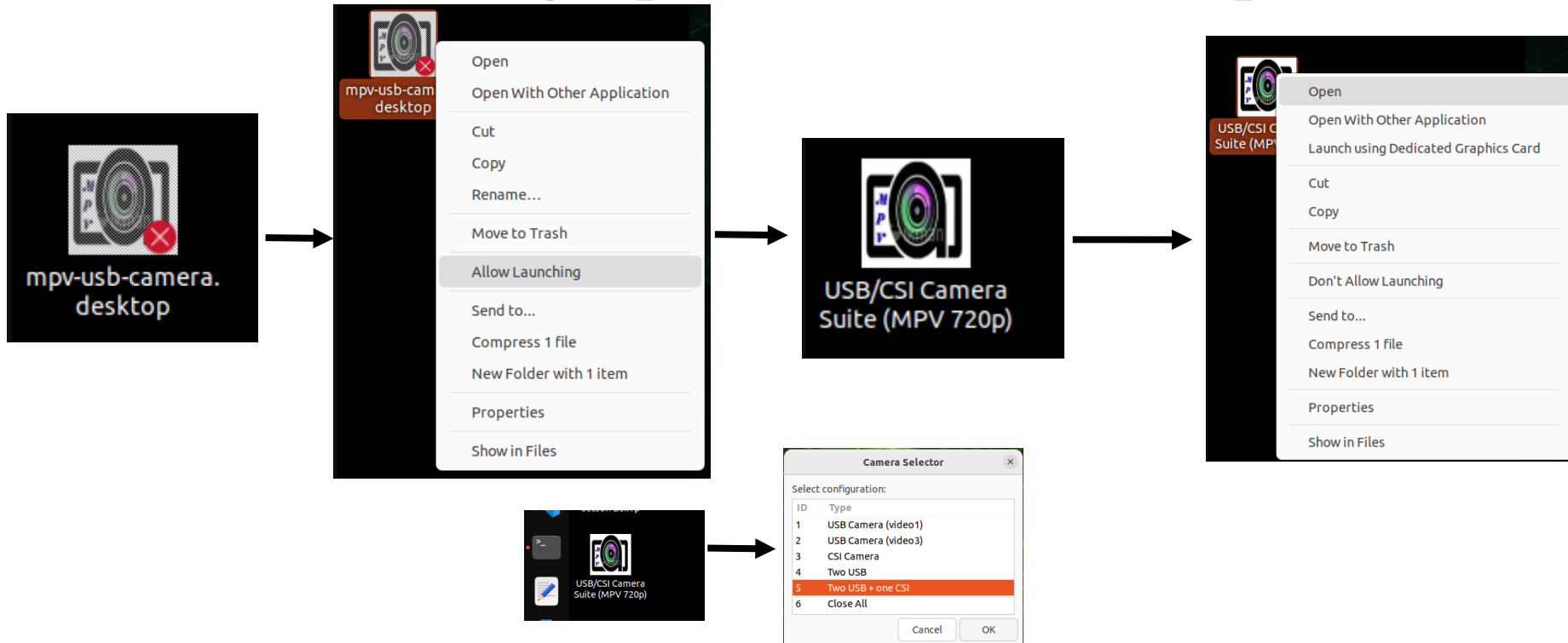


Note: Win2/3: Select win and enter q to close

Desktop Before/After Launching josSetup8-camApps.sh



Launching mpv-usb-camera.desktop



Script: jonSetup9-condaJl.sh

- Download a folder of useful files and install conda, opencv, jetson inference, ...

```
#!/bin/bash
```

```
# --- 1: Download a folder with various edited files ---
```

```
echo "[1/10] Setting up some path aliases..."
```

```
HOME_DIR="/home/vk-jn-or"
```

```
cd "$HOME_DIR"
```

```
DWNLD_PARENT="$HOME_DIR/Downloads/Jetson-Orin-Nano"
```

Alias for a download folder

```
# --- 2: CONDA ENVIRONMENT ---
```

```
# Create env and initialize conda for the current shell
```

```
echo "[2/10] Creating a conda env for jetson inference..."
```

```
~/miniforge3/bin/conda create -n vk2_env python=3.10.12 -y
```

```
source ~/miniforge3/etc/profile.d/conda.sh
```

A new conda env vk2_env

```
# --- 3: OPENCV & NUMPY ---
```

```
echo "[3/10] Installing opencv from a wheel..."
```

```
conda activate vk2_env
```

```
export PIP_INDEX_URL=https://pypi.jetson-ai-lab.io/jp6/cu128
```

```
pip install opencv-python
```

```
sudo apt install -y libtesseract-dev libatlas-base-dev
```

Opencv with CUDA and gstreamer support

```
# Downgrade Numpy to version < 2
```

```
echo "Downgrading numpy..."
```

```
pip uninstall -y numpy
```

```
pip install "numpy<2"
```

Downgrade numpy to meet jetson_inference requirements

```
# --- 4: JETSON-INFERENCE CLONE ---
```

```
echo "[4/10] Cloning jetson-inference..."
```

```
cd "$HOME_DIR"
```

```
git clone --recursive --depth=1 https://github.com/dusty-nv/jetson-inference
```

```
JET_INF_DIR="$HOME_DIR/jetson-inference"
```

Clone jetson_inference

```
# --- 5: NUMPY SYMBOLIC LINK ---
```

```
echo "[5/10] Creating a symbolic link for numpy..."
```

```
# Finding correct Numpy files and creating symbolic link
```

```
sudo ln -sf /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-  
packages/numpy/core/lib/libnpymath.a /usr/lib/libnpymath.a
```

Symbolic link for numpy

```
# --- 6: CMakeLists.txt REPLACEMENTS ---
```

```
# Update specific lines for CMakeLists for Conda path
```

```
# Following two CMakeLists.txt files need to be replaced:
```

```
# jetson-inference/python/bindings/CMakeLists.txt --> Edited Line 128
```

```
# jetson-utils/python/bindings/CMakeLists.txt --> Line 76
```

Replace in jetson_inference two
CMakeLists.txt files and confirm

```
echo "[6/10] Updating Cmakelists.txt files in two folders..."
```

```
JI_PY_BND_CMAKE="$DWNLD_PARENT/AM_JON_2SHARE/Edited-Files/updated/python_bindings"
```

```
JI_UT_PY_BND_CMAKE="$DWNLD_PARENT/AM_JON_2SHARE/Edited-Files/updated/utils_python_bindings"
```

```
cp "$JI_PY_BND_CMAKE/CMakeLists.txt" "$HOME_DIR/jetson-inference/python/bindings/CMakeLists.txt"
```

```
cp "$JI_UT_PY_BND_CMAKE/CMakeLists.txt" "$HOME_DIR/jetson-inference/utils/python/bindings/CMakeLists.txt"
```

```
# Verification check for Python Bindings CMake replacement
```

```
if diff -q "$JI_PY_BND_CMAKE/CMakeLists.txt" "$HOME_DIR/jetson-inference/python/bindings/CMakeLists.txt" >  
/dev/null; then
```

```
    echo "SUCCESS: $HOME_DIR/jetson-inference/python/bindings/CMakeLists.txt is identical to the edited  
source."
```

```
else
```

```
    echo "ERROR: $HOME_DIR/jetson-inference/python/bindings/CMakeLists.txt replacement failed!"
```

```
fi
```

```
if diff -q "$JI_UT_PY_BND_CMAKE/CMakeLists.txt" "$HOME_DIR/jetson-  
inference/utils/python/bindings/CMakeLists.txt" > /dev/null; then
```

```
    echo "SUCCESS: $HOME_DIR/jetson-inference/utils/python/bindings/CMakeLists.txt is identical to the  
edited source."
```

```
else
```

```
    echo "ERROR: $HOME_DIR/jetson-inference/utils/python/bindings/CMakeLists.txt replacement failed!"
```

```
fi
```

```
# --- 7: gstCamera.cpp and tensorNet.cpp REPLACEMENT ---
# Following two files need to be replaced:
# /home/vk-jn-or/jetson-inference/utils/camera/gstCamera.cpp
# /home/vk-jn-or/jetson-inference/c/tensorNet.cpp
```

**Replace in jetson_inference gstCamera.cpp
and tensorNet.cpp and confirm**

```
echo "[7/10] Updating gstCamera.cpp and tensorNet.cpp files..."
JI_UT_CAM_GST="$DWNLD_PARENT/AM_JON_2SHARE/Edited-Files/updated/utils_camera/Ver5-FinalWorking/ConstCaptRes-
UsrOutRes-Rev2"
JI_C_TNSR="$DWNLD_PARENT/AM_JON_2SHARE/Edited-Files/updated/c"

cp "$JI_UT_CAM_GST/gstCamera.cpp" "$HOME_DIR/jetson-inference/utils/camera/gstCamera.cpp"
cp "$JI_C_TNSR/tensorNet.cpp" "$HOME_DIR/jetson-inference/c/tensorNet.cpp"

# Verification check for gstCamera.cpp replacement
if diff -q "$JI_UT_CAM_GST/gstCamera.cpp" "$HOME_DIR/jetson-inference/utils/camera/gstCamera.cpp" > /dev/null;
then
    echo "SUCCESS: $HOME_DIR/jetson-inference/utils/camera/gstCamera.cpp is identical to the edited source."
else
    echo "ERROR: $HOME_DIR/jetson-inference/utils/camera/gstCamera.cpp replacement failed!"
fi

# Verification check for tensorNet.cpp replacement
if diff -q "$JI_C_TNSR/tensorNet.cpp" "$HOME_DIR/jetson-inference/c/tensorNet.cpp" > /dev/null; then
    echo "SUCCESS: $HOME_DIR/jetson-inference/c/tensorNet.cpp is identical to the edited source."
else
    echo "ERROR: $HOME_DIR/jetson-inference/c/tensorNet.cpp replacement failed!"
fi
```

```
# --- 8: BUILD & INSTALL JETSON INFERENCE ---
```

```
echo "[8/10] Building and installing jetson-inference..."
cd "$JET_INF_DIR"
mkdir -p build
cd build
cmake ../          # WAIT FOR USER INTERACTION: Select PyTorch 2.1 for Python 3.10
make -j$(nproc)
sudo make install
sudo ldconfig
```

Build and install jetson_inference

```
# --- 9: TENSORRT SYMBOLIC LINK ---
# Connect system-wide TensorRT to Conda
echo "[9/10] Connecting system-wide TensorRT to conda env..."
conda deactivate
ln -s /usr/lib/python3.10/dist-packages/tensorrt /home/vk-jn-
or/miniforge3/envs/vk2_env/lib/python3.10/site-packages
```

**Create symbolic link for
TENSORRT**

```
# --- 10: BASHRC & HELPER SCRIPTS ---
# Add LD_PRELOAD and Aliases
echo "[10/10] Adding LD-PRELOAD for a library and creating some aliases..."
if ! grep -q "libffi.so.7" ~/.bashrc; then
    echo "export LD_PRELOAD=/usr/lib/aarch64-linux-gnu/libffi.so.7" >> ~/.bashrc
fi

cat << 'EOF' > ~/condaActivate2.sh
#!/bin/bash
. "$HOME/miniforge3/etc/profile.d/conda.sh"
conda activate vk2_env
EOF

cat << 'EOF' > ~/condaDeactivate2.sh
#!/bin/bash
conda deactivate
source ~/.bashrc
EOF

chmod +x ~/condaActivate2.sh ~/condaDeactivate2.sh

if ! grep -q "alias condaon2" ~/.bashrc; then
    echo "alias condaon2='source ~/condaActivate2.sh'" >> ~/.bashrc
    echo "alias condaoff2='source ~/condaDeactivate2.sh'" >> ~/.bashrc
fi
```

**Scripts and aliases for activating and
deactivating newly created conda env**

```
echo "Setup Complete. Run 'source ~/.bashrc' and then 'condaon2' to begin."
```

Terminal O/P: jonSetup9-condaJI.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9
vk-jn-or@vkjnor-desktop:~$ cd /home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9/
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9$ ./jonSetup9-condaJI.sh
[1/10] Setting up some path aliases...
[2/10] Creating a conda env for jetson inference...
Channels:
- conda-forge
Platform: linux-aarch64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

environment location: /home/vk-jn-or/miniforge3/envs/vk2_env

added / updated specs:
- python=3.10.12

The following NEW packages will be INSTALLED:

 _openmp_mutex      conda-forge/linux-aarch64::_openmp_mutex-4.5-2_gnu
 bzip2              conda-forge/linux-aarch64::bzip2-1.0.8-h4777abc_8
 ca-certificates    conda-forge/noarch::ca-certificates-2026.1.4-hbd8a1cb_0
 icu                conda-forge/linux-aarch64::icu-78.2-hb1525cb_0
 ld_impl_linux-aar~ conda-forge/linux-aarch64::ld_impl_linux-aarch64-2.45.1-default_h1979696_101
 libffi             conda-forge/linux-aarch64::libffi-3.5.2-h376a255_0
 libgcc             conda-forge/linux-aarch64::libgcc-15.2.0-h8acb6b2_17
 libgcc-ng          conda-forge/linux-aarch64::libgcc-ng-15.2.0-he9431aa_17
 libgomp            conda-forge/linux-aarch64::libgomp-15.2.0-h8acb6b2_17
 liblzma            conda-forge/linux-aarch64::liblzma-5.8.2-he30d5cf_0
 liblzma-devel      conda-forge/linux-aarch64::liblzma-devel-5.8.2-he30d5cf_0
 libnsl             conda-forge/linux-aarch64::libnsl-2.0.1-h86ecc28_1
 libsqlite          conda-forge/linux-aarch64::libsqlite-3.51.2-h10b116e_0
 libstdcxx          conda-forge/linux-aarch64::libstdcxx-15.2.0-hef695bb_17
 libuuid            conda-forge/linux-aarch64::libuuid-2.41.3-h1022ec0_0
 libzlib            conda-forge/linux-aarch64::libzlib-1.3.1-h86ecc28_2
 ncurses            conda-forge/linux-aarch64::ncurses-6.5-ha32ae93_3
 openssl            conda-forge/linux-aarch64::openssl-3.6.1-h546c87b_1
 packaging          conda-forge/noarch::packaging-26.0-pyhcf101f3_0
 pip               conda-forge/noarch::pip-26.0.1-pyh8b19718_0
 python            conda-forge/linux-aarch64::python-3.10.12-hbbe8eec_0_cpython
 readline          conda-forge/linux-aarch64::readline-8.3-hb682ff5_0
 setuptools        conda-forge/noarch::setuptools-82.0.0-pyh332efcf_0
 tk                conda-forge/linux-aarch64::tk-8.6.13-noxft_h0dc03b3_103
 tzdata            conda-forge/noarch::tzdata-2025c-hc9c84f9_1
 wheel             conda-forge/noarch::wheel-0.46.3-pyhd8ed1ab_0
 xz                conda-forge/linux-aarch64::xz-5.8.2-hd704e39_0
 xz-gpl-tools       conda-forge/linux-aarch64::xz-gpl-tools-5.8.2-hd704e39_0
 xz-tools          conda-forge/linux-aarch64::xz-tools-5.8.2-he30d5cf_0
 zstd              conda-forge/linux-aarch64::zstd-1.5.7-h85ac4a6_6
```

Conda env

Terminal O/P: jonSetup9-condaJl.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9
Successfully installed numpy-2.2.6 opencv-contrib-python-4.11.0.86 opencv-python-4.11.0
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
libatlas-base-dev is already the newest version (3.10.3-12ubuntu1).
libtesseract-dev is already the newest version (4.1.1-2.1build1).
The following packages were automatically installed and are no longer required:
  libavcodec58 libavdevice58 libavfilter7 libavformat58 libavutil56 libpostproc55 libssh-gcrypt-4 libswresample3 libswscale5
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 308 not upgraded.
Downgrading numpy...
Found existing installation: numpy 2.2.6
Uninstalling numpy-2.2.6:
  Successfully uninstalled numpy-2.2.6
Looking in indexes: https://pypi.jetson-ai-lab.io/jp6/cu128
Collecting numpy<2
  Downloading https://pypi.jetson-ai-lab.io/root/pypi/%2Bf/d20/9d8969599b27a/numpy-1.26.4-cp310-cp310-manylinux_2_17_aarch64.manylinux2
  014_aarch64.whl (14.2 MB)
  14.2/14.2 MB 3.9 MB/s 0:00:03
Installing collected packages: numpy
Successfully installed numpy-1.26.4
[4/10] Cloning jetson-inference...
Cloning into 'jetson-inference'...
remote: Enumerating objects: 2154, done.
remote: Counting objects: 100% (2154/2154), done.
remote: Compressing objects: 100% (1668/1668), done.
remote: Total 2154 (delta 943), reused 1256 (delta 478), pack-reused 0 (from 0)
Receiving objects: 100% (2154/2154), 101.26 MiB | 21.27 MiB/s, done.
Resolving deltas: 100% (943/943), done.
Updating files: 100% (2223/2223), done.
Submodule 'plugins/pose' (https://github.com/dusty-nv/trt_pose) registered for path 'c/plugins/pose'
Submodule 'docker/containers' (https://github.com/dusty-nv/jetson-containers) registered for path 'docker/containers'
Submodule 'python/training/classification' (https://github.com/dusty-nv/pytorch-classification) registered for path 'python/training/cl
assification'
Submodule 'python/training/detection' (https://github.com/dusty-nv/pytorch-detection) registered for path 'python/training/detection'
Submodule 'python/training/segmentation' (https://github.com/dusty-nv/pytorch-segmentation) registered for path 'python/training/segmen
tation'
Submodule 'ros' (https://github.com/dusty-nv/ros_deep_learning) registered for path 'ros'
Submodule 'tools/camera-capture' (https://github.com/dusty-nv/camera-capture) registered for path 'tools/camera-capture'
Submodule 'utils' (https://github.com/dusty-nv/jetson-utils) registered for path 'utils'
Cloning into '/home/vk-jn-or/jetson-inference/c/plugins/pose'...
remote: Enumerating objects: 1478, done.
remote: Total 1478 (delta 0), reused 0 (delta 0), pack-reused 1478 (from 1)
Receiving objects: 100% (1478/1478), 2.01 MiB | 6.56 MiB/s, done.
Resolving deltas: 100% (821/821), done.
Cloning into '/home/vk-jn-or/jetson-inference/docker/containers'...
remote: Enumerating objects: 42211, done.
remote: Counting objects: 100% (540/540), done.
remote: Compressing objects: 100% (197/197), done.
remote: Total 42211 (delta 398), reused 343 (delta 343), pack-reused 41671 (from 4)
Receiving objects: 100% (42211/42211), 177.24 MiB | 13.85 MiB/s, done.
Resolving deltas: 100% (28878/28878), done.
```

Downgrading numpy

Cloning jetson_inference

Terminal O/P: jonSetup9-condaJl.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9
Hello AI World (jetson-inference)

PyTorch Installer (L4T R36.4.4)
If you want to train DNN models on your Jetson, this tool will download and
install PyTorch. Select the desired versions of pre-built packages below,
or see http://eLinux.org/Jetson\_Zoo for instructions to build from source.

You can skip this step and select Skip if you don't want to install PyTorch.

Keys:
↑↓ Navigate Menu
Space to Select
Enter to Continue

Packages to Install:
[ ] 1 PyTorch 2.1 for Python 3.10

< OK >      < Skip >
```

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9
Hello AI World (jetson-inference)

PyTorch Installer (L4T R36.4.4)
If you want to train DNN models on your Jetson, this tool will download and
install PyTorch. Select the desired versions of pre-built packages below,
or see http://eLinux.org/Jetson\_Zoo for instructions to build from source.

You can skip this step and select Skip if you don't want to install PyTorch.

Keys:
↑↓ Navigate Menu
Space to Select
Enter to Continue

Packages to Install:
[*] 1 PyTorch 2.1 for Python 3.10

< OK >      < Skip >
```

Select PyTorch Installation

Terminal O/P: jonSetup9-condaJI.sh

```
vk-jn-or@vkJnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9
+- Installing: /usr/local/bin/cuda-from-numpy.py
+- Installing: /usr/local/bin/cuda-from-pytorch.py
+- Installing: /usr/local/bin/cuda-streams.py
+- Installing: /usr/local/bin/cuda-to-cv.py
+- Installing: /usr/local/bin/cuda-to-numpy.py
+- Installing: /usr/local/bin/cuda-to-pytorch.py
+- Installing: /usr/local/bin/test-cuda.sh
+- Installing: /usr/local/bin/test-display.py
+- Installing: /usr/local/bin/test-logging.py
+- Installing: /usr/local/bin/test-video.py
+- Installing: /usr/local/bin/video-viewer.py
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson_utils_python.so
+- Set runtime path of "/home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson_utils_python.so" to ""
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/Jetson
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/Jetson/Utils
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/Jetson/Utils/__init__.py
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/Jetson/__init__.py
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson/Utils
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson/Utils/__init__.py
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson/__init__.py
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson_utils
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson_utils/__init__.py
+- Installing: /usr/local/bin/video-viewer
+- Set runtime path of "/usr/local/bin/video-viewer" to ""
+- Installing: /usr/local/bin/actionnet.py
+- Installing: /usr/local/bin/backgroundnet.py
+- Installing: /usr/local/bin/depthnet.py
+- Installing: /usr/local/bin/depthnet_utils.py
+- Installing: /usr/local/bin/detectnet-snap.py
+- Installing: /usr/local/bin/detectnet.py
+- Installing: /usr/local/bin/imagenet.py
+- Installing: /usr/local/bin/my-detection.py
+- Installing: /usr/local/bin/my-recognition.py
+- Installing: /usr/local/bin/posenet.py
+- Installing: /usr/local/bin/segnet.py
+- Installing: /usr/local/bin/segnet_utils.py
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson_inference_python.so
+- Set runtime path of "/home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson_inference_python.so" to ""
+- Up-to-date: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/Jetson
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/Jetson/Inference
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/Jetson/Inference/__init__.py
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/Jetson/__init__.py
+- Up-to-date: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson/inference
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson/inference/__init__.py
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson/__init__.py
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson_inference
+- Installing: /home/vk-jn-or/miniforge3/envs/vk2_env/lib/python3.10/site-packages/jetson_inference/__init__.py
[9/10] Connecting system-wide TensorRT to conda env...
[10/10] Adding LD-PRELOAD for a library and creating some aliases...
Setup Complete. Run 'source ~/.bashrc' and then 'condaon2' to begin.
vk-jn-or@vkJnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9$ source ~/.bashrc
vk-jn-or@vkJnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9$
```

jetson_inference build and install process

Terminal O/P: jonSetup9-condaJl.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9$ cat ~/.bashrc
# ~/.bashrc: executed by bash(1) for non-login shells.
# see /usr/share/doc/bash/examples/startup-files (in the package bash-doc)
# for examples

# If not running interactively, don't do anything
case $- in
  *i*) ;;
  *) return;;
esac

# don't put duplicate lines or lines starting with space in the history.
# See bash(1) for more options
HISTCONTROL=ignoreboth
```

⋮

```
# >>> conda initialize >>>
# !! Contents within this block are managed by 'conda init' !!
__conda_setup="$(('/home/vk-jn-or/miniforge3/bin/conda' 'shell.bash' 'hook' 2> /dev/null)"
if [ $? -eq 0 ]; then
    eval "$__conda_setup"
else
    if [ -f "/home/vk-jn-or/miniforge3/etc/profile.d/conda.sh" ]; then
        . "/home/vk-jn-or/miniforge3/etc/profile.d/conda.sh"
    else
        export PATH="/home/vk-jn-or/miniforge3/bin:$PATH"
    fi
fi
unset __conda_setup
# <<< conda initialize <<<

# Conda aliases
alias condaon1='source ~/condaActivate.sh'
alias condaoff1='source ~/condaDeactivate.sh'
export PATH=/usr/local/cuda/bin:$PATH
export LD_LIBRARY_PATH=/usr/local/cuda/lib64:$LD_LIBRARY_PATH

# Fix for libffi
export LD_PRELOAD=/usr/lib/aarch64-linux-gnu/libffi.so.8
export LD_PRELOAD=/usr/lib/aarch64-linux-gnu/libffi.so.7
alias condaon2='source ~/condaActivate2.sh'
alias condaoff2='source ~/condaDeactivate2.sh'
```

New additions to ~/.bashrc

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9$
```

Home: Before/After Install of jonSetup9-condaJl.sh

Name	Size	Modified
Desktop	4 items	19:29
Documents	0 items	18:37
Downloads	1 item	19:10
miniforge3	0 items	19:30
Music	0 items	18:37
Pictures	0 items	18:46
Public	0 items	18:37
Templates	0 items	18:37
Videos	0 items	18:37
condaActivate.sh	79 bytes	19:30
condaDeactivate.sh	46 bytes	19:30

Name	Size	Modified
Applications	2 items	Yesterday
Desktop	8 items	10:01
Documents	0 items	Yesterday
Downloads	1 item	Yesterday
jetson-inference	16 items	09:59
miniforge3	18 items	Yesterday
Music	0 items	Yesterday
Pictures	1 item	09:03
Public	0 items	Yesterday
snap	1 item	Yesterday
Templates	0 items	Yesterday
Videos	0 items	Yesterday
condaActivate.sh	79 bytes	Yesterday
condaDeactivate.sh	46 bytes	Yesterday
launch_cameras.sh	2.4 kB	09:03

Name	Size	Modified
Applications	2 items	Yesterday
Desktop	9 items	10:01
Documents	0 items	Yesterday
Downloads	1 item	Yesterday
jetson-inference	16 items	09:59
miniforge3	18 items	Yesterday
Music	0 items	Yesterday
Pictures	1 item	09:03
Public	0 items	Yesterday
snap	1 item	Yesterday
Templates	0 items	Yesterday
Videos	0 items	Yesterday
condaActivate.sh	79 bytes	Yesterday
condaActivate2.sh	79 bytes	10:08
condaDeactivate.sh	46 bytes	Yesterday
condaDeactivate2.sh	46 bytes	10:08
launch_cameras.sh	2.4 kB	09:03

Shell Script: jonSetup9-status.sh

- Check that replacement of four files done correctly

```
#!/bin/bash

echo "Setting up some path aliases..."
HOME_DIR="/home/vk-jn-or"
cd "$HOME_DIR"
DWNLD_PARENT="$HOME_DIR/Downloads/Jetson-Orin-Nano"

JI_PY_BND_CMAKE="$DWNLD_PARENT/AM_JON_2SHARE/Edited-Files/updated/python_bindings"
JI_UT_PY_BND_CMAKE="$DWNLD_PARENT/AM_JON_2SHARE/Edited-Files/updated/utils_python_bindings"

JI_UT_CAM_GST="$DWNLD_PARENT/AM_JON_2SHARE/Edited-Files/updated/utils_camera/Ver5-FinalWorking/ConstCaptRes-UsrOutRes-Rev2"
JI_C_TNSR="$DWNLD_PARENT/AM_JON_2SHARE/Edited-Files/updated/c"

# ***** Check that the files that needed to be replaced are correctly replaced *****

# Verification check for Python Bindings CMake replacement
if diff -q "$JI_PY_BND_CMAKE/CMakeLists.txt" "$HOME_DIR/jetson-inference/python/bindings/CMakeLists.txt" > /dev/null; then
    echo "SUCCESS: $HOME_DIR/jetson-inference/python/bindings/CMakeLists.txt is identical to the edited source."
else
    echo "ERROR: $HOME_DIR/jetson-inference/python/bindings/CMakeLists.txt replacement failed!"
fi

if diff -q "$JI_UT_PY_BND_CMAKE/CMakeLists.txt" "$HOME_DIR/jetson-inference/utils/python/bindings/CMakeLists.txt" > /dev/null; then
    echo "SUCCESS: $HOME_DIR/jetson-inference/utils/python/bindings/CMakeLists.txt is identical to the edited source."
else
    echo "ERROR: $HOME_DIR/jetson-inference/utils/python/bindings/CMakeLists.txt replacement failed!"
fi

# Verification check for gstCamera.cpp replacement
if diff -q "$JI_UT_CAM_GST/gstCamera.cpp" "$HOME_DIR/jetson-inference/utils/camera/gstCamera.cpp" > /dev/null; then
    echo "SUCCESS: $HOME_DIR/jetson-inference/utils/camera/gstCamera.cpp is identical to the edited source."
else
    echo "ERROR: $HOME_DIR/jetson-inference/utils/camera/gstCamera.cpp replacement failed!"
fi

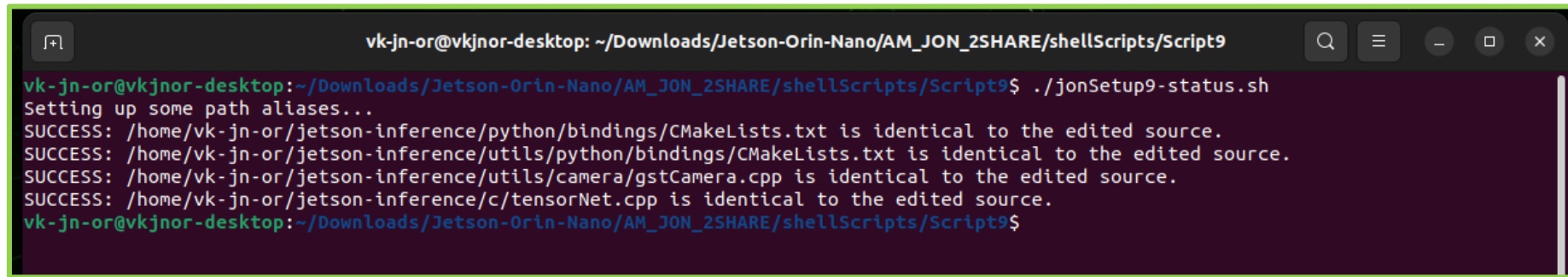
# Verification check for tensorNet.cpp replacement
if diff -q "$JI_C_TNSR/tensorNet.cpp" "$HOME_DIR/jetson-inference/c/tensorNet.cpp" > /dev/null; then
    echo "SUCCESS: $HOME_DIR/jetson-inference/c/tensorNet.cpp is identical to the edited source."
else
    echo "ERROR: $HOME_DIR/jetson-inference/c/tensorNet.cpp replacement failed!"
fi
```

Alias for source folders

CMakeLists.txt replacement checks

**Checks for replacements of
gstCamera.cpp and tensorNet.cpp**

Terminal O/P: jonSetup9-status.sh

A terminal window with a dark purple background and green text. The window title is 'vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9'. The terminal shows the execution of './jonSetup9-status.sh'. The output includes 'Setting up some path aliases...' followed by four 'SUCCESS' messages confirming that specific files are identical to the edited source. The prompt returns to the shell.

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9$ ./jonSetup9-status.sh
Setting up some path aliases...
SUCCESS: /home/vk-jn-or/jetson-inference/python/bindings/CMakeLists.txt is identical to the edited source.
SUCCESS: /home/vk-jn-or/jetson-inference/Utils/python/bindings/CMakeLists.txt is identical to the edited source.
SUCCESS: /home/vk-jn-or/jetson-inference/Utils/camera/gstCamera.cpp is identical to the edited source.
SUCCESS: /home/vk-jn-or/jetson-inference/c/tensorNet.cpp is identical to the edited source.
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script9$
```

- **Note:** The checks for these replacements are also done in jonSetup9-condaJl.sh but are easily missed with a flood of terminal messages

Shell Script: jonSetup10-conJI-Apps.sh

- Some applications for the conda env where jetson_inference is installed

```
#!/bin/bash
```

```
# --- INSTALL GSTREAMER/GI/XDOTOOL ---
```

```
# --- 1: ENVIRONMENT SETUP ---
```

```
echo "[1/5] Initial setup..."
```

```
HOME_DIR="/home/vk-jn-or"
```

```
CONDA_ENV_PATH="$HOME_DIR/miniforge3/envs/vk2_env"
```

```
echo "Starting installation of additional tools and GStreamer silencer..."
```

```
# Ensure we are working with the correct environment
```

```
source ~/miniforge3/etc/profile.d/conda.sh
```

```
conda activate vk2_env
```

```
# --- 2: GSTREAMER & GI INSTALLATION ---
```

```
echo "[2/5] Installing GStreamer GObject introspection and python-gi..."
```

```
sudo apt-get update
```

```
sudo apt-get install -y python3-gi gir1.2-gstreamer-1.0
```

```
# Link system 'gi' to the Conda environment
```

```
echo "Linking system-wide 'gi' to conda environment..."
```

```
# Path identified from manual notes: /usr/lib/python3/dist-packages/gi
```

```
ln -sf /usr/lib/python3/dist-packages/gi "$CONDA_ENV_PATH/lib/python3.10/site-packages"
```

```
# Verification of GStreamer inside Conda
```

```
echo "Verifying GStreamer connection in vk2_env..."
```

```
python3 -c "import gi; gi.require_version('Gst', '1.0'); from gi.repository import Gst; Gst.init(None);  
print('GStreamer connected successfully:', Gst.version_string())"
```

Conda env setup

gi and gstreamer pkgs
that will be used to
silence gstreamer
messages when running
python programs with
CSI camera

```

# --- 3: CONDA ENV VARIABLES (ACTIVATE/DEACTIVATE) ---
echo "[3/5] Configuring GStreamer environment variables for Conda..."

echo "Searching for GStreamer components..."

# DYNAMIC SEARCH: Find the Plugin Path (The folder containing hardware-accelerated plugins)
# We look for the folder containing 'libgstnvguscamerasrc.so' (common on Jetson)
GST_DIR=$(find /usr/lib/aarch64-linux-gnu/ -name "libgstnvguscamerasrc.so" -printf '%h\n' | head -n 1)
# Fallback to standard path if find fails
if [ -z "$GST_DIR" ]; then
    GST_DIR="/usr/lib/aarch64-linux-gnu/gstreamer-1.0"
fi

echo "Found Plugins at: $GST_DIR"

ACTIVATE_DIR="$CONDA_ENV_PATH/etc/conda/activate.d"
DEACTIVATE_DIR="$CONDA_ENV_PATH/etc/conda/deactivate.d"
mkdir -p "$ACTIVATE_DIR" "$DEACTIVATE_DIR"

# Create the Activation script
cat << EOF > "$ACTIVATE_DIR/env_vars.sh"
#!/bin/sh
# Points to where the Jetson hardware plugins live
export GST_PLUGIN_PATH=$GST_DIR
EOF

# Create the Deactivation script to keep the system clean
cat << EOF > "$DEACTIVATE_DIR/env_vars.sh"
#!/bin/sh
unset GST_PLUGIN_PATH
EOF

chmod +x "$ACTIVATE_DIR/env_vars.sh" "$DEACTIVATE_DIR/env_vars.sh"

```

gstreamer env setup for conda

```
# --- 4: WINDOW MANAGEMENT TOOLS ---  
echo "[4/5] Installing xdotool for window resizing and placement..."  
sudo apt-get install -y xdotool  
xdotool --version
```

**xdotool to manage camera image
windows created with jetson_utils**

```
# --- 5: FINAL VERIFICATION ---  
# Reactivate to apply the new env_vars.sh variables  
echo "[5/5] Reactivating environment to apply new GStreamer paths..."  
conda activate vk2_env
```

Applying env variable

```
echo "-----"  
echo "Setup Complete."  
echo "Testing GStreamer Inspection..."  
gst-inspect-1.0 --version
```

Testing setup

```
echo "Testing GStreamer Pipeline (videotestsrc)..."  
# This tests if the plugins are found and the display sink is working  
gst-launch-1.0 videotestsrc num-buffers=50 ! autovideosink  
  
echo "-----"  
echo "If you saw a test pattern window, GStreamer is correctly configured."
```

Terminal O/P: jonSetup10-conJI-Apps.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script10
vk-jn-or@vkjnor-desktop:~$ cd /home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script10/
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script10$ ./jonSetup10-conJI-Apps.sh
[1/5] Initial setup...
Starting installation of additional tools and GStreamer silencer...
[2/5] Installing GStreamer GObject introspection and python-gi...
Get:1 file:/var/cudnn-local-tegra-repo-ubuntu2204-9.3.0 InRelease [1,572 B]
Get:2 file:/var/l4t-cuda-tegra-repo-ubuntu2204-12-6-local InRelease [1,572 B]
Get:1 file:/var/cudnn-local-tegra-repo-ubuntu2204-9.3.0 InRelease [1,572 B]
Get:3 file:/var/nv-tensorrt-local-tegra-repo-ubuntu2204-10.3.0-cuda-12.5 InRelease [1,572 B]
Get:2 file:/var/l4t-cuda-tegra-repo-ubuntu2204-12-6-local InRelease [1,572 B]
Get:3 file:/var/nv-tensorrt-local-tegra-repo-ubuntu2204-10.3.0-cuda-12.5 InRelease [1,572 B]
Hit:4 http://ports.ubuntu.com/ubuntu-ports jammy InRelease
Hit:5 https://download.docker.com/linux/ubuntu jammy InRelease
Hit:6 http://ports.ubuntu.com/ubuntu-ports jammy-updates InRelease
Hit:7 http://ports.ubuntu.com/ubuntu-ports jammy-backports InRelease
Hit:8 https://repo.download.nvidia.com/jetson/common r36.4 InRelease
Hit:9 http://ports.ubuntu.com/ubuntu-ports jammy-security InRelease
Get:10 https://packages.microsoft.com/repos/code stable InRelease [3,590 B]
Hit:11 https://repo.download.nvidia.com/jetson/t234 r36.4 InRelease
Hit:12 https://repo.download.nvidia.com/jetson/ffmpeg r36.4 InRelease
Get:13 https://packages.microsoft.com/repos/code stable/main arm64 Packages [22.2 kB]
Fetched 25.8 kB in 2s (15.5 kB/s)
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gir1.2-gstreamer-1.0 is already the newest version (1.20.3-0ubuntu1.1).
gir1.2-gstreamer-1.0 set to manually installed.
python3-gi is already the newest version (3.42.1-0ubuntu1).
python3-gi set to manually installed.
The following packages were automatically installed and are no longer required:
  libavcodec58 libavdevice58 libavfilter7 libavformat58 libavutil56 libpostproc55 libssh-gcrypt-4 libswresample3 libswscale5
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 284 not upgraded.
Linking system-wide 'gi' to conda environment...
Verifying GStreamer connection in vk2_env...
GStreamer connected successfully: GStreamer 1.20.3
[3/5] Configuring GStreamer environment variables for Conda...
Searching for GStreamer components...
Found Plugins at: /usr/lib/aarch64-linux-gnu/gstreamer-1.0
[4/5] Installing xdotool for window resizing and placement...
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libavcodec58 libavdevice58 libavfilter7 libavformat58 libavutil56 libpostproc55 libssh-gcrypt-4 libswresample3 libswscale5
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libxdo3
The following NEW packages will be installed:
  libxdo3 xdotool
0 upgraded, 2 newly installed, 0 to remove and 284 not upgraded.
Need to get 57.0 kB of archives.
```

gi and gstreamer install;
gstreamer env setup

xdotool

Terminal O/P: jonSetup10-conJI-Apps.sh

```
Feb 19 19:29
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script10

libavcodec58 libavdevice58 libavfilter7 libavformat58 libavutil56 libpostproc55 libssh-gcrypt-4 libswresample3 libswscale5
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 292 not upgraded.
Linking system-wide 'gi' to conda environment...
Verifying GStreamer connection in vk2_env...
GStreamer connected successfully: GStreamer 1.20.3
[3/5] Configuring GStreamer environment variables for Conda...
Searching for GStreamer components...
Found Plugins at: /usr/lib/aarch64-linux-gnu/gstreamer-1.0
[4/5] Installing xdotool for window resizing and placement...
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libavcodec58 libavdevice58 libavfilter7 libavformat58 libavutil56 libpostproc55 libssh-gcrypt-4 libswresample3 libswscale5
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libxd3
The following NEW packages will be installed:
  libxd3 xdotool
0 upgraded, 2 newly installed, 0 to remove and 292 not upgraded.
Need to get 57.0 kB of archives.
After this operation, 195 kB of additional disk space will be used.
Get:1 http://ports.ubuntu.com/ubuntu-ports jammy/universe arm64 libxd3 arm64 1:3.20160805.1-4 [18.5 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports jammy/universe arm64 xdotool arm64 1:3.20160805.1-4 [38.5 kB]
Fetched 57.0 kB in 1s (108 kB/s)
debconf: delaying package configuration, since apt-utils is not installed
Selecting previously unselected package libxd3:arm64.
(Reading database ... 219833 files and directories currently installed.)
Preparing to unpack .../libxd3_1%3a3.20160805.1-4_arm64.deb ...
Unpacking libxd3:arm64 (1:3.20160805.1-4) ...
Selecting previously unselected package xdotool.
Preparing to unpack .../xdotool_1%3a3.20160805.1-4_arm64.deb ...
Unpacking xdotool (1:3.20160805.1-4) ...
Setting up libxd3:arm64 (1:3.20160805.1-4) ...
Setting up xdotool (1:3.20160805.1-4) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.9) ...
xdotool version 3.20160805.1
[5/5] Reactivating environment to apply new GStreamer paths...
.....
Setup Complete
Testing GStreamer Inspection...
gst-inspect-1.0 version 1.20.3
GStreamer 1.20.3
https://launchpad.net/distros/ubuntu/+source/gstreamer1.0
Testing GStreamer Pipeline (videotestsrc)...
Setting pipeline to PAUSED ...
Pipeline is PREROLLING ...
Pipeline is PREROLLED ...
Setting pipeline to PLAYING ...
Redistribute latency...
New clock: GstSystemClock
0:00:00.9 / 0:00:01.6 (59.9 %)

xdotool
```

Testing setup

This last line is displayed while test pattern is shown
and then the terminal o/p changes as below

```
New Clock: GstSystemClock
Got EOS from element "pipeline0".
Execution ended after 0:00:01.667038146
Setting pipeline to NULL ...
Freeing pipeline ...
.....
If you saw a test pattern window, GStreamer is correctly configured.
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script10$
```

Shell Script: jonSetup11-dckr.sh

- jetson_inference docker installation

```
#!/bin/bash
```

Aliases for folders

```
# --- 1: Preliminary Setup ---
```

```
HOME_DIR="/home/vk-jn-or"
```

```
JT_INF_DIR="$HOME_DIR/jetson-inference"
```

```
RVSD_DCKR_RUN_DIR="$HOME_DIR/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/Edited-Files/updated/docker"
```

```
# Replace default run.sh to enable custom CONTAINER_NAME="jetson-inference-session"
```

```
echo "Replacing run.sh with edited version for custom container naming..."
```

```
cd "$JT_INF_DIR"
```

```
cp "$RVSD_DCKR_RUN_DIR/run.sh" "$JT_INF_DIR/docker/run.sh"
```

```
chmod +x "$JT_INF_DIR/docker/run.sh"
```

Replace default run.sh
to launch docker with a
custom name

```
# --- 2: Launch Initial Container ---
```

```
# Note: This will download the r36.3.0 image and launch the container.
```

```
# Once inside, you must 'exit' to continue with the host permission setup.
```

```
echo "Launching Docker container. Please 'exit' once it has fully loaded to continue host setup."
```

```
./docker/run.sh --container dustynv/jetson-inference:r36.3.0
```

Launch a named container

Shell Script: jonSetup11-dckr.sh

- jetson_inference docker installation

```
1  #!/bin/bash
2
3  # --- 1: Preliminary Setup ---
4  HOME_DIR="/home/vk-jn-or"
5  JT_INF_DIR="$HOME_DIR/jetson-inference"
6  RVSD_DCKR_RUN_DIR="$HOME_DIR/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/Edited-Files/updated/docker"
7
8  # Replace default run.sh to enable custom CONTAINER_NAME="jetson-inference-session"
9  echo "Replacing run.sh with edited version for custom container naming..."
10 cd "$JT_INF_DIR"
11 cp "$RVSD_DCKR_RUN_DIR/run.sh" "$JT_INF_DIR/docker/run.sh"
12 chmod +x "$JT_INF_DIR/docker/run.sh"
13
14 # --- 2: Launch Initial Container ---
15 # Note: This will download the r36.3.0 image and launch the container.
16 # Once inside, you must 'exit' to continue with the host permission setup.
17 echo "Launching Docker container. Please 'exit' once it has fully loaded to continue host setup."
18 ./docker/run.sh --container dustynv/jetson-inference:r36.3.0
```

Aliases for folders

Replace default run.sh
to launch docker with a
custom name

Launch a named container

Terminal O/P: jonSetup11-dckr.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script11
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script11$ ./jonSetup11-dckr.sh
Replacing run.sh with edited version for custom container naming...
Launching Docker container. Please "exit" once it has fully loaded to continue host setup.
ARCH: aarch64
reading L4T version from /etc/nv_tegra_release
L4T BSP Version: L4T R36.4.4
localuser:root being added to access control list
CONTAINER_IMAGE: dustynv/jetson-inference:r36.3.0
DATA_VOLUME: --volume /home/vk-jn-or/jetson-inference/data:/jetson-inference/data --volume /home/vk-jn-or/jetson-inference/python/training/classification/data:/jetson-inference/python/training/classification/data --volume /home/vk-jn-or/jetson-inference/python/training/classification/models:/jetson-inference/python/training/classification/models --volume /home/vk-jn-or/jetson-inference/python/training/detection/ssd/data:/jetson-inference/python/training/detection/ssd/data --volume /home/vk-jn-or/jetson-inference/python/www/recognizer/data:/jetson-inference/python/www/recognizer/data
V4L2_DEVICES: --device /dev/video0 --device /dev/video1 --device /dev/video2 --device /dev/video3 --device /dev/video4
DISPLAY_DEVICE: -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix
Unable to find image 'dustynv/jetson-inference:r36.3.0' locally
r36.3.0: Pulling from dustynv/jetson-inference
d5a2ad729c09: Pull complete
2e6f5bdc8512: Extracting [=====] 95.81MB/225MB
ffba19198a9f: Download complete
2fd295528a0f: Download complete
28f84eed4302: Downloading [=====] 761.4MB/3.479GB
1d2d10eda001: Downloading [=====] 386.6MB/1.645GB
3e68d1f1f0a2: Download complete
d8cd7d9239d7: Download complete
411d140a42a6: Download complete
32f0b58253b2: Download complete
606e0a30f16f: Download complete
27e43713db75: Download complete
da940ea7ef5f: Download complete
a73aa8e537a7: Downloading [=====] 229.4MB/315.1MB
a46869cf885f: Waiting
f7e8de23fbce: Waiting
9a61ff882eb4: Waiting
219683df4e53: Waiting
f099086a1c25: Waiting
3282552dbecd: Waiting
f8bc360cc7f0: Waiting
42ffabca1e75: Waiting
c6a6207910ca: Waiting
2834479d4476: Waiting
790677d5943f: Waiting
```

jetson_inference container being downloaded

Terminal O/P: jonSetup11-dckr.sh

Terminal shell prompt

root@vkjnor-desktop: /opt/jetson-inference

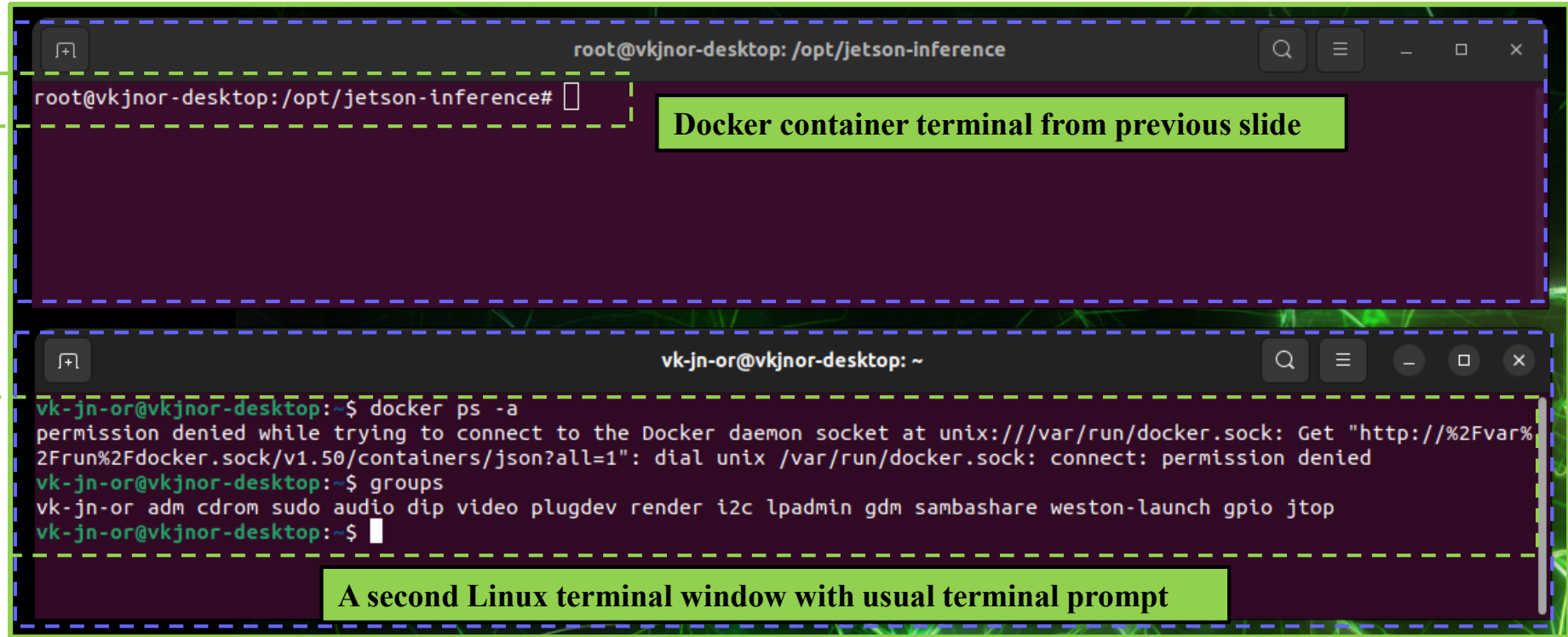
```
vk-jn-or@vkjnor-desktop:~$ cd /home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script11/
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script11$ ./jonSetup11-dckr.sh
Replacing run.sh with edited version for custom container naming...
Launching Docker container. Please 'exit' once it has fully loaded to continue host setup.
ARCH: aarch64
reading L4T version from /etc/nv_tegra_release
L4T BSP Version: L4T R36.4.4
localuser:root being added to access control list
CONTAINER_IMAGE: dustynv/jetson-inference:r36.3.0
DATA_VOLUME: --volume /home/vk-jn-or/jetson-inference/data:/jetson-inference/data --volume /home/vk-jn-or/jetson-inference/python/training/classif
ication/data:/jetson-inference/python/training/classification/data --volume /home/vk-jn-or/jetson-inference/python/training/classification/models:/jets
on-inference/python/training/classification/models --volume /home/vk-jn-or/jetson-inference/python/training/detection/ssd/data:/jetson-inference/python
/training/detection/ssd/data --volume /home/vk-jn-or/jetson-inference/python/training/detection/ssd/models:/jetson-inference/python/training/detection/
ssd/models --volume /home/vk-jn-or/jetson-inference/python/www/recognizer/data:/jetson-inference/python/www/recognizer/data
V4L2_DEVICES: --device /dev/video0 --device /dev/video1 --device /dev/video2 --device /dev/video3 --device /dev/video4
DISPLAY_DEVICE: -e DISPLAY=:0 -v /tmp/.X11-unix/:/tmp/.X11-unix
Unable to find image 'dustynv/jetson-inference:r36.3.0' locally
r36.3.0: Pulling from dustynv/jetson-inference
d5a2ad729c09: Pull complete
2e6f5bdc8512: Pull complete
ffba19198a9f: Pull complete
2fd295528a0f: Pull complete
28f84eed4302: Pull complete
1d2d10eda001: Pull complete
3e68d1f1f0a2: Pull complete
d8cd7d9239d7: Pull complete
411d140a42a6: Pull complete
32f0b58253b2: Pull complete
606e0a30f16f: Pull complete
27e43713db75: Pull complete
da940ea7ef5f: Pull complete
a73aa8e537a7: Pull complete
a46869cf885f: Pull complete
f7e8de23fbce: Pull complete
9a61ff882eb4: Pull complete
219683df4e53: Pull complete
f099086a1c25: Pull complete
3282552dbecd: Pull complete
f8bc360cc7f0: Pull complete
42ffabca1e75: Pull complete
c6a6207910ca: Pull complete
2834479d4476: Pull complete
790677d5943f: Pull complete
Digest: sha256:e00fd46a0465e5a8661e4ebf9580075a1fcaaa8e55a7cd2283f498f97037408f
Status: Downloaded newer image for dustynv/jetson-inference:r36.3.0
root@vkjnor-desktop: /opt/jetson-inference#
```

jetson_inference container pull completed and
the docker container has been launched

Docker container prompt

See the change in prompt → Now in
the docker container

Terminal O/P: jonSetup11-dckr.sh



- Having launched jetson_inference docker container, as in previous slide, leave it running as seen in the top panel above
- Launch a second Linux terminal as seen in the bottom panel above
- Now use the command `docker ps -a` to check if there are any running docker containers in the system
- See permission denied message and also see that the docker is not part of the current user “group”
- Can use `sudo docker ps -a` to find any running container; see jonSetup12-dckr.sh for a better approach

Shell Script: jonSetup12-dckr.sh

- In a Linux shell terminal (T1), launch jetson-inference docker container using:
`cd ~/jetson-inference && ./docker/run.sh --container dustynv/jetson-inference:r36.3.0`
- In another Linux shell terminal (T2), run `docker ps -a` followed by `groups`
- In T2, fix docker permission issue by running `./jonSetup12-dckr.sh`

```
#!/bin/bash
```

```
# --- 1: Fix Docker Permissions (Sudo-less Docker) ---  
echo "Configuring Docker group and user permissions..."  
sudo groupadd docker 2>/dev/null  
sudo usermod -aG docker $USER
```

Fix permission issue for docker
by adding docker to user group

```
# Apply the group changes to the current shell session to test immediately  
echo "Testing permissions in current session..."  
newgrp docker <<EONG  
echo "Current groups for this session:"  
groups  
docker ps -a  
EONG
```

```
echo "*****"  
echo "CLEANUP COMPLETE: Permissions have been applied to this terminal."  
echo "CRITICAL: You must REBOOT now to finalize group updates for all terminals."  
echo "After reboot, verify with 'groups' and then run any next scripts."  
echo "*****"
```

Terminal O/P: jonSetup12-dckr.sh

```
root@vkjnor-desktop: /opt/jetson-inference

vk-jn-or@vkjnor-desktop:~$ cd ~/jetson-inference && ./docker/run.sh --container dustynv/jetson-inference:r36.3.0
ARCH: aarch64
reading L4T version from /etc/nv_tegra_release
L4T BSP Version: L4T R36.4.4
localuser:root being added to access control list
CONTAINER_IMAGE: dustynv/jetson-inference:r36.3.0
DATA_VOLUME: --volume /home/vk-jn-or/jetson-inference/data:/jetson-inference/data --volume /home/vk-jn-or/jetson-inference/python/training/classification/data:/jetson-inference/python/training/classification/data --volume /home/vk-jn-or/jetson-inference/python/training/classification/models:/jetson-inference/python/training/classification/models --volume /home/vk-jn-or/jetson-inference/python/training/detection/ssd/data:/jetson-inference/python/training/detection/ssd/data --volume /home/vk-jn-or/jetson-inference/python/training/detection/ssd/models:/jetson-inference/python/training/detection/ssd/models --volume /home/vk-jn-or/jetson-inference/python/www/recognizer/data:/jetson-inference/python/www/recognizer/data
V4L2_DEVICES: --device /dev/video0 --device /dev/video1 --device /dev/video2 --device /dev/video3 --device /dev/video4
DISPLAY_DEVICE: -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix
root@vkjnor-desktop: /opt/jetson-inference#
```

T1: Running jetson-inference docker

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script12

vk-jn-or@vkjnor-desktop:~$ docker ps -a
permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Get "http://%2Fvar%2Frun%2Fdocker.sock/v1.50/containers/json?all=1": dial unix /var/run/docker.sock: connect: permission denied
vk-jn-or@vkjnor-desktop:~$ groups
vk-jn-or adm cdrom sudo audio dip video plugdev render i2c lpadmin gdm sambashare weston-launch gpio jtop
vk-jn-or@vkjnor-desktop:~$ cd /home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script12/
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script12$ ./jonSetup12-dckr.sh
Configuring Docker group and user permissions...
Testing permissions in current session...
Current groups for this session:
docker adm cdrom sudo audio dip video plugdev render i2c lpadmin gdm sambashare weston-launch gpio vk-jn-or jtop
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
9e0e0caf5eb8   dustynv/jetson-inference:r36.3.0    "/bin/bash"             4 minutes ago  Up 4 minutes                jetson-inference-session
*****
CLEANUP COMPLETE: Permissions have been applied to this terminal.
CRITICAL: You must REBOOT now to finalize group updates for all terminals.
After reboot, verify with 'groups' and then run any next scripts.
*****
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script12$
```

T2: Running jonSetup12-dckr.sh

Terminal O/P: jonSetup12-dckr.sh

- Reboot
- In a Linux shell terminal (T1), launch jetson-inference docker container using:
`cd ~/jetson-inference && ./docker/run.sh --container dustynv/jetson-inference:r36.3.0`
- In another Linux shell terminal (T2), run `groups` followed by `docker ps -a`
- Docker is now part of user group and jetson-inference-session is running docker session

The image shows two terminal windows side-by-side. The top window, labeled 'T1: Running jetson-inference docker', shows the execution of the script `./docker/run.sh --container dustynv/jetson-inference:r36.3.0`. The output includes system information like ARCH (aarch64), L4T version (R36.4.4), and a long list of Docker volume and device mappings. The bottom window, labeled 'T2: Running groups and docker ps -a', shows the user running `groups` and `docker ps -a`. The `groups` command output shows the user is in the `adm`, `cdrom`, `sudo`, `audio`, `dip`, `video`, `plugdev`, `render`, `i2c`, `lpadmin`, `gdm`, `sambashare`, `docker`, `weston-launch`, `gpio`, and `jtop` groups. The `docker ps -a` command output shows a single container named 'jetson-inference-session' in a running state.

```
root@vkjnor-desktop: /opt/jetson-inference

vk-jn-or@vkjnor-desktop:~$ cd ~/jetson-inference && ./docker/run.sh --container dustynv/jetson-inference:r36.3.0
ARCH: aarch64
reading L4T version from /etc/nv_tegra_release
L4T BSP Version: L4T R36.4.4
localuser:root being added to access control list
CONTAINER_IMAGE: dustynv/jetson-inference:r36.3.0
DATA_VOLUME: --volume /home/vk-jn-or/jetson-inference/data:/jetson-inference/data --volume /home/vk-jn-or/jetson-inference/python/traini
ng/classification/data:/jetson-inference/python/training/classification/data --volume /home/vk-jn-or/jetson-inference/python/training/classif
ication/models:/jetson-inference/python/training/classification/models --volume /home/vk-jn-or/jetson-inference/python/training/detection/ssd
/data:/jetson-inference/python/training/detection/ssd/data --volume /home/vk-jn-or/jetson-inference/python/training/detection/ssd/models:/jet
son-inference/python/training/detection/ssd/models --volume /home/vk-jn-or/jetson-inference/python/www/recognizer/data:/jetson-inference/pyth
on/www/recognizer/data
V4L2_DEVICES: --device /dev/video0 --device /dev/video1 --device /dev/video2 --device /dev/video3 --device /dev/video4
DISPLAY_DEVICE: -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix
root@vkjnor-desktop:/opt/jetson-inference#

vk-jn-or@vkjnor-desktop:~$ groups
vk-jn-or adm cdrom sudo audio dip video plugdev render i2c lpadmin gdm sambashare docker weston-launch gpio jtop
vk-jn-or@vkjnor-desktop:~$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS        NAMES
aac588469148   dustynv/jetson-inference:r36.3.0    "/bin/bash"            About a minute ago    Up About a minute                jetson-inference-session
vk-jn-or@vkjnor-desktop:~$
```

Shell Script: jonSetup13-dckr.sh

- Replace some files in jetson-inference docker container, install some apps, and recommit the docker to save changes/additions

```
#!/bin/bash
```

Create some aliases

```
# --- 1: Paths & Variables ---
echo "[1/11] Setting some aliases..."
HOME_DIR="/home/vk-jn-or"
JT_INF_DIR="$HOME_DIR/jetson-inference"
GST_CAM_RVSD_DIR="$HOME_DIR/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/Edited-Files/updated/Utils_Camera/Ver5-FinalWorking/ConstCaptRes-UsrOutRes-Rev2"
JT_INF_EDITED_DIR="$HOME_DIR/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/Edited-Files/jetson-inference"
OG_DCKR_RUN_DIR="$HOME_DIR/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/Edited-Files/original/docker"
CONTAINER_NAME="jetson-inference-session"
```

Copy some files

```
# --- 2: Initial Copy from "JT_INF_EDITED_DIR" to $JT_INF_DIR" ---
#Note: This needs to be copied NOW so that various files are available in jetson-inference folder so that they can be brought into docker container.
echo "[2/11] Copying some folders and files from the host to jetson-inference folder..."
cp -Rn "$JT_INF_EDITED_DIR/." "$JT_INF_DIR/"
echo "Make dockerOn.sh executable for jetson-inference-gstcam launch after it gets created as below..."
chmod +x "$JT_INF_DIR/dockerOn.sh"
```

```
# --- 3: User Verification ---
```

```
echo "[3/11] Before proceeding with this shell script execution, launch a terminal session."
echo "Run following commands to launch a docker container session jetson-inference-session."
echo "cd ~/jetson-inference && ./docker/run.sh --container dustynv/jetson-inference:r36.3.0 --volume ~/jetson-inference/usrfolder:/opt/jetson-inference/usrfolder"
read -p "Enter Y when you have launched the docker container jetson-inference-session: " user_input
```

```
if [[ ! "$user_input" =~ ^[Yy]$ ]]; then
    echo "Exiting script. Please launch the container and try again."
    exit 1
fi
```

Confirm from user that
jetson-inference-session
has been launched in another terminal

Replace gstCamera.cpp

```
# --- 4: Initial Host-to-Docker Copy ---
echo "[4/11] Copying revised gstCamera.cpp into the running container..."
docker cp "$GST_CAM_RVSD_DIR/gstCamera.cpp" ${CONTAINER_NAME}:/opt/jetson-inference/utils/camera/gstCamera.cpp
```

Install: gi, gstreamer, xdotool, zenity

```
# --- 5: Internal Package Installation ---
# This runs inside the container while it is active in your other terminal
echo "[5/11] Installing additional packages (xdotool, zenity, gstreamer utils)..."
docker exec -it $CONTAINER_NAME /bin/bash -c "
    apt-get update && \
    apt install -y python3-gi gir1.2-gstreamer-1.0 xdotool zenity
"
```

Replace some .cpp and .py files

```
# --- 6: Internal File Replacements ---
echo "[6/11] Updating internal .cpp and .py files inside the container..."
docker exec -it $CONTAINER_NAME /bin/bash -c "
    cp /opt/jetson-inference/usrfolder/backupcppFiles/updatedFiles/camera-capture.cpp /opt/jetson-inference/tools/camera-capture/camera-capture.cpp && \
    cp /opt/jetson-inference/usrfolder/backupcppFiles/updatedFiles/video-viewer.cpp /opt/jetson-inference/utils/video/video-viewer/video-viewer.cpp && \
    cp /opt/jetson-inference/usrfolder/backupcppFiles/updatedFiles/imagenet.cpp /opt/jetson-inference/examples/imagenet/imagenet.cpp && \
    cp /opt/jetson-inference/usrfolder/backupcppFiles/updatedFiles/detectnet.cpp /opt/jetson-inference/examples/detectnet/detectnet.cpp && \
    cp /opt/jetson-inference/usrfolder/backup_pyFiles/updatedFiles/video-viewer_BAB.py /opt/jetson-inference/build/aarch64/bin/video-viewer.py && \
    cp /opt/jetson-inference/usrfolder/backup_pyFiles/updatedFiles/video-viewer_UPE.py /opt/jetson-inference/utils/python/examples/video-viewer.py && \
    cp /opt/jetson-inference/usrfolder/backup_pyFiles/updatedFiles/imagenet_PE.py /opt/jetson-inference/python/examples/imagenet.py && \
    cp /opt/jetson-inference/usrfolder/backup_pyFiles/updatedFiles/imagenet_BAB.py /opt/jetson-inference/build/aarch64/bin/imagenet.py && \
    cp /opt/jetson-inference/usrfolder/backup_pyFiles/updatedFiles/detectnet_PE.py /opt/jetson-inference/python/examples/detectnet.py && \
    cp /opt/jetson-inference/usrfolder/backup_pyFiles/updatedFiles/detectnet_BAB.py /opt/jetson-inference/build/aarch64/bin/detectnet.py
"
```

**Rebuild and install docker container
with above changes to various files**

```
# --- 7: Internal Rebuild ---
echo "[7/11] Rebuilding jetson-inference inside the container..."
docker exec -it $CONTAINER_NAME /bin/bash -c "
    cd /opt/jetson-inference && \
    unset INDEX_HOST && \
    unset PIP_INDEX_URL && \
    rm -rf build && \
    mkdir build && \
    cd build && \
    export PIP_INDEX_URL=https://pypi.jetson-ai-lab.io/jp6/cu126 && \
    cmake ../ && \
    make -j$(nproc) && \
    make install && \
    python3 -c 'import jetson_utils; import jetson_inference' && \
    python3 -c 'import tensorrt; print(tensorrt.__version__)'
"
```

```
# --- 8: Image Commitment & Backup Tagging ---
echo "[8/11] Committing the updated container to image: jetson-inference-gstcam..."
docker commit $CONTAINER_NAME jetson-inference-gstcam
echo "Creating safe backup image..."
docker tag jetson-inference-gstcam jetson-inference-gstcam-safe
```

**Commit the updated docker
containing new apps as new image
jetson-inference-gstcam**

```
# --- 9: Revert Host Sync & Aliases ---
echo "[9/11] Reverting host run.sh to original and syncing host files..."
cp "$OG_DCKR_RUN_DIR/run.sh" "$JT_INF_DIR/docker/run.sh"
```

Revert to original run.sh

```
# --- 10: Bashrc Alias setup ---
echo "[10/11] Setting dockeron alias..."
if ! grep -q "alias dockeron=" ~/.bashrc; then
    echo "alias dockeron='source ~/jetson-inference/dockerOn.sh'" >> ~/.bashrc
    echo "Alias 'dockeron' added to .bashrc"
fi
```

dockeron alias

```
# --- 11: Take ownership of any newly created folders in jetson-inference ---
echo "[11/11] Fixing ownership of some folders..."
sudo chown -R $USER:$USER "$JT_INF_DIR"
```

folder ownership

```
echo "*****"
echo "FULL DOCKER SETUP COMPLETE."
echo "1. Close the other terminal window where the container was running."
echo "2. Run 'source ~/.bashrc'."
echo "3. Use the 'dockeron' command to launch your fully updated, custom container."
echo "*****"
```

Terminal O/P: jonSetup13-dckr.sh

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script13
vk-jn-or@vkjnor-desktop:~$ cd /home/vk-jn-or/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script13/
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script13$ ./jonSetup13-dckr.sh
[1/11] Setting some aliases...
[2/11] Copying some folders and files from the host to jetson-inference folder...
Make dockerOn.sh executable for jetson-inference-gstcam launch after it gets created as below...
[3/11] Before proceeding with this shell script execution, launch a terminal session.
Run following commands to launch a docker container session jetson-inference-session.
cd ~/jetson-inference && ./docker/run.sh --container dustynv/jetson-inference:r36.3.0 --volume ~/jetson-inference/usrfolder:/opt/jetson-inference/usrfolder
Enter Y when you have launched the docker container jetson-inference-session: ☐
```

T1: Running jonSetup13-dckr.sh

- In a Linux terminal, run from the directory containing jonSetup13-dckr.sh, `./jonSetup13-dckr.sh`
- In terminal T1, a message will appear to enter Y

- Before entering anything in terminal T1, open another Linux terminal T2 and run jetson-inference docker container using the below command that mounts a `usrfolder` in the jetson-inference container

```
cd ~/jetson-inference && ./docker/run.sh --container dustynv/jetson-inference:r36.3.0 --volume ~/jetson-inference/usrfolder:/opt/jetson-inference/usrfolder
```

Terminal O/P: jonSetup13-dckr.sh

T2: Running jetson-inference docker

Docker running

T1: Running jonSetup13-dckr.sh

- After launching the jetson-inference docker container in T2 and confirming that it is running (indicated by docker container prompt), enter Y in T1

Terminal O/P: jonSetup13-dckr.sh

```
root@vkjnor-desktop: /opt/jetson-inference
vk-jn-or@vkjnor-desktop:~$ cd ~/jetson-inference && ./docker/run.sh --container dustynv/jetson-inference:r36.3.0 --volume ~/jetson-inference/usrfolder:/opt/jetson-inference/usrfolder
ARCH: aarch64
reading L4T version from /etc/nv_tegra_release
L4T BSP Version: L4T R36.4.4
localuser:root being added to access control list
CONTAINER_IMAGE: dustynv/jetson-inference:r36.3.0
DATA_VOLUME: --volume /home/vk-jn-or/jetson-inference/data:/jetson-inference/data --volume /home/vk-jn-or/jetson-inference/python/training/classification/data:/jetson-inference/python/training/classification/data --volume /home/vk-jn-or/jetson-inference/python/training/detection/ssd/data:/jetson-inference/python/training/detection/ssd/data --volume /home/vk-jn-or/jetson-inference/python/www/recognizer/data:/jetson-inference/python/www/recognizer/data
USER_VOLUME: --volume /home/vk-jn-or/jetson-inference/usrfolder:/opt/jetson-inference/usrfolder
V4L2_DEVICES: --device /dev/video0 --device /dev/video1 --device /dev/video2 --device /dev/video3 --device /dev/video4
DISPLAY_DEVICE: -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix
root@vkjnor-desktop:/opt/jetson-inference#
```

T2: Running jetson-inference docker

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script13
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script13$ ./jonSetup13-dckr.sh
[1/11] Setting some aliases...
[2/11] Copying some folders and files from the host to jetson-inference folder...
Make dockerOn.sh executable for jetson-inference-gstcam launch after it gets created as below...
[3/11] Before proceeding with this shell script execution, launch a terminal session.
Run following commands to launch a docker container session jetson-inference-session.
cd ~/jetson-inference && ./docker/run.sh --container dustynv/jetson-inference:r36.3.0 --volume ~/jetson-inference/usrfolder:/opt/jetson-inference/usrfolder
Enter Y when you have launched the docker container jetson-inference-session: Y
[4/11] Copying revised gstCamera.cpp into the running container...
Successfully copied 27.1kB to jetson-inference-session:/opt/jetson-inference/Utils/camera/gstCamera.cpp
[5/11] Installing additional packages (xdotool, zenity, gstreamer utils)...
Get:1 http://ports.ubuntu.com/ubuntu-ports jammy InRelease [270 kB]
Get:2 http://ports.ubuntu.com/ubuntu-ports jammy-updates InRelease [128 kB]
Get:3 http://ports.ubuntu.com/ubuntu-ports jammy-backports InRelease [127 kB]
Get:4 http://ports.ubuntu.com/ubuntu-ports jammy-security InRelease [129 kB]
Get:5 http://ports.ubuntu.com/ubuntu-ports jammy/main arm64 Packages [1,758 kB]
Get:6 http://ports.ubuntu.com/ubuntu-ports jammy/restricted arm64 Packages [24.2 kB]
Get:7 http://ports.ubuntu.com/ubuntu-ports jammy/universe arm64 Packages [17.2 MB]
Get:8 http://ports.ubuntu.com/ubuntu-ports jammy/multiverse arm64 Packages [224 kB]
Get:9 http://ports.ubuntu.com/ubuntu-ports jammy-updates/restricted arm64 Packages [6,534 kB]
Get:10 http://ports.ubuntu.com/ubuntu-ports jammy-updates/main arm64 Packages [3,841 kB]
Get:11 http://ports.ubuntu.com/ubuntu-ports jammy-updates/multiverse arm64 Packages [47.7 kB]
Get:12 http://ports.ubuntu.com/ubuntu-ports jammy-updates/universe arm64 Packages [1,654 kB]
```

Replace gstcamera.cpp

App installs

T1: Running jonSetup13-dckr.sh

Terminal O/P: jonSetup13-dckr.sh

```
root@vkjnor-desktop: /opt/jetson-inference
vk-jn-or@vkjnor-desktop:~$ cd ~/jetson-inference && ./docker/run.sh --container dustynv/jetson-inference:r36.3.0 --volume ~/jetson-inference/usrfolder:/opt/jetson-inference/usrfolder
ARCH: aarch64
reading L4T version from /etc/nv_tegra_release
L4T BSP Version: L4T R36.4.4
localuser:root being added to access control list
CONTAINER_IMAGE: dustynv/jetson-inference:r36.3.0
DATA_VOLUME: --volume /home/vk-jn-or/jetson-inference/data:/jetson-inference/data --volume /home/vk-jn-or/jetson-inference/python/training/classification/data:/jetson-inference/python/training/classification/data --volume /home/vk-jn-or/jetson-inference/python/training/classification/models:/jetson-inference/python/training/classification/models --volume /home/vk-jn-or/jetson-inference/python/training/detection/ssd/data:/jetson-inference/python/training/detection/ssd/data --volume /home/vk-jn-or/jetson-inference/python/training/detection/ssd/models:/jetson-inference/python/training/detection/ssd/models --volume /home/vk-jn-or/jetson-inference/python/www/recognizer/data:/jetson-inference/python/www/recognizer/data
USER_VOLUME: --volume /home/vk-jn-or/jetson-inference/usrfolder:/opt/jetson-inference/usrfolder
V4L2_DEVICES: --device /dev/video0 --device /dev/video1 --device /dev/video2 --device /dev/video3 --device /dev/video4
DISPLAY_DEVICE: -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix
root@vkjnor-desktop:/opt/jetson-inference#
```

T2: Running jetson-inference docker

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script13
-- Up-to-date: /usr/local/bin/posenet.py
-- Up-to-date: /usr/local/bin/segnet.py
-- Up-to-date: /usr/local/bin/segnet_utils.py
-- Installing: /usr/lib/python3.10/dist-packages/jetson_inference_python.so
-- Set non-toolchain portion of runtime path of "/usr/lib/python3.10/dist-packages/jetson_inference_python.so" to ""
-- Up-to-date: /usr/lib/python3.10/dist-packages/Jetson
-- Up-to-date: /usr/lib/python3.10/dist-packages/Jetson/Inference
-- Up-to-date: /usr/lib/python3.10/dist-packages/Jetson/Inference/__init__.py
-- Installing: /usr/lib/python3.10/dist-packages/Jetson/__init__.py
-- Up-to-date: /usr/lib/python3.10/dist-packages/jetson
-- Up-to-date: /usr/lib/python3.10/dist-packages/jetson/inference
-- Up-to-date: /usr/lib/python3.10/dist-packages/jetson/inference/__init__.py
-- Installing: /usr/lib/python3.10/dist-packages/jetson/__init__.py
-- Up-to-date: /usr/lib/python3.10/dist-packages/jetson_inference
-- Up-to-date: /usr/lib/python3.10/dist-packages/jetson_inference/__init__.py
8.6.2
[8/11] Committing the updated container to image: jetson-inference-gstcam...
sha256:2691fd6a7dc596579ca02975d35fbee05f6ec497d1ea4d803347b61df33c6d8e
Creating safe backup image...
[9/11] Reverting host run.sh to original and syncing host files...
[10/11] Setting dockeron alias...
Alias 'dockeron' added to .bashrc
[11/11] Fixing ownership of some folders...
*****
FULL DOCKER SETUP COMPLETE.
1. Close the other terminal window where the container was running.
2. Run 'source ~/.bashrc'.
3. Use the 'dockeron' command to launch your fully updated, custom container.
*****
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script13$
```

**Commit docker
to a new image**

**T1: jonSetup13-dckr.sh
execution completed**

Terminal O/P: jonSetup13-dckr.sh

```
root@vkjnor-desktop: /opt/jetson-inference
vk-jn-or@vkjnor-desktop:~$ cd ~/jetson-inference && ./docker/run.sh --container dustynv/jetson-inference:r36.3.0 --volume ~/jetson-inference/usrfolder:/opt/jetson-inference/usrfolder
ARCH: aarch64
reading L4T version from /etc/nv_tegra_release
L4T BSP Version: L4T R36.4.4
localuser:root being added to access control list
CONTAINER_IMAGE: dustynv/jetson-inference:r36.3.0
DATA_VOLUME: --volume /home/vk-jn-or/jetson-inference/data:/jetson-inference/data --volume /home/vk-jn-or/jetson-inference/python/training/classification/data:/jetson-inference/python/training/classification/data --volume /home/vk-jn-or/jetson-inference/python/training/classification/models:/jetson-inference/python/training/classification/models --volume /home/vk-jn-or/jetson-inference/python/training/detection/ssd/data:/jetson-inference/python/training/detection/ssd/data --volume /home/vk-jn-or/jetson-inference/python/training/detection/ssd/models:/jetson-inference/python/training/detection/ssd/models --volume /home/vk-jn-or/jetson-inference/python/www/recognizer/data:/jetson-inference/python/www/recognizer/data
USER_VOLUME: --volume /home/vk-jn-or/jetson-inference/usrfolder:/opt/jetson-inference/usrfolder
V4L2_DEVICES: --device /dev/video0 --device /dev/video1 --device /dev/video2 --device /dev/video3 --device /dev/video4
DISPLAY_DEVICE: -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix
root@vkjnor-desktop:/opt/jetson-inference# exit
exit
vk-jn-or@vkjnor-desktop:~/jetson-inference$ source ~/.bashrc
vk-jn-or@vkjnor-desktop:~/jetson-inference$ dockeron
Granting X11 access to local root user...
non-network local connections being added to access control list
Restarting nvargus-daemon for camera stability...
Running System Memory Maintenance...
--- [System Maintenance] Starting Refresh ---
Cleaning system memory for GPU allocation...
Waiting 5 seconds for hardware stabilization...
--- [System Maintenance] Ready ---
Starting Docker container 'test-session'...
root@vkjnor-desktop:/opt/jetson-inference#
```

Exit from docker

dockeron alias

T2: Running test-session

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script13
-- Up-to-date: /usr/lib/python3.10/dist-packages/jetson/inference
-- Up-to-date: /usr/lib/python3.10/dist-packages/jetson/inference/__init__.py
-- Installing: /usr/lib/python3.10/dist-packages/jetson/__init__.py
-- Up-to-date: /usr/lib/python3.10/dist-packages/jetson_inference
-- Up-to-date: /usr/lib/python3.10/dist-packages/jetson_inference/__init__.py
8.6.2
[8/11] Committing the updated container to image: jetson-inference-gstcam...
sha256:2691fd6a7dc596579ca02975d35fbee05f6ec497d1ea4d803347b61df33c6d8e
Creating safe backup image...
[9/11] Reverting host run.sh to original and syncing host files...
[10/11] Setting dockeron alias...
Alias 'dockeron' added to .bashrc
[11/11] Fixing ownership of some folders...
*****
FULL DOCKER SETUP COMPLETE.
1. Close the other terminal window where the container was running.
2. Run 'source ~/.bashrc'.
3. Use the 'dockeron' command to launch your fully updated, custom container.
*****
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script13$
```

T1: jonSetup13-dckr.sh execution completed

Terminal O/P: jonSetup13-dckr.sh

```
# Conda aliases
alias condaon1='source ~/condaActivate.sh'
alias condaoff1='source ~/condaDeactivate.sh'
export PATH=/usr/local/cuda/bin:$PATH
export LD_LIBRARY_PATH=/usr/local/cuda/lib64:$LD_LIBRARY_PATH

# Fix for libffi
export LD_PRELOAD=/usr/lib/aarch64-linux-gnu/libffi.so.8
export LD_PRELOAD=/usr/lib/aarch64-linux-gnu/libffi.so.7
alias condaon2='source ~/condaActivate2.sh'
alias condaoff2='source ~/condaDeactivate2.sh'
alias dockeron='source ~/jetson-inference/dockerOn.sh'
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script13$
```

New addition to ~/.bashrc

Terminal O/P: jonSetup13-dckr.sh

```
root@vkjnor-desktop: /opt/jetson-inference

vk-jn-or@vkjnor-desktop:~$ cd ~/jetson-inference && ./docker/run.sh --container dustynv/jetson-inference:r36.3.0 --volume ~/jetson-inference/usrfolder:/opt/jetson-inference/usrfolder
ARCH: aarch64
reading L4T version from /etc/nv_tegra_release
L4T BSP Version: L4T R36.4.4
localuser:root being added to access control list
CONTAINER_IMAGE: dustynv/jetson-inference:r36.3.0
DATA_VOLUME: --volume /home/vk-jn-or/jetson-inference/data:/jetson-inference/data --volume /home/vk-jn-or/jetson-inference/python/training/classification/data:/jetson-inference/python/training/classification/data --volume /home/vk-jn-or/jetson-inference/python/training/classification/models:/jetson-inference/python/training/classification/models --volume /home/vk-jn-or/jetson-inference/python/training/detection/ssd/data:/jetson-inference/python/training/detection/ssd/data --volume /home/vk-jn-or/jetson-inference/python/training/detection/ssd/models:/jetson-inference/python/training/detection/ssd/models --volume /home/vk-jn-or/jetson-inference/python/www/recognizer/data:/jetson-inference/python/www/recognizer/data
USER_VOLUME: --volume /home/vk-jn-or/jetson-inference/usrfolder:/opt/jetson-inference/usrfolder
V4L2_DEVICES: --device /dev/video0 --device /dev/video1 --device /dev/video2 --device /dev/video3 --device /dev/video4
DISPLAY_DEVICE: -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix
root@vkjnor-desktop:/opt/jetson-inference# exit
exit
vk-jn-or@vkjnor-desktop:~/jetson-inference$ source ~/.bashrc
vk-jn-or@vkjnor-desktop:~/jetson-inference$ dockeron
Granting X11 access to local root user...
non-network local connections being added to access control list
Restarting nvargus-daemon for camera stability...
Running System Memory Maintenance...
--- [System Maintenance] Starting Refresh ---
Cleaning system memory for GPU allocation...
Waiting 5 seconds for hardware stabilization...
--- [System Maintenance] Ready ---
Starting Docker container 'test-session'...
root@vkjnor-desktop:/opt/jetson-inference#
```

T2: Running test-session

```
vk-jn-or@vkjnor-desktop: ~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script13
```

```
Creating safe backup image...
[9/11] Reverting host run.sh to original and syncing host files...
[10/11] Setting dockeron alias...
Alias 'dockeron' added to .bashrc
[11/11] Fixing ownership of some folders...
*****
FULL DOCKER SETUP COMPLETE.
1. Close the other terminal window where the container was running.
2. Run 'source ~/.bashrc'.
3. Use the 'dockeron' command to launch your fully updated, custom container.
*****
```

T1: docker ps -a and docker list

```
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script13$ docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS          NAMES
ede42b48a1e6   jetson-inference-gstcam             "/bin/bash"             2 minutes ago Up 2 minutes          test-session
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script13$ docker images
REPOSITORY          TAG         IMAGE ID      CREATED        SIZE
jetson-inference-gstcam-safe   latest     2691fd6a7dc5  3 minutes ago  15.2GB
jetson-inference-gstcam        latest     2691fd6a7dc5  3 minutes ago  15.2GB
dustynv/jetson-inference       r36.3.0    48d8c1f46455  20 months ago  14.8GB
vk-jn-or@vkjnor-desktop:~/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/shellScripts/Script13$
```

test-session based on jetson-inference-gstcam

Shell Script and Terminal O/P: jonSetup14-dckr.sh

- Get and place VSC json files in appropriate folders

```
#!/bin/bash
```

```
# ---1: Paths & Variables ---
```

```
HOME_DIR="/home/vk-jn-or"
```

```
# Source folders to download folders/files from
```

```
SRC_VK_DIR="$HOME_DIR/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/Edited-Files/VSC/VSC-VK"
```

```
SRC_JI_USR_DIR="$HOME_DIR/Downloads/Jetson-Orin-Nano/AM_JON_2SHARE/Edited-Files/VSC/VSC-JI/usrfolder"
```

```
# Destination folders to download folders/files into
```

```
DST_VK_DIR="$HOME_DIR"
```

```
DST_USR_DIR="$HOME_DIR/jetson-inference/usrfolder"
```

```
# --- 2: Copy from Folders/Files ---
```

```
echo "Copying some folders and files ..."
```

```
cp -Rn "$SRC_VK_DIR/." "$DST_VK_DIR/"
```

```
cp -Rn "$SRC_JI_USR_DIR/." "$DST_USR_DIR/"
```

VSC json files for openCV
(vk1_env and vk2_env)

VSC json files for jetson-
inference docker

vk-jn-or@vkjnor-desktop: ~/Desktop/Jetson-DskTp/Jet-Nano-Orin/5.SetupwithScripts/AM_JON_2SHARE/shellScripts/Script14

```
vk-jn-or@vkjnor-desktop:~$ cd /home/vk-jn-or/Desktop/Jetson-DskTp/Jet-Nano-Orin/5.SetupwithScripts/AM_JON_2SHARE/shellScripts/Script14/
vk-jn-or@vkjnor-desktop:~/Desktop/Jetson-DskTp/Jet-Nano-Orin/5.SetupwithScripts/AM_JON_2SHARE/shellScripts/Script14$ ./jonSetup14.sh
Copying some folders and files ...
vk-jn-or@vkjnor-desktop:~/Desktop/Jetson-DskTp/Jet-Nano-Orin/5.SetupwithScripts/AM_JON_2SHARE/shellScripts/Script14$
```

SD to SSD migration: jonSetup15Auto.sh

```
#!/bin/bash

# Jetson Orin Nano: SD to SSD Migration Script (Automated Version)
# Optimized to fix Superblock/Partition errors automatically

set -e # Exit on error

echo "-----"
echo "Starting Migration from SD Card to SSD"
echo "Ensure SSD is physically installed in the PCI slot before proceeding."
echo "-----"

# --- 1. Automated Hardware Prep (Replaces Manual Disks App) ---
echo "[STEP 1/8]: Rescanning PCI bus to find SSD..."
echo 1 | sudo tee /sys/bus/pci/rescan > /dev/null
sleep 2 # Give the kernel a moment to register the device

# Check if nvme0n1 exists
if lsblk | grep -q "nvme0n1"; then
    echo "Found NVMe SSD: $(lsblk -d -n -o SIZE /dev/nvme0n1) detected."

    echo "Wiping existing signatures and partition tables..."
    # This clears the 'ghost' partitions that cause Superblock errors
    sudo wipefs -a /dev/nvme0n1

    # Inform the kernel of the changes
    sudo partprobe /dev/nvme0n1
    sudo sync && sleep 2
    echo "SSD is now clean and ready."
else
    echo "ERROR: NVMe SSD (/dev/nvme0n1) not found!"
    echo "Please check the physical connection and try again."
    exit 1
fi
```

**For SD 2 SSD migration,
bootup JON from the SD
card as prepared using 14
earlier shell scripts**

**Ensure SSD exists and wipe out
any partitions that may cause
superblock errors
DO NOT use Disks Application
as it can cause issues**

```
# --- 2. Setup Directory and Clone Repo ---
DWNLD_DIR="/home/vk-jn-or/Downloads/Jetson-Orin-Nano"
echo "[STEP 2/8]: Setting up directory at $DWNLD_DIR"
mkdir -p "$DWNLD_DIR"
cd "$DWNLD_DIR"

if [ ! -d "migrate-jetson-to-ssd" ]; then
    echo "Cloning migration repository..."
    git clone https://github.com/jetsonhacks/migrate-jetson-to-ssd
fi
cd migrate-jetson-to-ssd

# --- 3. Copy Partition Structure ---
echo "[STEP 3/8]: Creating partition structure on SSD..."
# Since we used wipefs, this script will now see a truly blank disk
sudo bash make_partitions.sh

# --- 4. Copy Data ---
echo "[STEP 4/8]: Copying data from SD to SSD (This may take a while)..."
sudo bash copy_partitions.sh

# --- 5. Configure Boot ---
echo "[STEP 5/8]: Modifying SSD configuration for boot..."
sudo bash configure_ssd_boot.sh

# --- 6 Integrity Check ---
echo "[STEP 6/8]: Verifying file integrity..."
# Mount the SSD temporarily to check it
sudo mkdir -p /mnt/ssd_check
sudo mount /dev/nvme0n1p1 /mnt/ssd_check

# Count files in /usr on both drives
SD_COUNT=$(sudo find /usr -xdev | wc -l)
SSD_COUNT=$(sudo find /mnt/ssd_check/usr -xdev | wc -l)
```

**Setup directory and clone
migrate-jetson-to-ssd repo**

**Run the three scripts of repo: copy
partition structure, copy data, and
configure boot from SSD**

**Check file counts in one large folder
on SD and on SSD to ensure
integrity of the migration operation**

```
echo "SD Card (/usr) file count: $SD_COUNT"
echo "SSD (/usr) file count: $SSD_COUNT"
```

```
# Clean up
```

```
sudo umount /mnt/ssd_check
```

```
if [ "$SD_COUNT" -eq "$SSD_COUNT" ]; then
    echo "☑ Integrity Check Passed: Program files match exactly."
```

```
else
    # A small difference (1-5 files) is sometimes okay due to temp files,
    # but a large gap is a problem.
    DIFF=$((SD_COUNT - SSD_COUNT))
    echo "⚠ Warning: File count mismatch. Difference: ${DIFF#-} files."
    echo "This can happen if files were deleted/added during the copy."
fi
```

Display results of integrity check

```
# --- 7. Partition Expansion ---
```

```
echo "[STEP 7/8]: Installing GParted to expand SSD storage..."
```

```
sudo apt-get update && sudo apt-get install -y gparted
```

```
echo "-----"
```

```
echo "MANUAL ACTION REQUIRED IN GPARTED:"
```

```
echo "1. Select /dev/nvme0n1 from the top-right menu."
```

```
echo "2. Right-click the 'APP' (root) partition."
```

```
echo "3. Select 'Resize/Move' and drag the arrow to fill unallocated space."
```

```
echo "4. Click the 'Checkmark' icon to Apply Changes."
```

```
echo "-----"
```

```
sudo gparted
```

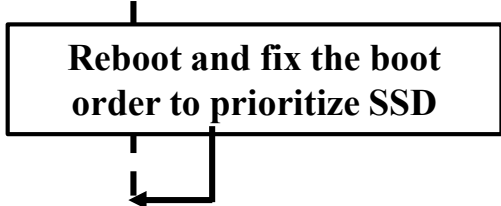
```
read -p "Press [Enter] once you have finished resizing in GParted..."
```

Expand App partition to
use full storage on SSD

```
# --- 8. Final Instructions ---
echo "-----"
echo "[STEP 8/8]: MIGRATION COMPLETE!"
echo "To finish, we will reboot. During boot:"
echo "1. Repeatedly press [ESC] to enter BIOS/UEFI."
echo "2. Go to: Boot Maintenance Manager -> Boot Options -> Change Boot Order."
echo "3. Move SSD to the top using the '+' key."
echo "4. Save and Exit."
echo "-----"

read -p "Ready to reboot? (y/n): " confirm
if [[ $confirm == [yY] ]]; then
    sudo reboot now
fi
```

**Reboot and fix the boot
order to prioritize SSD**

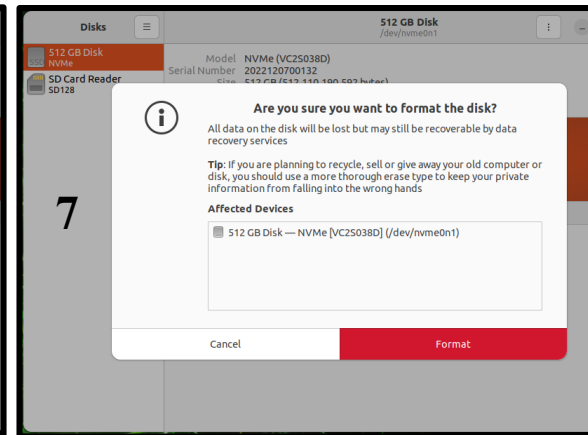
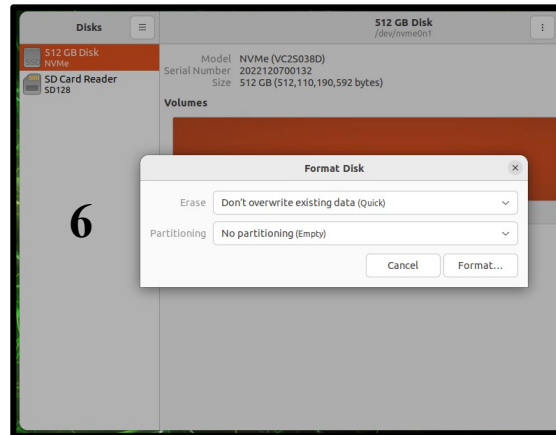
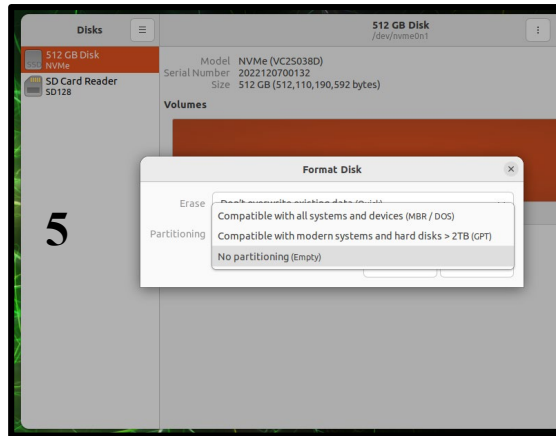
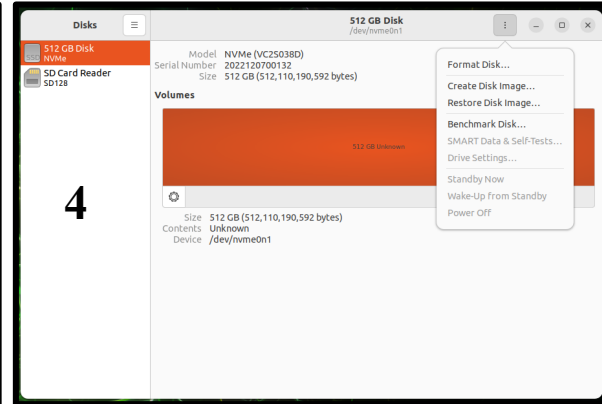
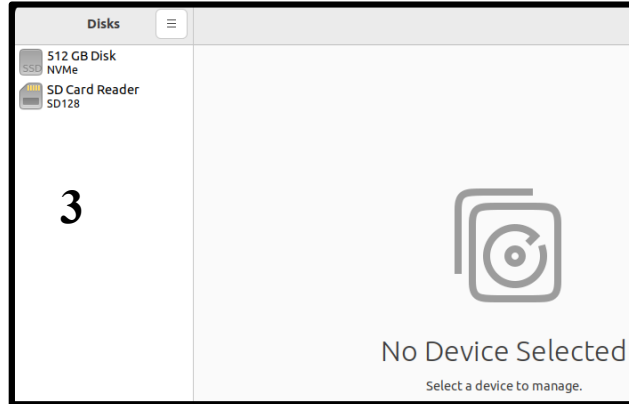
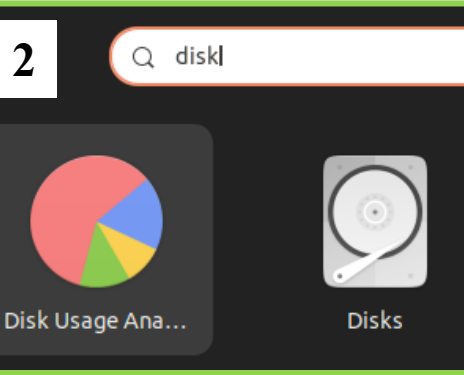


Disks App to Format SSD with No Partition

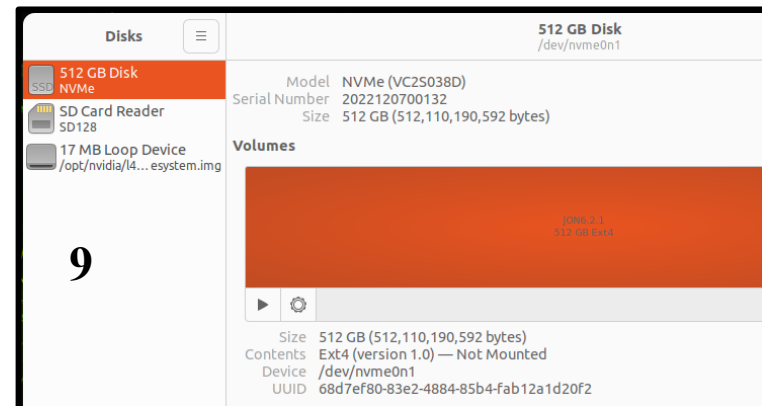
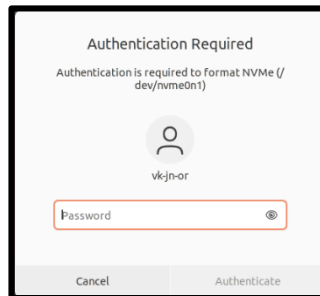
1



Look for
Apps



8



Error: Migration Script without Step 1

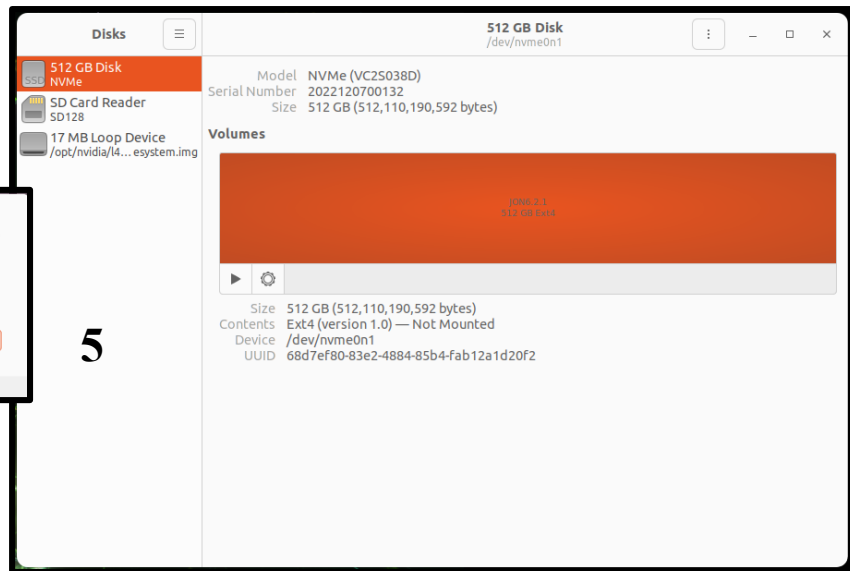
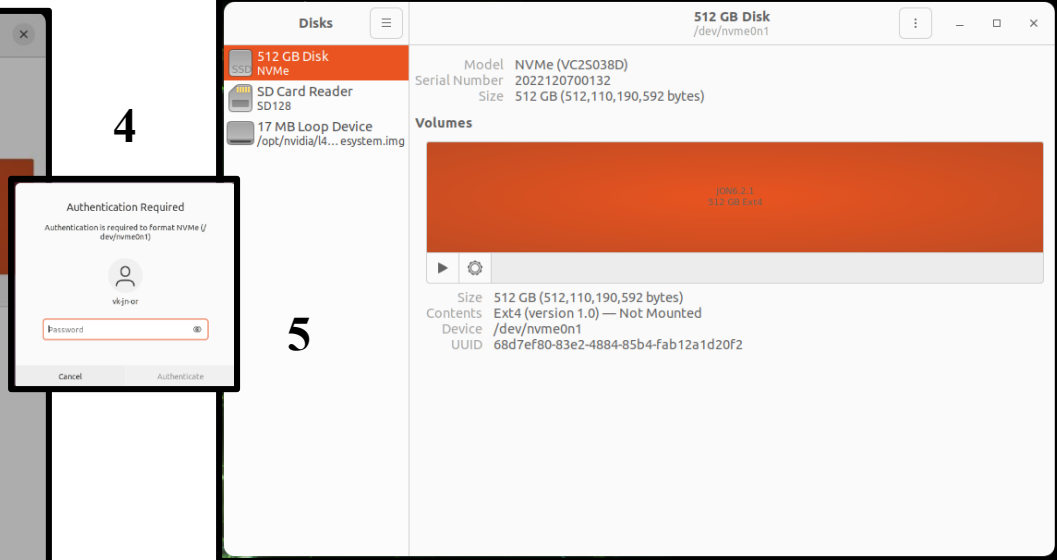
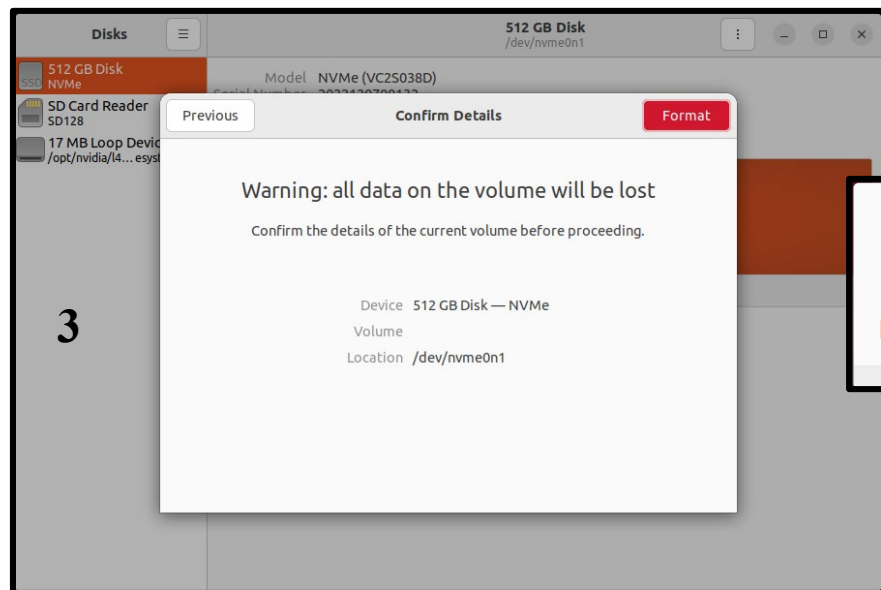
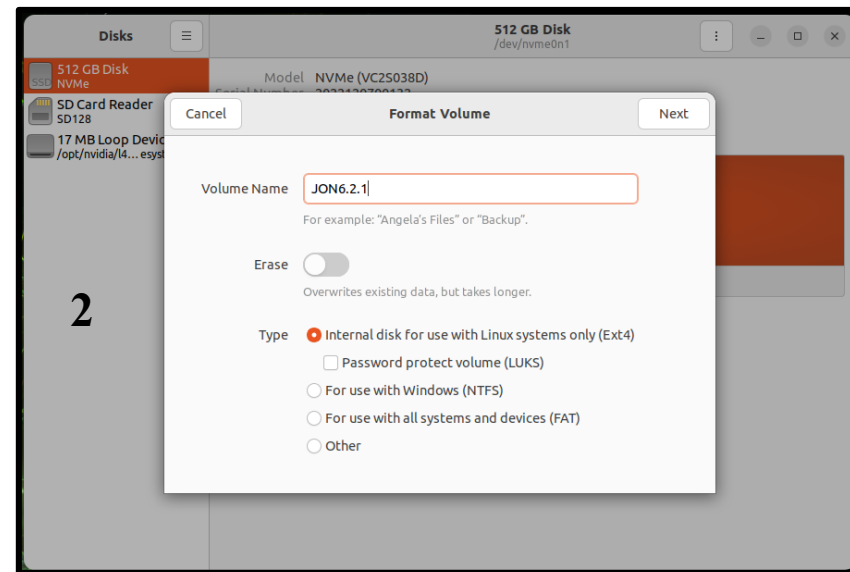
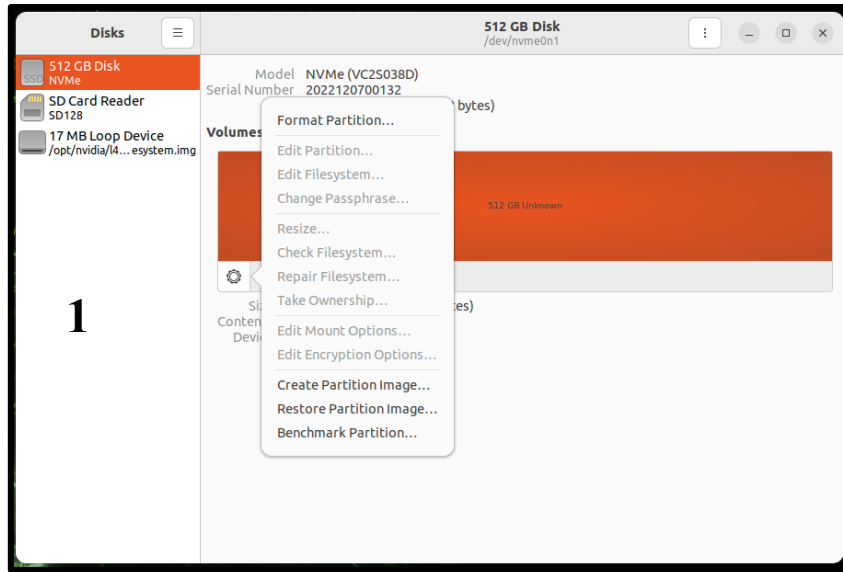
```
vk-jn-or@vkjnor-desktop: ~/Desktop/Jetson-DskTp/Jet-Nano-Orin/5.SetupwithScripts/AM_JON_2SHARE/shellScripts/Script15
mkfs.fat 4.2 (2021-01-31)
FAT32 filesystem created successfully on /dev/nvme0n1p10.
Processing destination partition: /dev/nvme0n1p11
Corresponding source partition: /dev/mmcblk0p11
Source partition: /dev/mmcblk0p11, Detected filesystem: None
Source partition /dev/mmcblk0p11 has no filesystem. Leaving /dev/nvme0n1p11 empty.
Processing destination partition: /dev/nvme0n1p12
Corresponding source partition: /dev/mmcblk0p12
Source partition: /dev/mmcblk0p12, Detected filesystem: None
Source partition /dev/mmcblk0p12 has no filesystem. Leaving /dev/nvme0n1p12 empty.
Processing destination partition: /dev/nvme0n1p13
Corresponding source partition: /dev/mmcblk0p13
Source partition: /dev/mmcblk0p13, Detected filesystem: None
Source partition /dev/mmcblk0p13 has no filesystem. Leaving /dev/nvme0n1p13 empty.
Processing destination partition: /dev/nvme0n1p14
Corresponding source partition: /dev/mmcblk0p14
Source partition: /dev/mmcblk0p14, Detected filesystem: None
Source partition /dev/mmcblk0p14 has no filesystem. Leaving /dev/nvme0n1p14 empty.
Processing destination partition: /dev/nvme0n1p15
Corresponding source partition: /dev/mmcblk0p15
Source partition: /dev/mmcblk0p15, Detected filesystem: None
Source partition /dev/mmcblk0p15 has no filesystem. Leaving /dev/nvme0n1p15 empty.
Flushing all writes before UUID adjustments...
Adjusting filesystem UUIDs...
Processing destination partition: /dev/nvme0n1p1 (type: ext4)
Corresponding source partition: /dev/mmcblk0p1
Checking filesystem on /dev/nvme0n1p1...
e2fsck 1.46.5 (30-Dec-2021)
Pass 1: Checking inodes, blocks, and sizes
Pass 2: Checking directory structure
Pass 3: Checking directory connectivity
Pass 4: Checking reference counts
Pass 5: Checking group summary information
/dev/nvme0n1p1: 11/7708672 files (0.0% non-contiguous), 630352/30834688 blocks
tune2fs 1.46.5 (30-Dec-2021)
Setting the UUID on this filesystem could take some time.
Proceed anyway (or wait 5 seconds to proceed) ? (y,N) <proceeding>
Updated UUID for /dev/nvme0n1p1.
Processing destination partition: /dev/nvme0n1p2 (type: ext4)
Corresponding source partition: /dev/mmcblk0p2
Checking filesystem on /dev/nvme0n1p2...
e2fsck 1.46.5 (30-Dec-2021)
Error reading block 62423039 (Invalid argument). Ignore error<y>? yes
Force rewrite<y>? yes
Superblock has an invalid journal (inode 8).
Clear<y>? yes
*** journal has been deleted ***

The filesystem size (according to the superblock) is 125026560 blocks
The physical size of the device is 32768 blocks
Either the superblock or the partition table is likely to be corrupt!
Abort<y>?
```

SSD formatted using Disks App
“Automated Hardware Prep” of shell script skipped

Various error messages,
ignore to continue migration
but may face problems with
filesystem

Run Disks App and Fix in Shell Script



Terminal O/P: jonSetup15Auto.sh

```
vk-jn-or@vkjnor-desktop: ~/Desktop/Jetson-DskTp/Jet-Nano-Orin/5.SetupwithScripts/AM_JON_2SHARE/shellScripts/Script15
vk-jn-or@vkjnor-desktop:~/Desktop/Jetson-DskTp/Jet-Nano-Orin/5.SetupwithScripts/AM_JON_2SHARE/shellScripts/Script15$ ./jonSetup15Auto.sh

-----
Starting Migration from SD Card to SSD
Ensure SSD is physically installed in the PCI slot before proceeding.
-----
[STEP 1/8]: Rescanning PCI bus to find SSD...
Found NVMe SSD: 476.9G detected.
Wiping existing signatures and partition tables...
/dev/nvme0n1: 2 bytes were erased at offset 0x00000438 (ext4): 53 ef
SSD is now clean and ready.
[STEP 2/8]: Setting up directory at /home/vk-jn-or/Downloads/Jetson-Orin-Nano
[STEP 3/8]: Creating partition structure on SSD...
Source disk: /dev/mmcblk0
Destination disk: /dev/nvme0n1
Are you sure you want to clone the partition structure and file systems? This will overwrite /dev/nvme0n1. (y/N): y
Clearing all partition and filesystem metadata from /dev/nvme0n1...
Creating new GPT entries in memory.
GPT data structures destroyed! You may now partition the disk using fdisk or
other utilities.
Backing up partition table from /dev/mmcblk0...
The operation has completed successfully.
Restoring partition table to /dev/nvme0n1...
Creating new GPT entries in memory.
Warning! Current disk size doesn't match that of the backup!
Adjusting sizes to match, but subsequent problems are possible!
The operation has completed successfully.
Randomizing GUIDs for /dev/nvme0n1...
The operation has completed successfully.
Flushing disk writes...
Reloading partition table on /dev/nvme0n1...
Replicating file systems...
Processing destination partition: /dev/nvme0n1p1
Corresponding source partition: /dev/mmcblk0p1
Source partition: /dev/mmcblk0p1, Detected filesystem: ext4
Creating ext4 filesystem on /dev/nvme0n1p1...
mke2fs 1.46.5 (30-Dec-2021)
Discarding device blocks: done
Creating filesystem with 30834688 4k blocks and 7708672 inodes
Filesystem UUID: af234525-91f3-43c8-8f3a-9962d34a87cb
Superblock backups stored on blocks:
    32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632, 2654208,
    4096000, 7962624, 11239424, 20480000, 23887872

Allocating group tables: done
Writing inode tables: done
Creating journal (131072 blocks): done
Writing superblocks and filesystem accounting information: done
}
ext4 filesystem created successfully on /dev/nvme0n1p1.
Processing destination partition: /dev/nvme0n1p2
Corresponding source partition: /dev/mmcblk0p2
Source partition: /dev/mmcblk0p2, Detected filesystem: None
Source partition /dev/mmcblk0p2 has no filesystem. Leaving /dev/nvme0n1p2 empty.
Processing destination partition: /dev/nvme0n1p3
```

**No error in Step 3,
superblocks and filesystem
created, ext4 filesystem
done, ...**

Terminal O/P: jonSetup15Auto.sh

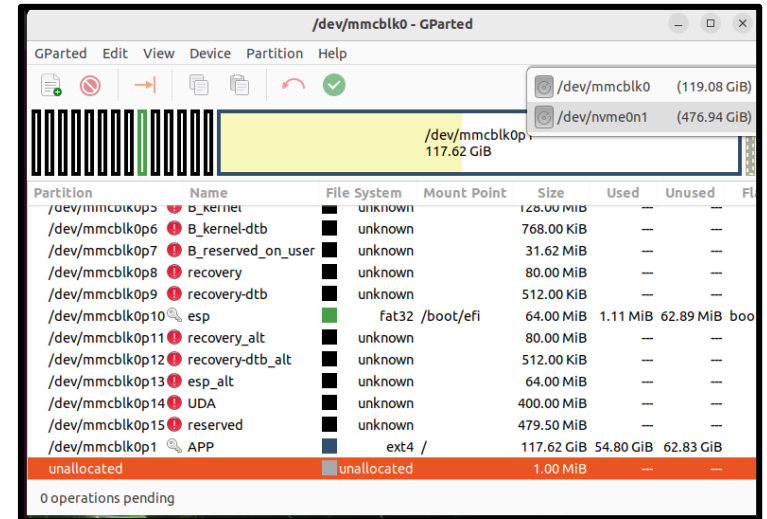
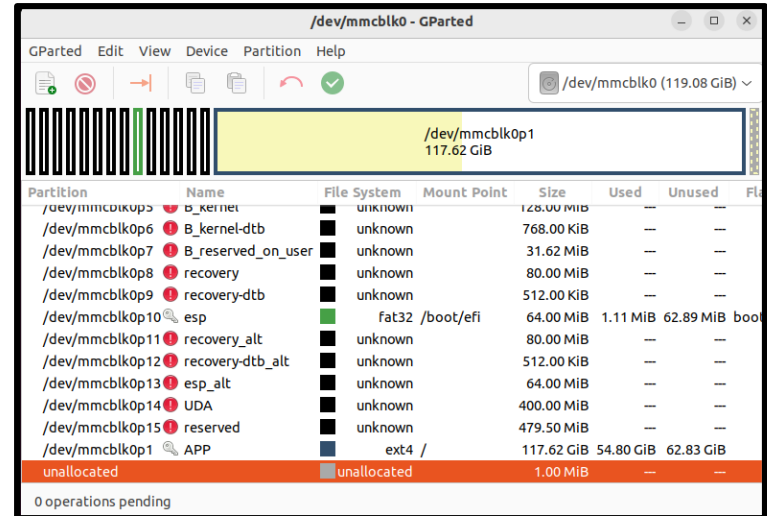
```
vk-jn-or@vkjnor-desktop: ~/Desktop/Jetson-DskTp/Jet-Nano-Orin/5.SetupwithScripts/AM_JON_2SHARE/shellScripts/Script15
Processing destination partition: /dev/nvme0n1p2 (type: )
Corresponding source partition: /dev/mmcblk0p2
Skipping /dev/nvme0n1p2: No filesystem detected.
Processing destination partition: /dev/nvme0n1p3 (type: )
Corresponding source partition: /dev/mmcblk0p3
Skipping /dev/nvme0n1p3: No filesystem detected.
Processing destination partition: /dev/nvme0n1p4 (type: )
Corresponding source partition: /dev/mmcblk0p4
Skipping /dev/nvme0n1p4: No filesystem detected.
Processing destination partition: /dev/nvme0n1p5 (type: )
Corresponding source partition: /dev/mmcblk0p5
Skipping /dev/nvme0n1p5: No filesystem detected.
Processing destination partition: /dev/nvme0n1p6 (type: )
Corresponding source partition: /dev/mmcblk0p6
Skipping /dev/nvme0n1p6: No filesystem detected.
Processing destination partition: /dev/nvme0n1p7 (type: )
Corresponding source partition: /dev/mmcblk0p7
Skipping /dev/nvme0n1p7: No filesystem detected.
Processing destination partition: /dev/nvme0n1p8 (type: )
Corresponding source partition: /dev/mmcblk0p8
Skipping /dev/nvme0n1p8: No filesystem detected.
Processing destination partition: /dev/nvme0n1p9 (type: )
Corresponding source partition: /dev/mmcblk0p9
Skipping /dev/nvme0n1p9: No filesystem detected.
Processing destination partition: /dev/nvme0n1p10 (type: vfat)
Corresponding source partition: /dev/mmcblk0p10
Source FAT32 label for /dev/mmcblk0p10: <none>
Source FAT32 partition /dev/mmcblk0p10 has no label. Skipping label assignment.
Source PARTLABEL for /dev/mmcblk0p10: esp
Setting PARTLABEL: esp on /dev/nvme0n1p10...
Setting name!
partNum is 9
The operation has completed successfully.
Updated PARTLABEL for /dev/nvme0n1p10 to esp.
Processing destination partition: /dev/nvme0n1p11 (type: )
Corresponding source partition: /dev/mmcblk0p11
Skipping /dev/nvme0n1p11: No filesystem detected.
Processing destination partition: /dev/nvme0n1p12 (type: )
Corresponding source partition: /dev/mmcblk0p12
Skipping /dev/nvme0n1p12: No filesystem detected.
Processing destination partition: /dev/nvme0n1p13 (type: )
Corresponding source partition: /dev/mmcblk0p13
Skipping /dev/nvme0n1p13: No filesystem detected.
Processing destination partition: /dev/nvme0n1p14 (type: )
Corresponding source partition: /dev/mmcblk0p14
Skipping /dev/nvme0n1p14: No filesystem detected.
Processing destination partition: /dev/nvme0n1p15 (type: )
Corresponding source partition: /dev/mmcblk0p15
Skipping /dev/nvme0n1p15: No filesystem detected.
Forcing partition table reread after UUID updates...
Partition cloning, file system replication, and UUID adjustment complete.
[STEP 4/8]: Copying data from SD to SSD (This may take a while)...
Source disk: /dev/mmcblk0
Destination disk: /dev/nvme0n1
Are you sure you want to copy contents from /dev/mmcblk0 to /dev/nvme0n1? (Y/N): y
```

Ready to copy SD content to SSD

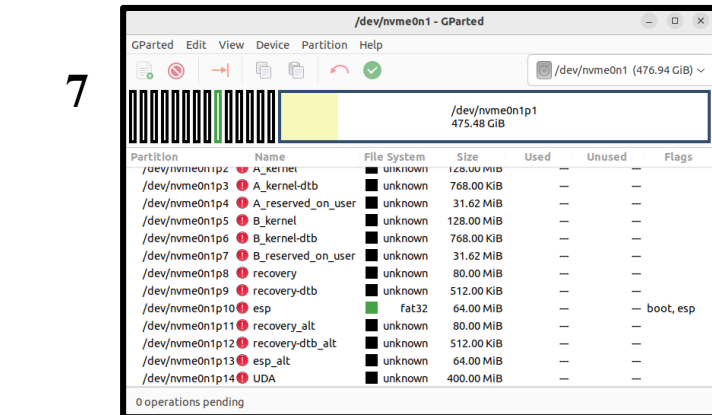
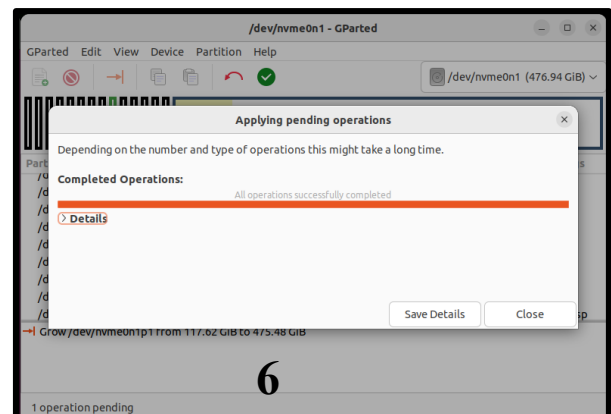
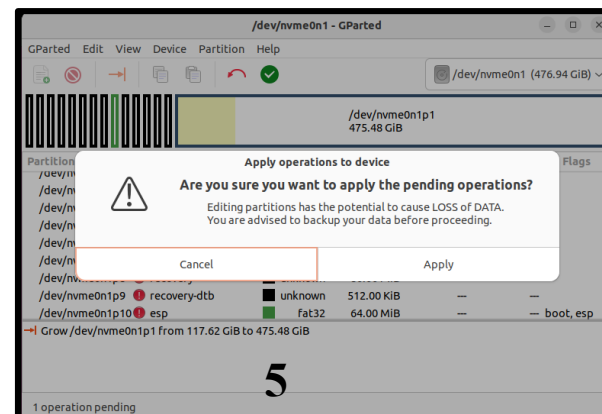
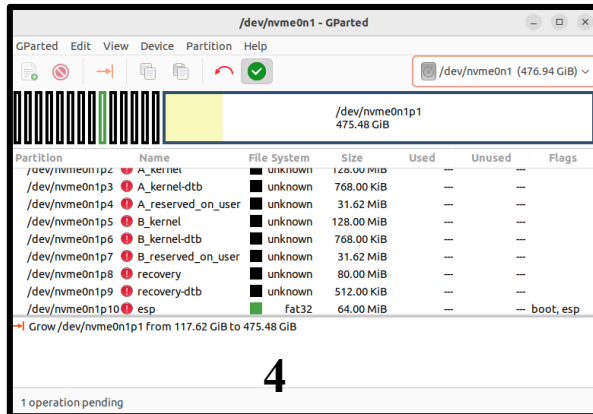
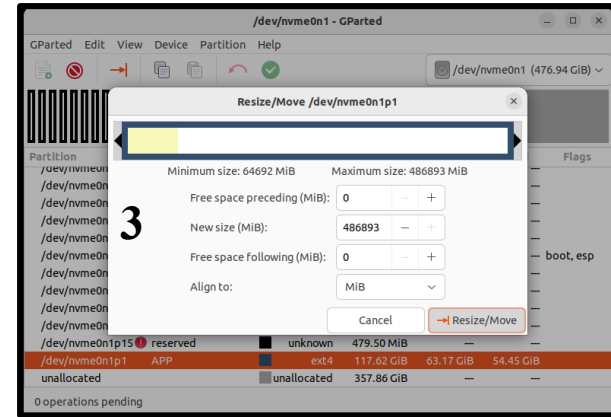
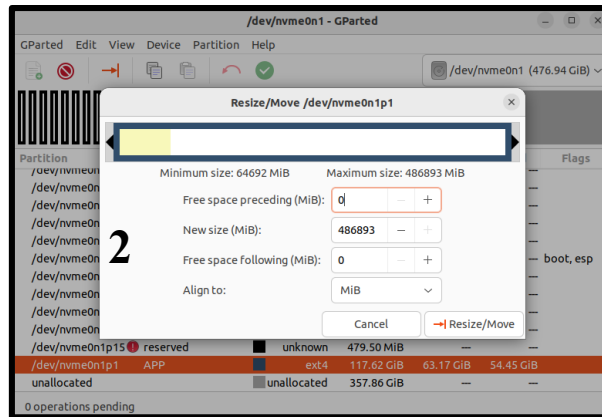
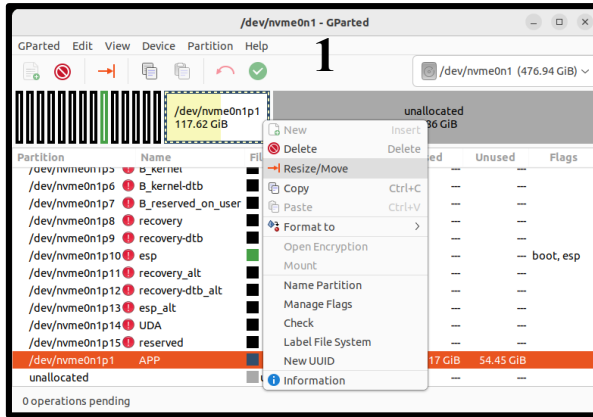
Terminal and Gparted Output: jonSetup15Auto.sh

```
vk-jn-or@vkjnor-desktop: ~/Desktop/Jetson-DskTp/Jet-Nano-Orin/5.SetupwithScripts/AM_JON_2SHARE/shellScripts/Script15

Checking filesystem compatibility from /dev/mmcblk0p15 () to /dev/nvme0n1p15 ()...
Source partition /dev/mmcblk0p15 has no filesystem. Performing raw copy using dd...
444596224 bytes (445 MB, 424 MiB) copied, 5 s, 88.2 MB/s
119+1 records in
119+1 records out
502792192 bytes (503 MB, 480 MiB) copied, 6.28967 s, 79.9 MB/s
Raw copy from /dev/mmcblk0p15 to /dev/nvme0n1p15 completed.
Flushing disk caches...
Partition content copy complete.
[STEP 5/8]: Modifying SSD configuration for boot...
Detected EFI partition: /dev/nvme0n1p10
EFI partition UUID: 3C08-7481
Updated /mnt/ssh/boot/extlinux/extlinux.conf with root=/dev/nvme0n1p1
Updated /mnt/ssh/etc/fstab with EFI UUID=3C08-7481
System configurations updated on the SSD.
You may need to change the boot order to prioritize the SSD.
[STEP 6/8]: Verifying file integrity...
SD Card (/usr) file count: 233210
SSD (/usr) file count: 233210
✓ Integrity Check Passed: Program files match exactly.
[STEP 7/8]: Installing GParted to expand SSD storage...
Get:1 file:/var/cudnn-local-tesra-repo-ubuntu2204-9.3.0 InRelease [1,572 B]
Get:2 file:/var/l4t-cuda-tesra-repo-ubuntu2204-12.6-local InRelease [1,572 B]
Get:1 file:/var/cudnn-local-tesra-repo-ubuntu2204-9.3.0 InRelease [1,572 B]
Get:2 file:/var/l4t-cuda-tesra-repo-ubuntu2204-12.6-local InRelease [1,572 B]
Get:3 file:/var/nv-tensorrt-local-tesra-repo-ubuntu2204-10.3.0-cuda-12.5 InRelease [1,572 B]
Get:3 file:/var/nv-tensorrt-local-tesra-repo-ubuntu2204-10.3.0-cuda-12.5 InRelease [1,572 B]
Hit:4 https://repo.download.nvidia.com/jetson/common r36.4 InRelease
Hit:5 https://download.docker.com/linux/ubuntu jammy InRelease
Hit:6 http://ports.ubuntu.com/ubuntu-ports jammy InRelease
Hit:7 http://ports.ubuntu.com/ubuntu-ports jammy-updates InRelease
Hit:8 http://ports.ubuntu.com/ubuntu-ports jammy-security InRelease
Hit:9 https://repo.download.nvidia.com/jetson/t234 r36.4 InRelease
Hit:10 https://packages.microsoft.com/repos/code stable InRelease
Hit:11 https://repo.download.nvidia.com/jetson/ffmpeg r36.4 InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
gparted is already the newest version (1.3.1-1ubuntu1).
The following packages were automatically installed and are no longer required:
  libavcodec58 libavdevice58 libavfilter7 libavformat58 libavutil56 libpostproc55 libssh-gcrypt-4 libswresample3 libswscale5
Use 'sudo apt autoremove' to remove them.
0 upgraded, 0 newly installed, 0 to remove and 293 not upgraded.
-----
MANUAL ACTION REQUIRED IN GPARTED:
1. Select /dev/nvme0n1 from the top-right menu.
2. Right-click the 'APP' (root) partition.
3. Select 'Resize/Move' and drag the arrow to fill unallocated space.
4. Click the 'Checkmark' icon to Apply Changes.
-----
GParted 1.3.1
configuration --enable-libparted-dmraid --enable-online-resize
libparted 3.4
□
```



Gparted and Terminal O/P: jonSetup15Auto.sh



8

MANUAL ACTION REQUIRED IN GPARTED:

1. Select /dev/nvme0n1 from the top-right menu.
2. Right-click the 'APP' (root) partition.
3. Select 'Resize/Move' and drag the arrow to fill unallocated space.
4. Click the 'Checkmark' icon to Apply Changes.

GParted 1.3.1 configuration --enable-libparted-dmraid --enable-online-resize libparted 3.4

Press [Enter] once you have finished resizing in GParted...

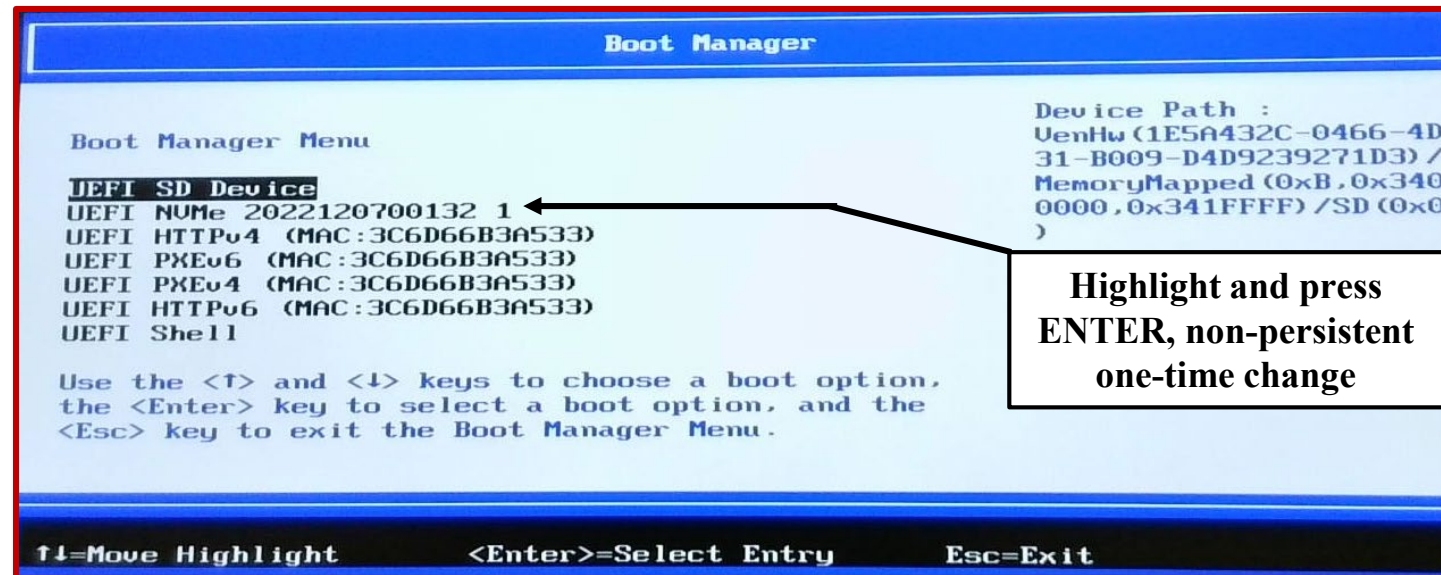
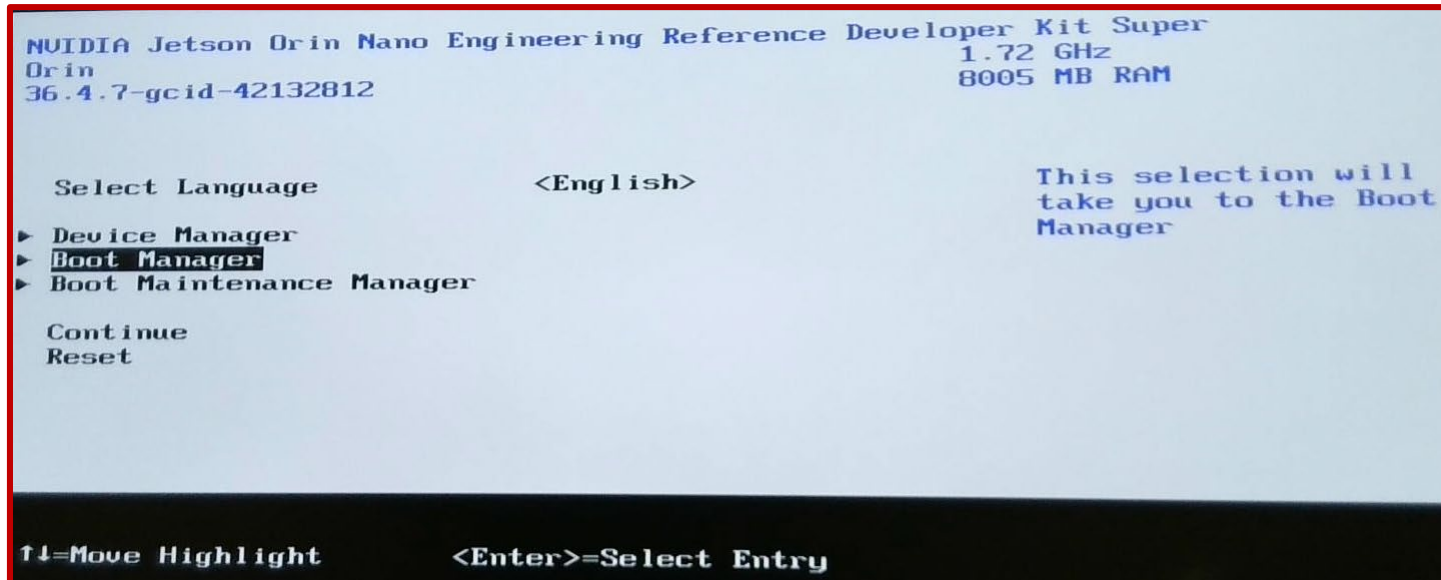
[STEP 8/8]: MIGRATION COMPLETE!

To finish, we will reboot. During boot:

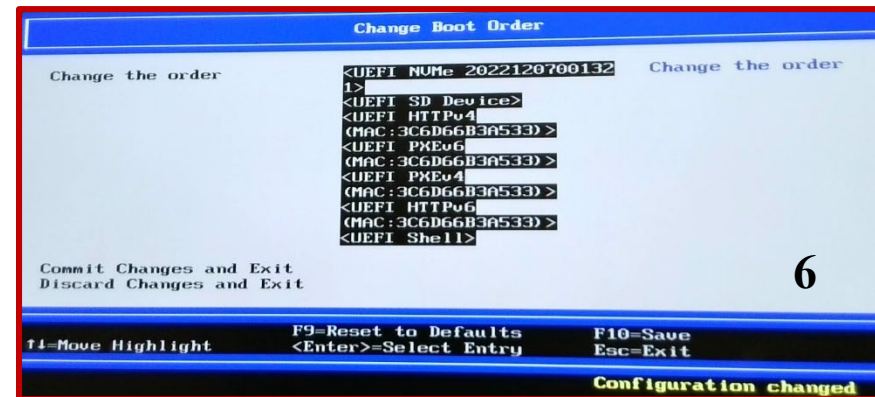
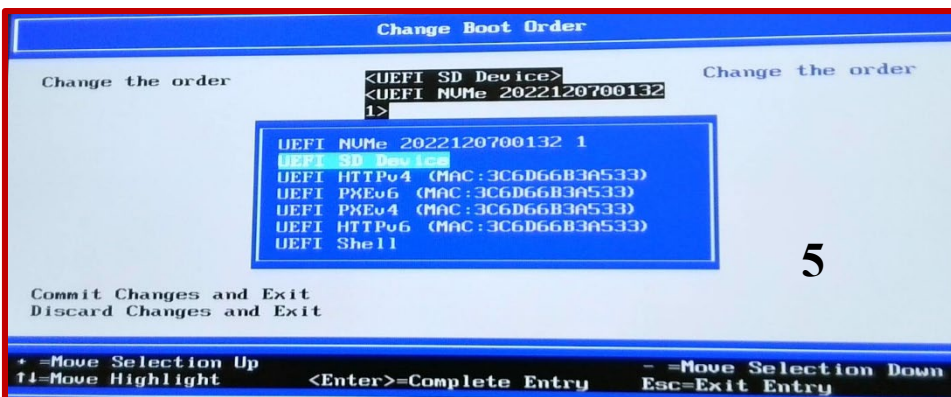
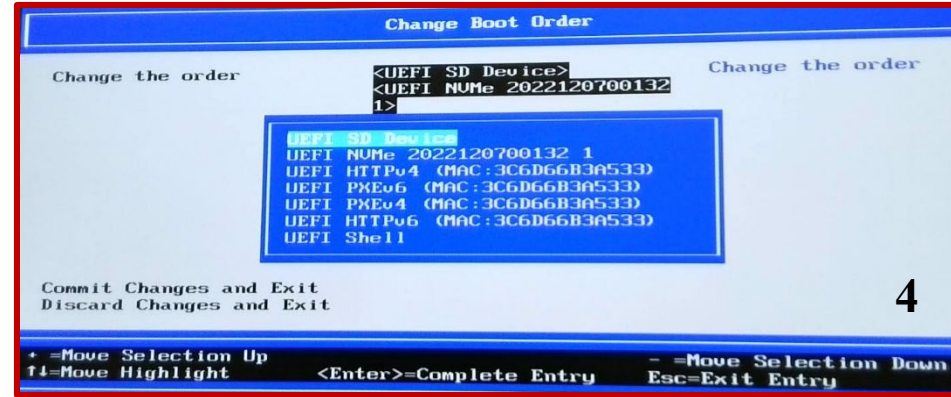
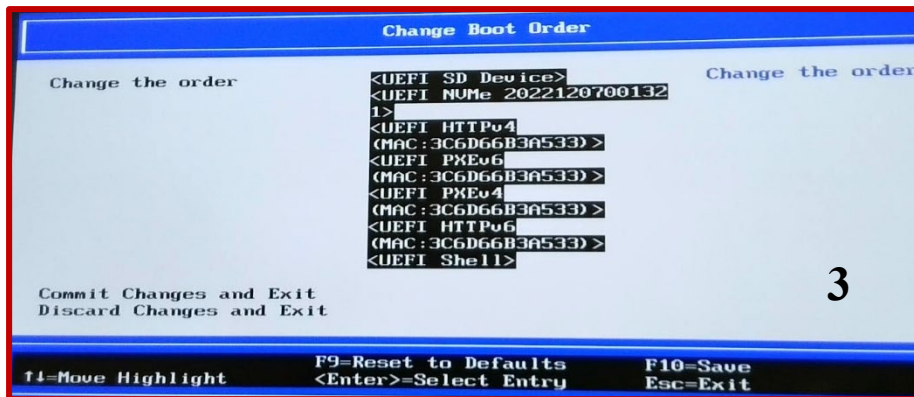
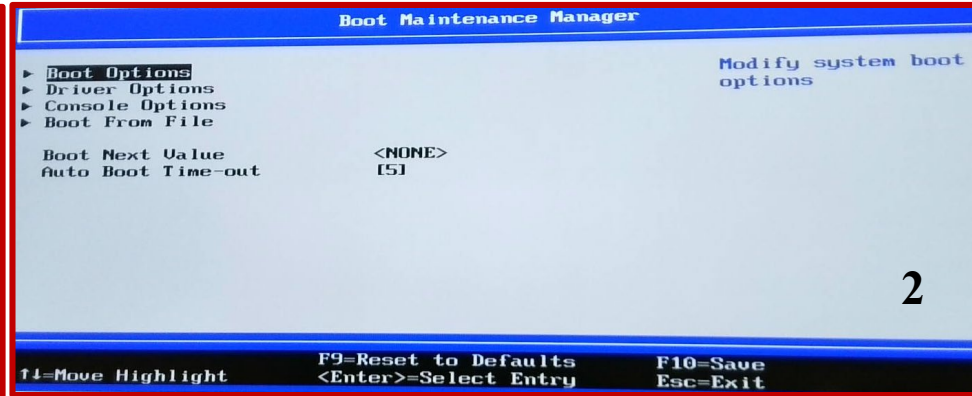
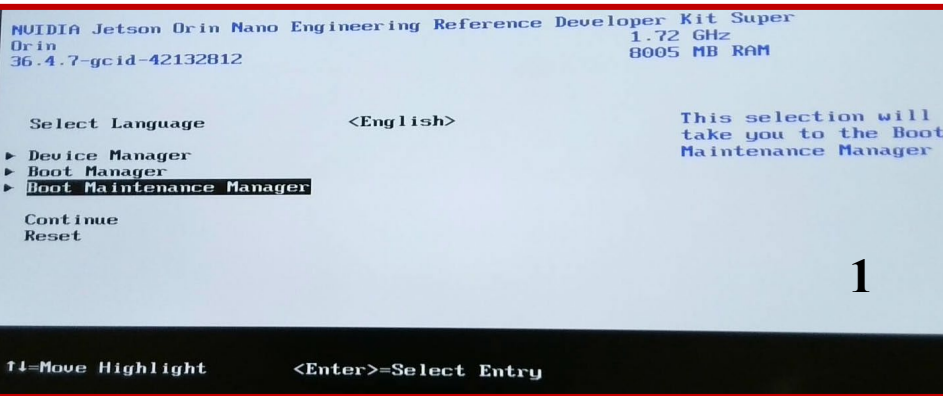
1. Repeatedly press [ESC] to enter BIOS/UEFI.
2. Go to: Boot Maintenance Manager -> Boot Options -> Change Boot Order.
3. Move SSD to the top using the '+' key.
4. Save and Exit.

Ready to reboot? (y/n): y

Boot Order: Repeated ESC on Bootup



Fix Boot Order



Fix Boot Order

Change Boot Order

Change the order

<UEFI NUMe 2022120700132
1>
<UEFI SD Device>
<UEFI HTTPv4>

Change the order

1

Save configuration changes?
Press 'Y' to confirm, 'N'/'ESC' to ignore.

(MAC:3C6D66B3A533)
<UEFI Shell>

Commit Changes and Exit
Discard Changes and Exit

↑↓=Move Highlight

F9=Reset to D
<Enter>=Select

NUIDIA Jetson Orin Nano Engineering Reference Developer Kit Super
Orin
36.4.7-gcid-42132812
1.72 GHz
8005 MB RAM

2

Select Language

<English>

This selection will
take you to the Boot
Manager

- ▶ Device Manager
- ▶ **Boot Manager**
- ▶ Boot Maintenance Manager

Continue
Reset

Boot Manager

Boot Manager Menu

3

UEFI NUMe 2022120700132 1
UEFI SD Device
UEFI HTTPv4 (MAC:3C6D66B3A533)
UEFI PXEv6 (MAC:3C6D66B3A533)
UEFI PXEv4 (MAC:3C6D66B3A533)
UEFI HTTPv6 (MAC:3C6D66B3A533)
UEFI Shell

Use the <↑> and <↓> keys to choose a boot o
the <Enter> key to select a boot option, an
<Esc> key to exit the Boot Manager Menu.

Device Path :
UenHw (1E5A432C-0466-4D
31-B009-D4D9239271D3) /
MemoryMapped (0xB,0x141

NUIDIA Jetson Orin Nano Engineering Reference Developer Kit Super
Orin
36.4.7-gcid-42132812
1.72 GHz
8005 MB RAM

Select Language

<English>

Continue

- ▶ Device Manager
- ▶ **Boot Manager**
- ▶ Boot Maintenance Manager

Continue
Reset

4

↑↓=Move Highlight

<Enter>=Select Entry

↑↓=Move Highlight

<Enter>=Select Entry